Universität Bonn

Mathematisches Institut
Prof. Dr. Shuntaro Yamagishi
Number Theory I
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Lecture Notes as of 12th July 2026

LaTeX by Hannes Heimberger and Carl Jontza

Corrections and Remarks

If you find a mistake in this file or have a remark concerning this document, I would be glad if you let me know. It would be great if you wrote a mail. I can be reached through the address hannes.heimberger@uni-bonn.de.

Solutions to exercises

I will add solutions to the exercises after their correction, at least if my tutor marked them correct. These may, however, not be written down in the most optimal way. The proofs left as exercsises will be marked as such. If you have suggestions about how to improve these proofs, comments are welcome.

Colours

My annotations to the notes are blue. Additional details that are neither part of the lecture nor relevant to the exam will be coloured in orange.

Thanks

I thank Daniel Emse for providing his template, which I adapted for these notes.

Contents

Number Theory I

Notations and conventions

Number sets

- The natural numbers, denoted by $\mathbb{N}$, contain 0 (which is in accord with *DIN 5473* (*German Industrial Norm*) – Welcome to Germany! xD).

Subsets and -structures

- If we write $A \subset B$, we mean $A \subseteq B$ and $A \neq B$ (see *DIN 5473*).
- We write $\sqsubset$ or $\sqsubseteq$ for substructures (subfields, linear subspaces of vector spaces, etc.) of algebraic structures (didn't yet make it into *DIN 5473* :().

For $f : \mathbb{C} \times \cdots \to \mathbb{C}$, we define

$$N(f) = \{s \in \mathbb{C} : f(s, -) = 0\}$$

and

$$N^*(f) = \{s \in [0, 1] \times \mathbb{R} \subset \mathbb{C} : f(s, -) = 0\}.$$

Quantors

- If we write $\forall\!\!\!\forall_{\dots}$, we mean something holds for almost all ...

Functions

- We have $\{x\} = x - \lfloor x \rfloor \;\; \forall\!\!\!\forall_{x \in \mathbb{R}}$.

Fields

- We write $\mathbb{A}$ for $\overline{\mathbb{Q}}$.

Chapter 1

Geometry of Numbers

1.1 Introduction

Definition 1.1.1 – $\mathbb{Z}^n_{\text{prim}}$

We denote

$$\mathbb{Z}^n_{\text{prim}} = \{(x_1, \ldots, x_n) \in \mathbb{Z}^n : \gcd(x_1, \ldots, x_n) = 1\}.$$

Theorem 1.1.2 (Dirichlet, 1842)

Let $\alpha \in \mathbb{R}$. Then, for every integer $Q \geq 2$, there exists $(x, y) \in \mathbb{Z}^2_{\text{prim}}$ such that $0 < y \leq Q$ and $|x - \alpha y| \leq 1/Q$.

Proof. Consider the $Q+1$ numbers $0, \alpha - \lfloor \alpha \rfloor, 2\alpha - \lfloor 2\alpha \rfloor, \ldots, Q\alpha - \lfloor Q\alpha \rfloor$. Each of these is in the interval $[0, 1)$. Partition the interval $[0, 1)$ into Q subintervals of length $1/Q$. Now, we can apply the pigeonhole principle to deduce that

$$\exists_{0 \leq i \leq j \leq Q} : |(j\alpha - \lfloor j\alpha \rfloor) - (i\alpha - \lfloor i\alpha \rfloor)| \leq \frac{1}{Q}.$$

This implies

$$|(j - i)\alpha - (\lfloor j\alpha \rfloor - \lfloor i\alpha \rfloor)| \leq \frac{1}{Q}.$$

Let $\tilde{y} = j - i$ and $\tilde{x} = \lfloor j\alpha \rfloor - \lfloor i\alpha \rfloor$. Then $|\tilde{y}\alpha - \tilde{x}| \leq 1/Q$. Let $d = (\tilde{y}, \tilde{x})$, $y = \tilde{y}/d$ and $x = \tilde{x}/d$ and divide through by d to get that

$$|y\alpha - x| \leq \frac{1}{dQ} \leq \frac{1}{Q}.$$ □

A very simple consequence of this:

Corollary 1.1.3 (Irrational approximation)

Let $\alpha \in \mathbb{R} \setminus \mathbb{Q}$. Then there exist infinitely many $(x, y) \in \mathbb{Z}^2_{\text{prim}}$ such that $y > 0$ and

$$\left|\alpha - \frac{x}{y}\right| \leq \frac{1}{y^2}.$$

Proof. For each $Q \geq 2$, let $(x_Q, y_Q) \in \mathbb{Z}^2_{\text{prim}}$ be such that $0 < y_Q \leq Q$ and $|x_Q - y_Q\alpha| \leq {}^1/_Q$ (Theorem 1.1.2 ensures the existence of such a pair). This implies

$$\left|\alpha - \frac{x_Q}{y_Q}\right| \leq \frac{1}{Qy_Q} \leq \frac{1}{y_Q^2}.$$

Suppose the set

$$S = \{(x_Q, y_Q) : Q \geq 2\}$$

is finite. Then there exists $(x_0, y_0) \in \mathbb{Z}^2_{\text{prim}}$ such that $(x_Q, y_Q) = (x_0, y_0)$ $\forall_{Q \geq Q_0}$ and some Q_0. This means

$$\left|\alpha - \frac{x_Q}{y_Q}\right| = \left|\alpha - \frac{x_0}{y_0}\right| \leq \frac{1}{Q} \quad \forall_{Q \geq Q_0}.$$

Taking the limit as $Q \to \infty$ yields

$$\alpha = \frac{x_0}{y_0},$$

which is a contradiction due to the fact that α was supposed to be irrational. ↯ □

Question 1.1.4 (Is this error term optimal?)

Can we do better than ${}^1/_{y^2}$ for some $\alpha \in \mathbb{R}$? More explicitly, do there exist infinitely many $(x, y) \in \mathbb{Z}^2_{\text{prim}}$ such that $y > 0$ and

$$\exists_{\delta > 0} : \left|\alpha - \frac{x}{y}\right| < \frac{1}{y^{2+\delta}}.$$

Answer. Yes, but it is rare. However, this is not the case if α is algebraic over $\mathbb{Q}$. □

Theorem 1.1.5 (Roth, 1955)

Let $\alpha \in \mathbb{A}$ and $\delta > 0$. Then

$$\#\{(x, y) \in \mathbb{Z}^2_{\text{prim}} : y > 0, |\alpha - {}^x/_y| \leq {}^1/_{y^{2+\delta}}\} < \aleph_0.$$

Proving this is one of the main goals of this course. Another important result we will prove is Thue's theorem.

Theorem 1.1.6 (Thue, 1909)

Let $F \in \mathbb{Z}[X, Y]$ be a binary form (hence homogenious) of degree d. Suppose $\deg F(X, 1) = d$ and that $F(X, 1)$ has at least three distinct zeroes in $\mathbb{C}$. Let $m \in \mathbb{Z} \setminus \{0\}$. Then

$$\#\{(x, y) \in \mathbb{Z}^2 : F(x, y) = m\} < \aleph_0.$$

Also, time permitted, Professor Yamagishi would like to go through the details of Stanley

Yao Xiao's proof of Büchi's problem.[1]

Problem 1.1.7 (Büchi)

Let $0 < x_1 < \cdots < x_5$ be integers such that

$$x_{i+1}^2 - 2x_i^2 + x_{i-1}^2 = 2 \ \forall_{i \in \{2,\dots,4\}}.$$

Then there exists $x_0 \in \mathbb{Z}$ such that $x_i = x_0 + i \ \forall_{1 \le i \le 5}$.

This leads to the following refinement of Hilbert's tenth problem: There is no algorithm which answers the question about the existence of integer solutions to an arbitrary system of diagonal quadratic equations:

$$A \begin{pmatrix} x_1^2 \\ \vdots \\ x_n^2 \end{pmatrix} = b.$$

Professor Yamagishi recommends the book *An introduction to the geometry of numbers*[1] by Cassels. He also recommends a course on **Prime Numbers** taught next semester (Selected Topics in Number Theory). Tutorial times are **Monday at 14:00** and **Tuesday at 12:00**. You can register for these **Tomorrow at 16:00**.

Lecture 1 (15th April 2026 - W1L1)

Lecture 2 (17th April 2026 - W1L2)

1.2 Lattices and discrete subgroups

Let $n \geq 2$. Given $x_1, \dots, x_n \in \mathbb{R}^n$, we say that they are linearly independent over $\mathbb{R}$, if $c_1x_1 + \cdots + c_rx_r \neq 0 \ \forall_{(c_1,\dots,c_r)^t \in \mathbb{R}^r \setminus \{0\}}$. We consider the usual topology on $\mathbb{R}^n$ generated by the Euclidean metric. For subsets of $\mathbb{R}^n$, we consider the subspace topology. We say a subset $\Lambda \subseteq \mathbb{R}^n$ is **discrete** if $\#\Lambda \cap B < \aleph_0$ for every bounded subset $B \subseteq \mathbb{R}^n$. Furthermore, Λ is a **discrete subgroup** if it is discrete and $c_1x_1 + c_2x_2 \in \Lambda \ \forall_{c_1,c_2 \in \mathbb{Z}, x_1,x_2 \in \Lambda}$. In this case, we define $\operatorname{rank}(\Lambda) = r$ as the maximal number r such that Λ contains r linearly independent vectors over $\mathbb{R}$. A **basis** of Λ is a set $\{v_1, \dots, v_r\}$ of linearly independent vectors over $\mathbb{R}$ such that $\Lambda = \{c_1v_1 + \cdots + c_rv_r : c_1, \dots, c_r \in \mathbb{Z}\}$.

Lemma 1.2.1 (Lattice Basis Construction)

Let Λ be a discrete subgroup of $\mathbb{R}^n$ and $\operatorname{rank}(\Lambda) = r \geq 1$. Also assume that $\{w_1, \dots, w_r\} \subseteq \Lambda$ is linearly independent over $\mathbb{R}$. Then there exists a basis $\{v_1, \dots, v_r\}$ of Λ such that for all $1 \leq k \leq r$, we have

$$v_k = \alpha_{k,1}w_1 + \cdots + \alpha_{k,k}w_k \text{ for some } \alpha_{k,1}, \dots, \alpha_{k,k} \in [0,1] \text{ with } \alpha_{k,k} \neq 0.$$

Proof. Let $1 \leq k \leq r$. We define

$$S_k = \{\alpha_1 w_1 + \cdots + \alpha_k w_k : \alpha_1, \dots, \alpha_k \in [0,1], \alpha_k \neq 0\}.$$

[1] Professor Yamagishi: This is a paper by a friend of mine. Actually, it has nothing to do with geometry of numbers. But I want to do it anyway.

Since Λ is discrete and S_k is bounded,

$$\underbrace{0 <}_{w_k \in \Lambda \cap S_k} \#(\Lambda \cap S_k) \ll 1.$$

We choose v_k to be a vector from $\Lambda \cap S_k$ with minimal α_k. Let us write

$$v_k = \alpha_{k,1} w_1 + \cdots + \alpha_{k,k} w_k.$$

Then for any

$$\alpha_1 w_1 + \cdots + \alpha_k w_k \in \Lambda \cap S_k,$$

we have $0 < \alpha_{k,k} \leq \alpha_k \leq 1$. It is clear that $\{v_1, \ldots, v_r\}$ is linearly independent over $\mathbb{R}$. For each $0 \leq k \leq r$, let

$$M_k = \Lambda \cap \left\{ \sum_{j=1}^{k} \alpha_j w_j : \alpha_1, \ldots, \alpha_k \in \mathbb{R} \right\} \text{ with } M_0 = \{0\}.$$

Note that $M_r = \Lambda$. We now use induction to show that $M_k = \{c_1 v_1 + \cdots + c_k v_k : c_1, \ldots, c_k \in \mathbb{Z}\}\ \forall_{0 \leq k \leq r}$. When $k = 0$, the statement is trivially true. Suppose the statement holds for $k-1$.
Let

$$x = \sum_{j=1}^{k} \beta_j w_j \in M_k \subseteq \Lambda \text{ and set } z_k = \left\lfloor \frac{\beta_k}{\alpha_{k,k}} \right\rfloor \in \mathbb{Z}.$$

Then, $0 \leq \beta_k - z_k \alpha_{k,k} < \alpha_{k,k}$. Consider $x - z_k \underbrace{v_k}_{\in \Lambda} \in \Lambda$. Also,

$$\begin{aligned} x - z_k v_k &= \sum_{j=1}^{k} (\beta_j - z_k \alpha_{k,j}) w_j \\ &= \underbrace{\sum_{j=1}^{k-1} \lfloor \beta_j - z_k \alpha_{k,j} \rfloor w_j}_{\in \Lambda} + u, \end{aligned}$$

where

$$u = \sum_{j=1}^{k-1} \{\beta_j - z_k \alpha_{k,j}\} w_j + (\beta_k - z_k \alpha_{k,k}) w_k.$$

It is clear that $u \in \Lambda$. Also, if $\beta_k - z_k \alpha_{k,k} > 0$, then $u \in S_k$ (because $\{\beta_j - z_k \alpha_{k,j}\} \in [0,1]$ and $0 < \beta_k - z_k \alpha_{k,k} < \alpha_{k,k} \leq 1$), which implies $u \in \Lambda \cap S_k$. But this contradicts the minimality of $\alpha_{k,k}$ ↯. Hence, $\beta_k = z_k \alpha_{k,k}$. Then,

$$x - z_k v_k = \sum_{j=1}^{k-1} (\beta_j - z_k \alpha_{k,j}) w_j \in M_{k-1}.$$

This concludes the induction. $\square$

Definition 1.2.2 – (Full) Lattice

A **(full) lattice** in $\mathbb{R}^n$ is a discrete sugroup of $\mathbb{R}^n$ which has rank n.

Remark: For our intents and purposes, a **lattice** is always **full**.

Remark (Bases of Lattices)**:** If Λ is a lattice in $\mathbb{R}^n$, then by the previous lemma, there exists a basis $\{v_1, \ldots, v_n\}$ such that $\Lambda = \{c_1 v_1 + \cdots + c_n v_n : c_1, \ldots, c_n \in \mathbb{Z}\}$.

Definition 1.2.3 – Determinant of a Lattice

We define the **determinant** of Λ to be

$$\det(\Lambda) = |\det(v_1 \cdots v_n)|\,.$$

Recall that $\mathrm{GL}_n(\mathbb{Z})$ is the multiplicative group of $n \times n$-matrices over $\mathbb{Z}$ with determinant ± 1.

Lemma 1.2.4 (Independence of the Determinant of a Lattice from the Basis Choice)

Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice. Suppose that $\{v_1, \ldots, v_n\}$ and $\{w_1, \ldots, w_n\}$ are two bases of Λ. Then there exists $u = (u_{i,j})_{i,j \in \{1,\ldots,n\}} \in \mathrm{GL}_n(\mathbb{Z})$ such that

$$w_i = \sum_{j=1}^{n} u_{i,j} v_j \ \forall_{1 \le i \le n}.$$

In particular,

$$|\det(w_1 \cdots w_n)| = |\det(v_1 \cdots v_n)|.$$

Proof. Let us define $U = (u_{i,j})_{i,j \in \{1,\ldots,n\}}$ by

$$w_i = \sum_{j=1}^{n} u_{i,j} v_j \ \forall_{1 \le i \le n}.$$

Since $w_i \in \Lambda$ and $\{v_1, \ldots, v_n\}$ is a basis of Λ, we get $u_{i,j} \in \mathbb{Z}\ \forall_{1 \le i,j \le n}$, which ensures $\det(U) \in \mathbb{Z}$. By the same argument, we define $\widetilde{U} = (\tilde{u}_{i,j})_{i,j \in \{1,\ldots,n\}}$ analogously. Note that $U^{-1} = \widetilde{U}$. Hence, $\det(\widetilde{U}) = \det(U) = \pm 1$. $\square$

Let M, L be two lattices in $\mathbb{R}^n$ with $M \subseteq L$. Further let $\{v_1, \ldots, v_n\}$ be a basis of L and $\{w_1, \ldots, w_n\}$ be a basis of M. Define U_0 as the $n \times n$ matrix that fulfils

$$(w_1 \cdots w_n) = (v_1 \cdots v_n) U_0.$$

Then, we have $U_0 \in M_n(\mathbb{Z})$. We define the **index** of M in L by

$$[L : M] = |\det U_0| = \left| \frac{\det M}{\det L} \right|.$$

Note: This does not depend on the choice of bases of M and L.

Example. Let

$$\Lambda_0 = \mathrm{Span}_{\mathbb{Z}}\left\{\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix}\right\} \text{ and } \Lambda = \mathrm{Span}_{\mathbb{Z}}\left\{\begin{pmatrix}2\\0\end{pmatrix}, \begin{pmatrix}0\\2\end{pmatrix}\right\}.$$

It is clear that

$$U_0 = \begin{pmatrix}2 & 0\\0 & 2\end{pmatrix}$$

here. We thus have

$$[\Lambda : \Lambda_0] = |\det U_0| = 4,$$

which aligns nicely with the fact that every fourth point $\lambda \in \Lambda_0$ is also in Λ. □

Lecture 2 (17th April 2026 - W1L2)

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Lecture 3 (22nd April 2026 - W2L1)

1.3 Convex Bodies

Definition 1.3.1 – Convex Subset of $\mathbb{R}^n$

A set $C \subseteq \mathbb{R}^n$ is **convex** if given any $x, y \in C$, we have

$$\{tx + (1-t)y : t \in [0,1]\} \subseteq C.$$

Definition 1.3.2 – Central Symmetric Convex Body in $\mathbb{R}^n$

A **central symmetric convex body** in $\mathbb{R}^n$ is a closed bounded convex $C \subseteq \mathbb{R}^n$ fulfilling $0 \in C^\circ$ and $x \in C \iff -x \in C$.

Example. Consider any norm on $\mathbb{R}^n$. Then the set

$$B_1(0) = \{x \in \mathbb{R}^n : \|x\| \leq 1\}$$

is a central symmetric convex body. □

Let $C \subseteq \mathbb{R}^n$ be a central symmetric convex body. Define

$$\|x\|_C = \min\{\lambda \in \mathbb{R}_{\geq 0} : x \in \lambda C\},$$

where $\lambda C = \{\lambda x : x \in C\}$.

Lemma 1.3.3 (Norm properties of $\|\cdot\|_C$)

The map $\|\cdot\|_C$ is

(i) well-defined,

(ii) a norm on $\mathbb{R}^n$ and

(iii) $\lambda C = \{x \in \mathbb{R}^n : \|x\|_C \leq \lambda\}$.

Proof. We'll prove (i). Proving the other two statements is left to the reader as an exercise. Clearly, $\|0\|_C = 0$. Let $x \in \mathbb{R}^n \setminus \{0\}$. Consider the set

$$S = \{\lambda \in \mathbb{R}_{\geq 0} : x \in \lambda C\}.$$

Since 0 is in the interior of C, there exists $\delta > 0$ such that

$$\{z \in \mathbb{R}^n : \|z\|_2 \leq \delta\} \subseteq C.$$

Hence, we have

$$x \in \frac{\|x\|_2}{\delta} C, \text{ which implies } S \neq \varnothing.$$

Let $0 \leq \mu = \inf_{\lambda \in S} \lambda$. By the definition of the infimum, we have

$$x \in \left(\mu + \frac{1}{m}\right) C \ \forall_{m \in \mathbb{N}_{\geq 1}}.$$

Thus, we have

$$\left(\mu + \frac{1}{m}\right)^{-1} x \in C.$$

Since C is bounded, it must be that $\mu > 0$. Furthermore, since C is closed and

$$\left(\mu + \frac{1}{m}\right)^{-1} x \in C \ \forall_{m \in \mathbb{N}_{\geq 1}},$$

we have

$$\lim_{m \to \infty} \left(\mu + \frac{1}{m}\right)^{-1} x = \frac{1}{\mu} x \in C,$$

which implies $x \in \mu C$, i.e. $\mu \in S$. $\square$

1.4 Minkowski's First Theorem

Given a measurable $S \subseteq \mathbb{R}^n$, we define the n-dimensional volume $\mathrm{vol}(S) \in \overline{\mathbb{R}_{\geq 0}}$ as $\lambda^n(S)$, the n-dimensional Lebesgue measure of S.
Some key properties:

- All open and closed sets are measurable.
- Bounded measurable sets have finite volume.
- For measurable S,

  $$\mathrm{vol}(S) = \int_S 1 \, \mathrm{d}x_1 \cdots \mathrm{d}x_n,$$

  if this integral (the n-fold Riemann integral) is defined.
- The set of all measurable sets in $\mathbb{R}^n$ forms a σ-algebra over $\mathbb{R}$.

- If $\varphi : \mathbb{R}^n \to \mathbb{R}^n$ is a linear transformation and $S \subseteq \mathbb{R}^n$ is measurable, then $\varphi(S)$ is measurable with

$$\mathrm{vol}(\varphi(S)) = |\det \varphi| \mathrm{vol}(S).$$

- The function vol is σ-additive.

Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice with basis $\{v_1, \dots, v_n\}$. We call

$$F = \{\alpha_1 v_1 + \cdots + \alpha_n v_n : 0 \leq \alpha_1, \dots, \alpha_n < 1\}$$

a **fundamental parallelepiped** of Λ (or a **fundamental domain**). We have

$$\mathbb{R}^n = \bigcup_{u \in \Lambda} u + F,$$

where the $u + F$ are pairwise disjoint. Also,

$$\mathrm{vol}(F) = \int_{[0,1)^n} |\det[v_1 \cdots v_n]| \, \mathrm{d}\alpha_1 \cdots \mathrm{d}\alpha_n = \det \Lambda.$$

Theorem 1.4.1 (Minkowski's First Convex Body Theorem, 1896)

Let C be a central symmetric convex body in $\mathbb{R}^n$. Also assume that $\Lambda \subseteq \mathbb{R}^n$ is a lattice. Suppose we have

$$\mathrm{vol}(C) \geq 2^n \det(\Lambda).$$

Then $C \cap (\Lambda \setminus \{0\}) \neq \varnothing$.

Proof. First assume that $\mathrm{vol}(C) > 2^n \det(\Lambda)$. Then the set ${}^1/_2 C$ satisfies

$$\mathrm{vol}\left(\frac{1}{2}C\right) = \frac{1}{2^n}\mathrm{vol}(C) > \det \Lambda.$$

Let F be a fundamental parallelepiped of Λ and

$$S_a = \frac{1}{2}C \cap (a + F) \ \forall_{a \in \Lambda}.$$

Then,

$$\sum_{a \in \Lambda} \mathrm{vol}(S_a) = \mathrm{vol}\left(\frac{1}{2}C\right) > \det \Lambda.$$

Let us define

$$\begin{aligned} T_a = & \; -a + S_a \\ = & \left(-a + \frac{1}{2}C\right) \cap F \subseteq F. \end{aligned}$$

Now,

$$\sum_{a\in\Lambda} \mathrm{vol}(T_a) = \sum_{a\in\Lambda} \mathrm{vol}(S_a) > \det\Lambda = \mathrm{vol}(F).$$

We proved the existence of a collection of subsets of F whose sum of volumes is greater than $\mathrm{vol}(F)$. Therefore, there exists $a \neq b \in \Lambda$ such that $T_a \cap T_b \neq \varnothing$. Let $u \in T_a \cap T_b$. There exist $x, y \in {}^1\!/_2 C$ such that $x - a = u = y - b$.
This implies that $x - y = a - b \in \Lambda \setminus \{0\}$. We also know that $2x, 2y \in C$. Hence, we have $-2y \in C$, which implies

$$\frac{1}{2}(2x) + \frac{1}{2}(-2y) = x - y \in C.$$

Next, let us suppose $\mathrm{vol}(C) = 2^n \det(\Lambda)$. Let $m \in \mathbb{N}_{\geq 1}$. Since

$$\mathrm{vol}\left(\left(1 + \frac{1}{m}\right) C\right) > 2^n \det(\Lambda),$$

it follows from the previous case that there exists

$$0 \neq x_m \in (1 + {}^1\!/_m)C \cap \Lambda \subseteq 2C \cap \Lambda \ \forall_{m\in\mathbb{N}_{\geq 1}}.$$

We know that $2C$ is bounded, so $2C \cap \Lambda$ is finite. Therefore, there exists an infinite sequence $m_1 < \cdots$ such that $0 \neq x_0 = x_{m_1} = x_{m_2} = \ldots$. Since $x_0 = x_{m_i} \in ((1 + {}^1\!/_{m_i})C) \cap \Lambda$, we have

$$\left(1 + \frac{1}{m_i}\right)^{-1} x_0 \in C \ \forall_{i\in\mathbb{N}_{\geq 1}}.$$

Hence, $0 \neq x_0 \in \Lambda \cap C$. □

Lemma 1.4.2 (Cardinality Formula for Lattice-Body Intersections)

Let C be a central symmetric convex body and Λ a full lattice in $\mathbb{R}^n$. Then

$$\#(xC \cap \Lambda) = \frac{\mathrm{vol}(C)}{\det(\Lambda)} x^n + \mathcal{O}(x^{n-1}) \ \forall_{x\geq 1}.$$

Proof. Let $N(x) = \#(xC \cap \Lambda)$. Consider

$$S = \bigcup_{u\in xC\cap\Lambda} u + F,$$

where F is the fundamental domain of Λ. Since this is a disjoint union, we have

$$\mathrm{vol}(S) = N(x)\mathrm{vol}(F).$$

As F is bounded,

$$\exists_{\sigma>0} : \|y\|_C \leq \sigma \ \forall_{y\in F}.$$

Let $x > \sigma$.

Claim 1.4.3

We have

$$(x-\sigma)C \subseteq S \subseteq (x+\sigma)C.$$

The result follows from this claim by taking the volume

$$(x-\sigma)^n \mathrm{vol}(C) \leq \mathrm{vol}(S) \leq (x+\sigma)^n \mathrm{vol}(C),$$

which can be rewritten as

$$\mathrm{vol}(S) = x^n \mathrm{vol}(C) + \mathcal{O}(x^{n-1}).$$

Then just divide by $\mathrm{vol}(F) = \det \Lambda$ (see above).

Proof of the claim. Given any $\widehat{x} \in \mathbb{R}^n$, let us write $\widehat{x} = u + y$ with $u \in \Lambda$ and $y \in F$. Then $(x-\sigma)C \subseteq S$, because if $\|\widehat{x}\|_C \leq x - \sigma$, then this implies $\|u\|_C \leq x$ (use the triangle inequality $\|u\|_C \leq \|\widehat{x} - u\|_C + \|\widehat{x}\|_C = \|y\|_C + \|\widehat{x}\|_C \leq \sigma + x - \sigma = x$). Similarly, $S \subseteq (x+\sigma)C$ follows from $\|u\|_C \leq x$, which implies $\|\widehat{x}\|_C \leq x + \sigma$ (again triangle inequality). □

This completes the proof. □

Lecture 3 (22nd April 2026 - W2L1)

Lecture 4 (24th April 2026 - W2L2)

We will study another proof of Minkowski's first theorem 1.4.1 using Lemma 1.4.2.

Second proof of Theorem 1.4.1. Let $m \in \mathbb{N}_{\geq 1}$. We divide Λ into congruence classes modulo $2m$. Each $x \in \Lambda$ is written as

$$x = x_1 v_1 + \cdots + x_n v_n,$$

where $\{v_1, \ldots, v_n\}$ is a basis of Λ. Consider

$$\Lambda_a = \{x_1 v_1 + \cdots + x_n v_n : x_i \equiv a_i \mod 2m \ \forall_{i \in \{1,\ldots,n\}}\},$$

for each $a \in \{0, \ldots, 2m-1\}^n$. Using Lemma 1.4.2, we have

$$\begin{aligned} \#(mC \cap \Lambda) &= \frac{\mathrm{vol}(C)}{\det(\Lambda)} m^n + \mathcal{O}(m^{n-1}) \\ &> 2^n(1+\varepsilon)m^n, \end{aligned}$$

provided that m is sufficiently large. Hence, there exists $a \in \{0, \ldots, 2m-1\}^n$ such that $\Lambda_a \cap mC$ contains at least two distinct vectors u, v. Then $0 \neq {}^1\!/_{2m}(u-v) \in \Lambda$. Since we have $u, v \in mC$, we also have $u, -v \in mC$, which implies ${}^1\!/_2 u - {}^1\!/_2 v \in mC$. Finally, ${}^1\!/_{2m}(u-v) \in C$, which completes the proof. □

Corollary 1.4.4 (Bound for Norm of Linear Combinations)

Let $\alpha_{ij} \in \mathbb{R}$ for $1 \leq i, j \leq n$ and $L_i(x) = \alpha_{i1}x_1 + \cdots + \alpha_{in}x_n$ for $x \in \mathbb{Z}^n$. Suppose that $0 < |\det(\alpha_{ij})_{1 \leq i,j \leq n}| \leq A_1 \cdots A_n$, where $A_1, \ldots, A_n > 0$.

Then, there exists $x \in \mathbb{Z}^n \setminus \{0\}$ such that

$$|L_i(x)| \le A_i \ \forall_{i \in \{1,\dots,n\}}.$$

Proof. Let $\mathfrak{B} = [-1,1]^n$ and $\Lambda = \left\{ \left(\frac{L_1(x)}{A_1}, \dots, \frac{L_n(x)}{A_n} \right) : x \in \mathbb{Z}^n \right\}$. Then this is a lattice inside $\mathbb{R}^n$. Also,

$$\Lambda = \left\{ \begin{pmatrix} \frac{\alpha_{11}}{A_1} & \cdots & \frac{\alpha_{1n}}{A_1} \\ \vdots & & \vdots \\ \frac{\alpha_{n1}}{A_n} & \cdots & \frac{\alpha_{nn}}{A_n} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} : x \in \mathbb{Z}^n \right\}.$$

We have

$$\det \Lambda = \frac{|\det(\alpha_{ij})_{1 \le i,j \le n}|}{A_1 \cdots A_n} \le 1,$$

which implies

$$\frac{\operatorname{vol}(\mathfrak{B})}{\det(\Lambda)} = \frac{2^n}{\det \Lambda} \ge 2^n.$$

By Minkowski's first theorem, there exists $0 \ne x \in \Lambda \cap \mathfrak{B}$. □

Corollary 1.4.5 (Simultaneous Irrational Approximation)

Let $(\alpha_1, \dots, \alpha_n) \in \mathbb{R}^n \setminus \mathbb{Q}^n$. Then,

$$\# \left\{ (x_1, \dots, x_n, y) \in \mathbb{Z}^{n+1}_{\text{prim}} : y > 0, \left| \alpha_i - \frac{x_i}{y} \right| \le y^{-1-1/n} \ \forall_{1 \le i \le n} \right\} = \aleph_0.$$

Proof. Let $Q > 1$. Consider the set

$$\{(x_1, \dots, x_n, y) \in \mathbb{Z}^{n+1} : |x_i - \alpha_i y| \le Q^{-1/n}, 0 < |y| \le Q\}.$$

Let

$$\begin{pmatrix} L_1(x,y) \\ \vdots \\ L_{n+1}(x,y) \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & \cdots & 0 & -\alpha_1 \\ 0 & \ddots & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & 0 & \vdots \\ \vdots & & \ddots & \ddots & -\alpha_n \\ 0 & \cdots & \cdots & 0 & 1 \end{pmatrix}}_{:= \widehat{A}} \begin{pmatrix} x_1 \\ \vdots \\ x_n \\ y \end{pmatrix}.$$

We use Corollary 1.4.4 (with $A_1 = \cdots = A_n = Q^{-1/n}$ and $A_{n+1} = Q$). Then there exists $(x,y) \in \mathbb{Z}^{n+1} \setminus \{0\}$ such that

$$\begin{aligned} |x_i - \alpha_i y| \le &\ Q^{-1/n} \\ |y| \le &\ Q \ \forall_{1 \le i \le n}. \end{aligned}$$

(since $A_1 \cdots A_{n+1} = 1 = \det \widehat{A}$). We know that $y \ne 0$ because $y = 0$ would imply $(x,y) = 0$. If $\gcd(x_1, \dots, x_n, y) > 1$, then divide through by the gcd. Also, if $y < 0$, divide through by (-1). Now, for any $Q > 1$, there exists $(x_1, \dots, x_n, y) \in \mathbb{Z}^{n+1}_{\text{prim}} \setminus \{0\}$ such that $0 < y \leq Q$ and $|x_i - \alpha_i y| \leq Q^{-1/n}$ $\forall_{1 \leq i \leq n}$. The assertion now follows through reasoning analogously to Corollary 1.1.3. $\square$

Given $\theta \in \mathbb{R}$, we denote

$$\|\theta\| = \min\{|\theta - n| : n \in \mathbb{Z}\}.$$

From Corollary 1.4.5, for $\alpha_1, \alpha_2 \in \mathbb{R}$, not both in $\mathbb{Q}$, we get

$$\begin{aligned} &\#\{y \in \mathbb{N}_{\geq 1} : \|\alpha_1 y\| \leq y^{-1/2}, \|\alpha_2 y\| \leq y^{-1/2}\} = \aleph_0 \\ \implies &\#\{y \in \mathbb{N}_{\geq 1} : y\|\alpha_1 y\|\|\alpha_2 y\| \leq 1\} = \aleph_0 \end{aligned}$$

(note that this also holds for $\alpha_1, \alpha_2 \in \mathbb{Q}$).

Hypothesis 1.4.6 (Littlewood)

Let $\alpha_1, \alpha_2 \in \mathbb{R}$. For any $\varepsilon > 0$, there exists $y \in \mathbb{N}_{\geq 1}$ such that

$$y\|\alpha_1 y\|\|\alpha_2 y\| < \varepsilon.$$

1.5 Minkowski's Second Theorem

Let Λ be a lattice and C a central symmetric convex body in $\mathbb{R}^n$. The **successive minima** $0 < \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$ of Λ with respect to C are defined as

$$\lambda_i = \min\{\lambda \in \mathbb{R}_{\geq 0} : \dim \operatorname{Span}_{\mathbb{R}}(\lambda C \cap \Lambda) \geq i\} \ \forall_{1 \leq i \leq n}.$$

Lemma 1.5.1 (Well-definedness and Construction of the Successive Minima)

The successive $0 < \lambda_1 \leq \cdots \leq \lambda_n < \infty$ are well-defined. Also, there are $v_1, \dots, v_n \in \Lambda$, which are linearly independent over $\mathbb{R}$, such that $v_i \in \lambda_i C$ $\forall_{1 \leq i \leq n}$.

Note: The set $\{v_1, \dots, v_n\}$ might not be a basis of Λ.

Proof. Recall that

$$\|x\|_C = \min\{\lambda \in \mathbb{R}_{\geq 0} : x \in \lambda C\} \text{ and } \lambda C = \{x \in \mathbb{R}^n : \|x\|_C \leq \lambda\}.$$

Let $\Lambda \setminus \{0\} = \{x_1, \dots\}$, where $0 < \|x_1\|_C \leq \dots$. Also note that given any $m \in \mathbb{N}_{\geq 1}$, there are finitely many i such that $m - 1 \leq \|x_i\|_C \leq m$ (because mC is bounded and λ is discrete). Let $\lambda_1 = \|x_1\|_C$. For $i \in \{2, \dots, n\}$, define

$$k_i = \min\{k \in \mathbb{N}_{\geq 1} : \dim \operatorname{Span}_{\mathbb{R}}\{x_1, \dots, x_k\} = i\}, \text{ set } v_i = x_{k_i} \text{ and let } \lambda_i = \|x_i\|_C.$$

Clearly, $v_i \in \lambda_i C \cap \Lambda$ and $0 < \lambda_1 \leq \cdots \leq \lambda_n < \infty$. Take a basis $\{v_1, \dots, v_n\} \subseteq \Lambda$ of $\mathbb{R}^n$. We have $\max_{i \in \{1,\dots,n\}} |v_i| \in \mathbb{R}$. There are only finitely many vectors with a smaller norm than this. Hence, the finiteness claim holds. Finally, we show that these are indeed the successive minima: By construction, the

$$v_1, \dots, v_i \in \lambda_i C \cap \Lambda \text{ are linearly independent over } \mathbb{R}.$$

Take $0 < \lambda < \lambda_i$ and let ℓ be the largest number such that $x_\ell \in \lambda C \cap \Lambda$. Then, we have $\ell < k_i$, and by the definition of the k_i, it follows that

$$\dim \operatorname{Span}_{\mathbb{R}}(\lambda C \cap \Lambda) = \dim \operatorname{Span}_{\mathbb{R}}\{x_1, \ldots, x_\ell\} < i.$$

This shows that λ_i is indeed the successive minimum. $\square$

Theorem 1.5.2 (Minkowski's Second Convex Body Theorem)

Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice and $C \subseteq \mathbb{R}^n$ a central symmetric convex body. Further assume that $\lambda_1, \ldots, \lambda_n$ are the successive minima of Λ with respect to C. Then, we have

$$\frac{2^n}{n!}\frac{\det \Lambda}{\operatorname{vol}(C)} \le \lambda_1 \cdots \lambda_n \le 2^n \frac{\det \Lambda}{\operatorname{vol}(C)}.$$

Before we prove this, let us study a couple of examples to highlight the importance of this result.

Example. Let $n \ge 2$ and

$$\mathfrak{B} = B_1^n(0).$$

Further let $\Lambda \subseteq \mathbb{R}^n$ be a lattice and $\lambda_1, \ldots, \lambda_n$ be the successive minima of Λ with respect to $\mathfrak{B}$. It follows from Lemma 1.5.1 that there exist $x_1, \ldots, x_n \in \Lambda$ which are linearly independent over $\mathbb{R}$ and fulfil $x_i \in \lambda_i \mathfrak{B} \cap \Lambda$. By construction, we have $\|x_i\|_{\mathfrak{B}} = \lambda_i \iff |x_i| = \lambda_i$. Theorem 1.5.2 implies

$$\prod_{i=1}^n |x_i| \le \frac{2^n \det \Lambda}{V_n},$$

where

$$V_n = \operatorname{vol}(\mathfrak{B}) = \begin{cases} 1 & \text{if } n = 1, \\ \pi & \text{if } n = 2, \\ \frac{2\pi}{n} V_{n-2} & \text{if } n \ge 3. \end{cases}$$ $\square$

Both bounds are sharp. To prove this, we shall consider two more examples.

Example. Let $\Lambda = \mathbb{Z}^n$. Take any real $0 < \lambda_1 < \cdots < \lambda_n$. Define $C = \{x \in \mathbb{R}^n : |x_i| \le {}^1/_{\lambda_i}\ \forall_{1 \le i \le n}\}$. Then C is a central symmetric convex body with

$$\operatorname{vol}(C) = \frac{2^n}{\lambda_1 \cdots \lambda_n}. \tag{$*$}$$

Let us now show that the λ_j are the successive minima of Λ with respect to C. For $\lambda \ge 0$, we have

$$\lambda C = \{x \in \mathbb{R}^n : |x_i| \le {}^{\lambda}/_{\lambda_i}\ \forall_{1 \le i \le n}\}.$$

Then $e_1, \ldots, e_j \in \lambda_j C$ are linearly independent over $\mathbb{R}$. Suppose that $0 < \lambda < \lambda_j$ and $y \in \mathbb{Z}^n \cap \lambda C$. Necessarily, $y_j = \cdots = y_n = 0$. Therefore,

$$\mathbb{Z}^n \cap \lambda C \subseteq \mathbb{Z}^{j-1} \times \{(0, \ldots, 0) \in \mathbb{Z}^{n-j+1}\}.$$

This cannot contain j linearly independent vectors. Hence, λ_j is the successive minimum. Thus,

$$\lambda_1 \cdots \lambda_n \overset{(*)}{=} \frac{2^n}{\mathrm{vol}(C)} = \frac{2^n \det \Lambda}{\mathrm{vol}(C)}.$$

This proves that the upper bound is sharp. □

Example. Take $\Lambda = \mathbb{Z}^n$,

$$C = \left\{ x \in \mathbb{R}^n : \sum_{i=1}^{n} |x_i| \le 1 \right\}.$$

Exercise 1.1

Show that $\mathrm{vol}(C) = \frac{2^n}{n!}$ and $\lambda_1 = \cdots = \lambda_n = 1$.

Proof. We shall prove a stronger claim about the volume of such shapes. Let

$$C_r = \left\{ x \in \mathbb{R}^n : \sum_{i=1}^{n} |x_i| \le r \right\}.$$

Then $\mathrm{vol}(C_r) = \frac{(2r)^n}{n!}$. We will prove this by induction. The statement is clearly true for $n = 1$. Assume that the statement holds for $n-1$. For n, we use Fubini's theorem with

$$E_t = \left\{ x \in \mathbb{R}^{n-1} : \sum_{i=1}^{n-1} |x_i| \le r - |t| \right\}.$$

We have

$$\begin{aligned}
\mathrm{vol}(C) &= \int_{[-r,r]} \mathrm{vol}(E_t)\, \mathrm{d}\lambda(t) \\
&\overset{\mathrm{IH}}{=} \frac{2^n}{(n-1)!} \int_{[0,r]} (r-t)^{n-1}\, \mathrm{d}\lambda(t) \\
&= \frac{2^n}{n!}\, t^n\big|_0^r \\
&= \frac{(2r)^n}{n!}.
\end{aligned}$$

For the other statement, just check that we have $\{e_1, \ldots, e_n\} \subseteq C$. □

This proves that the lower bound in Theorem 1.5.2 is sharp. □

Lecture 4 (24th April 2026 - W2L2)

Lecture 5 (29th April 2026 - W3)

Lemma 1.5.3 (Convex Hull)

Let $w_1, \ldots, w_r \in \mathbb{R}^n$ and define

$$C(w_1, \ldots, w_r) = \left\{ \sum_{i=1}^{r} x_i w_i : (x_1, \ldots, x_r) \in \mathbb{R}^r, \sum_{i=1}^{r} |x_i| \le 1 \right\}.$$

Then $C(w_1, \ldots, w_r)$ is the smallest convex subset in $\mathbb{R}^n$ which contains $w_1, \ldots, w_r$ and is symmetric about 0. We call $C(w_1, \ldots, w_r)$ the **convex hull** of $w_1, \ldots, w_r$.

Proof. This is an easy exercise.
We shall prove this in two steps:

(i) **The set $C(w_1, \ldots, w_r)$ is convex and symmetric about 0.**
The symmetry is clear. Let $x, y \in C(w_1, \ldots, w_r)$. Then $[x, y] = t\sum_{i=1}^r y_i w_i + (1 - t)\sum_{i=1}^r x_i w_i \subseteq C$. We have

$$\begin{aligned}\sum_{i=1}^r |ty_i + (1-t)x_i| &\overset{\triangle}{\leq} t\sum_{i=1}^r |y_i| + (1-t)\sum_{i=1}^r |x_i| \\ &\leq t + (1-t) \\ &= 1.\end{aligned}$$

(ii) **The set $C(w_1, \ldots, w_r)$ is minimal with this property.**
We shall prove this by induction over r. Let T_r be the minimal convex subset of $\mathbb{R}^n$ containing $w_1, \ldots, w_r$. For $r = 1$, we have $C(w_1) = [-w_1, w_1] \subseteq T_1$. Assume that the statement holds for $r - 1$. For $x = \sum_{i=1}^r x_i w_i \in C(w_1, \ldots, w_r)$, we have

$$\widehat{x} = \frac{1}{1 - |x_r|}\sum_{i=1}^{r-1} x_i w_i \in C(w_1, \ldots, w_{r-1}) \subseteq T_{r-1} \subseteq T_r$$

and $w_r \in T$, which implies that $[\widehat{x}, w_r] \subseteq T$ and $[\widehat{x}, -w_r] \subseteq T$, one of which gives us the claim for $t = |x_r|$. □

Lemma 1.5.4 (Lattice-Base Ordering Compatible with Successive Minima)

Let $\lambda_1, \ldots, \lambda_n$ be the successive minima of Λ with respect to C. Then there exists a basis $\{b_1, \ldots, b_n\}$ of Λ such that if

$$x = u_1 b_1 + \cdots + u_n b_n$$

with $\|x\|_C < \lambda_j$ for some $j \in \{1, \ldots, n\}$, then

$$x = u_1 b_1 + \cdots + u_{j-1} b_{j-1}.$$

Proof. Let $\mu_1 < \cdots < \mu_s$ be such that $\mu_1 = \lambda_1 = \cdots = \lambda_{k_1}$, $\mu_2 = \lambda_{k_1+1} = \cdots = \lambda_{k_2}$ and so on with $k_s = n$.
By the definition of the successive minima, $a = 0 \ \forall_{a \in \Lambda : \|a\|_C < \mu_1}$. Since $\mu_2 > \lambda_{k_1}$, by Lemma 1.5.1, there exist linearly independent $a_1, \ldots, a_{k_1} \in \Lambda$ such that $\|a_1\|_C, \ldots, \|a_{k_1}\|_C \leq \lambda_{k_1} < \mu_2$. And given any $x \in \Lambda$ with $\|x\|_C < \mu_2$, then

$$\exists_{u_1, \ldots, u_{k_1} \in \mathbb{R}} : x = u_1 a_1 + \cdots + u_{k_1} a_{k_1}$$

(again use the definition of successive minima). We may continue in this manner (extend $a_1, \ldots, a_{k_1}$) to obtain linearly independent $a_1, \ldots, a_n \in \Lambda$ with the following property:

$$\forall_{1 \leq t \leq s} : a_1, \ldots, a_{k_{t-1}} \text{ are linearly independent over } \mathbb{R} \text{ and } \|a_j\|_C < \mu_t \ \forall_{1 \leq j \leq k_{t-1}}.$$

Given any $x \in \Lambda$ with $\|x\|_C < \mu_t$, we have

$$x = u_1 a_1 + \cdots + u_{k_{t-1}} a_{k_{t-1}}$$

for some $u_i \in \mathbb{R}$ (!).
Then, by Lemma 1.2.1, we obtain a basis $\{b_1, \ldots, b_n\}$ of Λ such that

$$\begin{aligned} b_1 &= \xi_{1,1} a_1 \\ b_2 &= \xi_{2,1} a_1 + \xi_{2,2} a_2 \\ &\vdots \\ b_n &= \xi_{n,1} a_1 + \cdots + \xi_{n,n} a_n \end{aligned}$$

with $\xi_{i,j} \in \mathbb{R}, \xi_{i,i} \neq 0 \ \forall_{1 \le i,j \le n}$. This basis clearly satisfies the desired property. $\square$

Before we can prove Theorem 1.5.2, we need a technical theorem.

Theorem 1.5.5 (Volume of $S(t)$)

We have

$$\mathrm{vol}(S(t)) \ge \begin{cases} t^n \mathrm{vol}(C) & \text{if } t \le \frac{\lambda_1}{2}, \\ \frac{\lambda_1}{2} \cdots \frac{\lambda_j}{2} t^{n-j} \mathrm{vol}(C) & \text{if } \frac{\lambda_j}{2} \le t \le \frac{\lambda_{j+1}}{2}, \\ \frac{\lambda_1}{2} \cdots \frac{\lambda_n}{2} \mathrm{vol}(C) & \text{if } t \ge \frac{\lambda_n}{2}. \end{cases}$$

with

$$S(t) = S_F(t) = \{\eta \in F \mid \exists_{x \in \Lambda} : \eta + x \in tC\},$$

where $\Lambda \subseteq \mathbb{R}^n$ is a lattice, $C \subseteq \mathbb{R}^n$ is a central symmetric convex body, $\lambda_1, \ldots, \lambda_n$ are the successive minima of Λ with respect to C and F is the fundamental parallelepiped of Λ.

Proof. We can reduce the problem to the case where the basis $\{b_1, \ldots, b_n\}$ given in Lemma 1.5.4 is $\{e_1, \ldots, e_n\}$ (Exercise sheet 3). In particular, $\Lambda = \mathbb{Z}^n$. Let $1 \le j \le n-1$ and $x_1 = (x_{1,1}, \ldots, x_{1,n}), x_2 = (x_{2,1}, \ldots, x_{2,n})$ be two vectors in tC with $t < \lambda_{j+1}/2$, such that $x_1 - x_2 \in \Lambda$. Then

$$\|\underbrace{x_1 - x_2}_{\in \mathbb{Z}^n}\|_C \le \|x_1\|_C + \|x_2\|_C \le 2t < \lambda_{j+1}. \tag{1}$$

This implies $x_1 - x_2 \in \mathbb{Z}^j \times \{(0, \ldots, 0)\}$. Denote

$$S_j(t, z) = \{(\eta_1, \ldots, \eta_j) \in [0,1)^j \mid \exists_{(m_1, \ldots, m_j) \in \mathbb{Z}^j} : \|(\eta_1 + m_1, \ldots, \eta_j + m_j, z)\|_C \le t\}.$$

Recall that

$$S(t) = \{\eta \in [0,1)^n \mid \exists_{m \in \mathbb{Z}^n} : \|\eta + m\|_C \le t\}.$$

Let $\eta \in [0,1)^n$ and $t < \lambda_{j+1}/2$. Supppose that we have $m, m' \in \mathbb{Z}^n$ such that

$$\|\eta + m\|_C \le t \text{ and } \|\eta + m'\|_C \le t.$$

Then, by (1), we get $(m_{j+1}, \dots, m_n) = (m'_{j+1}, \dots, m'_n)$. Thus, for each $\eta \in [0,1)^n$ from the definition of $S(t)$, the last $n-j$ coordinates of m are unique. We partition $S(t)$ based on this. Write

$$S(t) = \bigsqcup_{m_{j+1},\dots,m_n \in \mathbb{Z}} \bigsqcup_{\eta_{j+1},\dots,\eta_n \in [0,1)} \{\eta \in [0,1)^n : \exists_{(m_1,\dots,m_j)\in\mathbb{Z}^j} : \|\eta + m\|_C \le t\}.$$

(Here $m = (m_1, \dots, m_n)$ and $\eta = (\eta_1, \dots, \eta_j, \eta_{j+1}, \dots, \eta_n)$). Therefore,

$$\mathrm{vol}(S(t)) = \int_{\mathbb{R}^{n-j}} \mathrm{vol}^j(S_j(t,z))\,\mathrm{d}z \ \forall_{t < \frac{\lambda_{j+1}}{2}}. \tag{2}$$

Lecture 5 (29th April 2026 - W3)

Lecture 6 (6th May 2026 - W4L1)

Now, take any $\alpha \ge 1$.

Claim 1.5.6

We have $\mathrm{vol}(S_j(\alpha t, \alpha z)) \ge \mathrm{vol}(S_j(t,z))\ \forall_{z\in\mathbb{R}^{n-j}}$.

If the right hand side is 0, then there is nothing to prove. Suppose otherwise that there is $y_0 = (y_{0,1}, \dots, y_{0,j}) \in \mathbb{R}^j$ such that $\|(y_0, z)\|_C \le t$. Given any $y \in \mathbb{R}^j$ such that $\|(y,z)\|_C \le t$, then

$$\begin{aligned}
& \|(y + (\alpha-1)y_0, \alpha z)\|_C \\
= \ & \|(y,z) + (\alpha-1)(y_0,z)\|_C \\
\le \ & \|(y,z)\|_C + (\alpha-1)\|(y_0,z)\|_C \\
\le \ & t + (\alpha-1)t \\
= \ & \alpha t.
\end{aligned} \tag{3}$$

Let us define a notation: given $y \in \mathbb{R}^j$, let

$$\{y\} = (y_1 - \lfloor y_1 \rfloor, \dots, y_j - \lfloor y_j \rfloor) \in [0,1)^j.$$

Then

$$S_j(t,z) = \{\{y\} \in [0,1)^j \mid y \in \mathbb{R}^j, \|(y,z)\|_C \le t\}.$$

Now, we define

$$W = \{\{y + (\alpha-1)y_0\} \in [0,1)^j \mid y \in \mathbb{R}^j, \|(y,z)\|_C \le t\}.$$

Then $\mathrm{vol}(S_j(t,z)) = \mathrm{vol}(W)$ (by the translation invariance of the Lebesgue measure). Furthermore, by equation (3), we have

$$\begin{aligned}
W \subseteq \ & \{\{y + (\alpha-1)y_0\} \in [0,1)^n \mid y \in \mathbb{R}^j, \|(y + (\alpha-1)y_0, \alpha z)\|_C \le \alpha t\} \\
= \ & \{\{y\} \in [0,1)^j \mid y \in \mathbb{R}^j, \|(y, \alpha z)\|_C \le \alpha t\} \\
= \ & S_j(\alpha t, \alpha z).
\end{aligned}$$

Take the volume to obtain

$$\mathrm{vol}(S_j(t,z)) = \mathrm{vol}(W) \le \mathrm{vol}(S_j(\alpha t, \alpha z)),$$

proving the claim.
Suppose that $0 < t \leq \alpha t < \lambda_{j+1}/2$. Then by equation (2) and the claim,

$$\begin{aligned}\mathrm{vol}(S(\alpha t)) &= \int_{\mathbb{R}^{n-j}} \mathrm{vol}(S_j(\alpha t, z))\,\mathrm{d}z \\ &= \alpha^{n-j}\int_{\mathbb{R}^{n-j}} \mathrm{vol}(S_j(\alpha t, \alpha z))\,\mathrm{d}z \\ &\geq \alpha^{n-j}\int_{\mathbb{R}^{n-j}} \mathrm{vol}(S_j(t, z))\,\mathrm{d}z.\end{aligned}$$

This gives

$$\mathrm{vol}(S(\alpha t)) \geq \alpha^{n-j}\mathrm{vol}(S(t)). \tag{4}$$

We are ready to prove the theorem. When $t < \lambda_1/2$, we have

$$\mathrm{vol}(S(t)) = \mathrm{vol}(tC) = t^n \mathrm{vol}(C).$$

This is because if $\eta + m, \eta + m' \in tC$ with $m, m' \in \mathbb{Z}^n$, then

$$\|m - m'\|_C = \|(\eta + m) - (\eta + m')\|_C \leq \|\eta + m\|_C + \|\eta + m'\|_C \leq 2t < \lambda_1,$$

so $m = m'$. Therefore, $\mathrm{vol}(S(t)) = \mathrm{vol}(tC) = t^n\mathrm{vol}(C)$ if $t < \lambda_1/2$. Since $S(t) \subseteq S(\lambda_1/2)$ for any $t \leq \lambda_1/2$, we have $\mathrm{vol}(S(\lambda_1/2)) \geq (\lambda_1/2)^n\mathrm{vol}(C)$.
We now use induction to prove the statement for $1 \leq j \leq n-1$. Suppose that the statement holds for any $t \leq \lambda_j/2$ for some j. Consider $\lambda_j/2 \leq t < \lambda_{j+1}/2$. Set $t_0 = \lambda_j/2$ and $\alpha \in [1, \lambda_{j+1}/\lambda_j)$ such that $t = \alpha t_0$. Then

$$\begin{aligned}\mathrm{vol}(S(t)) &= \mathrm{vol}(S(\alpha t_0)) \\ &\geq \alpha^{n-j}\mathrm{vol}(S(t_0)) \\ &\geq \alpha^{n-j}\left(\frac{\lambda_1}{2}\cdots\frac{\lambda_{j-1}}{2}\right)t_0^{n-j+1}\mathrm{vol}(C) \\ &= t^{n-j}\left(\frac{\lambda_1}{2}\cdots\frac{\lambda_{j-1}}{2}\frac{\lambda_j}{2}\right)\mathrm{vol}(C).\end{aligned}$$

And for $t = \lambda_{j+1}/2$, this holds by taking the limit since $S(t) \subseteq S(\lambda_{j+1}/2)$ if $t \leq \lambda_{j+1}/2$. This completes induction and gives us the statement for all $t \leq \lambda_n/2$.
Finally, suppose that $t \geq \lambda_n/2$. Then $S(\lambda_n/2) \subseteq S(t)$, and so

$$\begin{aligned}\mathrm{vol}(S(t)) &\geq \mathrm{vol}(S(\lambda_n/2)) \\ &\geq \frac{\lambda_1}{2}\cdots\frac{\lambda_n}{2}\mathrm{vol}(C).\end{aligned}$$

(This follows from the case $t = \lambda_n/2$.) □

We shall now prove Minkowski's Second Theorem:

Proof of Theorem 1.5.2. We prove the correctness of the lower and upper bound separately.

(i) **Lower Bound**
Let $v_1, \ldots, v_n \in \Lambda$ be linearly independent over $\mathbb{R}$ such that $v_i \in \lambda_i C$ $\forall_{1\leq i\leq n}$ (Lemma 1.5.1). By definition, we have $\lambda_i^{-1}v_i \in C$ $\forall_{1\leq i\leq n}$.

Define

$$D = C(\lambda_1^{-1}v_1, \ldots, \lambda_n^{-1}v_n) \subseteq C$$

and note that this has the properties stated in Lemma 1.5.3. Now, let $\varphi : \mathbb{R}^n \to \mathbb{R}^n$ be a linear transformation by $\varphi(e_i) = \lambda_i^{-1}v_i \ \forall_{1\le i\le n}$. Then $D = \varphi(\mathfrak{B}_1)$, where

$$\mathfrak{B}_1 = \left\{ x \in \mathbb{R}^n : \sum_{i=1}^{n} |x_i| \le 1 \right\}.$$

Hence,

$$\begin{aligned} \mathrm{vol}(D) = & \ |\det\varphi|\mathrm{vol}(\mathfrak{B}_1) \\ = & \ \frac{|\det[v_1 \cdots v_n]|}{\lambda_1 \cdots \lambda_n} \frac{2^n}{n!} \\ = & \ \frac{\det\widetilde{\Lambda}}{\lambda_1 \cdots \lambda_n} \frac{2^n}{n!}, \end{aligned}$$

where $\widetilde{\Lambda}$ is a lattice with basis $\{v_1, \ldots, v_n\}$. Since $\{v_1, \ldots, v_n\} \subseteq \Lambda$, $\widetilde{\Lambda}$ is a sublattice of Λ. This implies that

$$\det\widetilde{\Lambda} = [\Lambda : \widetilde{\Lambda}] \det\Lambda \ge \det\Lambda.$$

We combine everything to get

$$\mathrm{vol}(C) \ge \mathrm{vol}(D) = \frac{\det\widetilde{\Lambda}}{\lambda_1 \cdots \lambda_n} \frac{2^n}{n!} \ge \frac{\det\Lambda}{\lambda_1 \cdots \lambda_n} \frac{2^n}{n!}.$$

Thus,

$$\lambda_1 \cdots \lambda_n \ge \frac{\det\Lambda}{\mathrm{vol}(C)} \frac{2^n}{n!}.$$

(ii) **Upper Bound**
Let Λ be a lattice inside $\mathbb{R}^n$ and C a central symmetric convex body inside $\mathbb{R}^n$. Given $t > 0$, let

$$S(t) = S_F(t) = \{\eta \in F \mid \ \exists_{x\in\Lambda} : \eta + x \in tC\}.$$

where F is the fundamental parallelepiped of Λ. The assertion follows from Theorem 1.5.5 by applying it with $t = \lambda_n/2$. We have

$$\det\Lambda = \mathrm{vol}(F) \ge \mathrm{vol}(S_F(\lambda_n/2)) \ge \frac{\lambda_1}{2} \cdots \frac{\lambda_n}{2} \mathrm{vol}(C),$$

which can be transformed into the statement. □

This proof is, largely exaggerated, just a very technical induction. The proof of Roth's theorem will also be, largely exaggerated, a very technical induction.

1.6 Dual lattice

Let Λ be a lattice in $\mathbb{R}^n$. We define the dual lattice Λ^* to be

$$\Lambda^* = \{x \in \mathbb{R}^n \mid \langle x, y\rangle \in \mathbb{Z} \ \forall_{y \in \Lambda}\}.$$

Let $\{v_1, \dots, v_n\}$ be a basis of Λ and let $A = \begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix}$. Then

$$\begin{aligned}\Lambda^* &= \{x \in \mathbb{R}^n \mid \langle x, y\rangle \in \mathbb{Z} \ \forall_{y \in \Lambda}\} \\ &= \{x \in \mathbb{R}^n \mid \langle x, Az\rangle \in \mathbb{Z} \ \forall_{z \in \mathbb{Z}^n}\} \\ &= \{x \in \mathbb{R}^n \mid \langle A^t x, z\rangle \in \mathbb{Z} \ \forall_{z \in \mathbb{Z}^n}\} \\ &= \{x \in \mathbb{R}^n \mid A^t x \in \mathbb{Z}^n\} \\ &= \{(A^t)^{-1} w \mid w \in \mathbb{Z}^n\}.\end{aligned}$$

Hence,

$$\det \Lambda^* = |\det (A^t)^{-1}| = \frac{1}{|\det A|} = \frac{1}{\det \Lambda}.$$

Theorem 1.6.1 (Estimate for Products of Successive Minima and Dual Successive Minima)

Let Λ be a lattice in $\mathbb{R}^n$ and let $\lambda_1, \dots, \lambda_n$ be its successive minima with respect to $B = \{x \in \mathbb{R}^n \mid \|x\|_2 \le 1\}$. Let Λ^* be its dual, and let $\lambda_1^*, \dots, \lambda_n^*$ be the successive minima of Λ^* with respect to B. Then

$$1 \le \lambda_i \lambda_{n-i+1}^* \le c(n)$$

for all $1 \le i \le n$ where $c(n)$ is a constant that depends only on n.

Successive minima for dual lattices are only considered with respect to the unit ball.

Proof. We begin with the lower bound. Let $v_1, \dots, v_n \in \Lambda$ be linearly independent over $\mathbb{R}$ and $\|v_i\|_2 = \lambda_i$ for all $1 \le i \le n$ (from Lemma 1.5.1). Similarly, take $v_1^*, \dots, v_n^* \in \Lambda^*$ linearly independent over $\mathbb{R}$ with $\|v_i^*\|_2 = \lambda_i^*$ for all $1 \le i \le n$. Take $1 \le i \le n$. Define

$$V_i = \{x \in \mathbb{R}^n \mid \langle x, v_k\rangle = 0 \text{ for each } 1 \le k \le i\}.$$

Since $v_1, \dots, v_i$ are linearly independent over $\mathbb{R}$, we obtain $\dim V_i = n - i$. Therefore, at least one of $v_1^*, \dots, v_{n+1-i}^*$ is not in V_i. Thus, there is $1 \le i_0 \le i$ and $1 \le k_0 \le n + 1 - i$ such that $\langle v_{i_0}, v_{k_0}^*\rangle \ne 0$. From the definition of Λ^*, we have $\left|\langle v_{i_0}, v_{i_0}^*\rangle\right| \in \mathbb{Z}$ and

$$1 \le \left|\langle v_{i_0}, v_{i_0}^*\rangle\right| \overset{\text{C.-S. I.}}{\le} \|v_{i_0}\|_2 \cdot \|v_{k_0}^*\|_2 = \lambda_{i_0}\lambda_{k_0}^* \le \lambda_i \lambda_{n+1-i}^*.$$

Next, the upper bound. By Theorem 1.5.2, we have

$$\lambda_1 \cdots \lambda_n \le \frac{2^n}{\mathrm{vol}(B)} \det \Lambda, \qquad \lambda_1^* \cdots \lambda_n^* \le \frac{2^n}{\mathrm{vol}(B)} \det \Lambda^*.$$

This gives

$$\prod_{i=1}^{n} \lambda_i \lambda_{n+1-i}^* \leq \frac{4^n}{\mathrm{vol}(B)^2} \det \Lambda \det \Lambda^* = \frac{4^n}{\mathrm{vol}(B)^2}.$$

Take any $1 \leq i_0 \leq n$. Then

$$\lambda_{i_0} \lambda_{n+1-i_0}^* = \frac{4^n}{\mathrm{vol}(B)^2} \frac{1}{\prod_{i \neq i_0} \underbrace{\lambda_i \lambda_{n+1-i}^*}_{\geq 1}} \leq \frac{4^n}{\mathrm{vol}(B)^2}. \qquad \square$$

As an application, we will show that there is no „small“ vector in Λ^*. This gives a good approximation of vectors in $\mathbb{R}^n$ by Λ.

Corollary 1.6.2 (Approximation of Vectors in $\mathbb{R}^n$ with Lattice Vectors)

Let us use the same notation as in Theorem 1.6.1. Suppose that $0 < R \leq \lambda_1^*$. Then, given any $b \in \mathbb{R}^n$,

$$\exists_{x \in \Lambda} : \|b - x\|_2 \leq \frac{nc(n)}{2R}.$$

Proof. Since $\lambda_1^* \geq R$, by Theorem 1.6.1, we obtain

$$\lambda_n \leq \frac{c(n)}{\lambda_1^*} \leq \frac{c(n)}{R}.$$

Let $v_1, \ldots, v_n \in \Lambda$ be linearly independent over $\mathbb{R}$ with $\|v_i\|_2 = \lambda_i \leq \lambda_n \leq c(n)/R$ for all $1 \leq i \leq n$. Take any $b \in \mathbb{R}^n$, and write $b = \xi_1 v_1 + \cdots + \xi_n v_n$. Let $m_1, \ldots, m_n \in \mathbb{Z}$ such that $|\xi_i - m_i| \leq \frac{1}{2}$. Then $x = m_1 v_1 + \cdots + m_n v_n \in \Lambda$. We have

$$\|x - b\|_2 = \|(\xi_1 - m_1)v_1 + \cdots + (\xi_n - m_n)v_n\|_2 \leq \frac{1}{2}(\|v_1\|_2 + \cdots + \|v_n\|_2) \leq \frac{n}{2}\frac{c(n)}{R}. \square$$

Lecture 6 (29th April 2026 - W3)

Lecture 7 (8th May 2026 - W4L2)

Recall from last time...
Given a lattice Λ, we defined the dual lattice

$$\Lambda^* = \{x \in \mathbb{R}^n : \langle x, y \rangle \in \mathbb{Z}, y \in \Lambda\}.$$

If $\Lambda = A\mathbb{Z}^n$, then $\Lambda^* = (A^t)^{-1}\mathbb{Z}^n$.

Corollary

Using the same notation as in Theorem 1.6.1, suppose $0 < R \leq \lambda_1^*$. Then given any $b \in \mathbb{R}^n$, there exists $x \in \Lambda$ such that

$$\|b - x\|_2 \leq \frac{nC(n)}{2R}.$$

Theorem 1.6.3 (Shifted Simultaneous Diophantine Approximation)

Let $\alpha_1, \dots, \alpha_n, \theta_1, \dots, \theta_n \in \mathbb{R}$. Suppose that $\{1, \alpha_1, \dots, \alpha_n\}$ is linearly independent over $\mathbb{Q}$. Then, for any constant $\varepsilon > 0$, there exist infinitely many $(x_1, \dots, x_n, y) \in \mathbb{Z}^{n+1}$ such that

$$|\alpha_i y - x_i - \theta_i| \leq \varepsilon \ \forall_{1 \leq i \leq n}.$$

Remark: Recall Corollary 1.4.5, which was a multi-dimensional Dirichlet's theorem. This is a version with a shift θ (inhomogeneous), and a nice application of the concept of a dual lattice. Also note that the condition of the linear independence is necessary (exercise).

Proof. Let M be a sufficiently large integer. Also let

$$A = \begin{pmatrix} & & -\alpha_1 \\ & \mathbb{1}_n & \vdots \\ & & -\alpha_n \\ 0 & \cdots \quad 0 & M^{-1} \end{pmatrix}$$

and

$$\begin{aligned} \Lambda_M &= \{Az : z = (x_1, \dots, x_n, y) \in \mathbb{Z}^{n+1}\} \\ &= \{(x_1 - \alpha_1 y, \dots, x_n - \alpha_n y, M^{-1} y) : (x_1, \dots, x_n, y) \in \mathbb{Z}^{n+1}\}. \end{aligned}$$

Write

$$b = \begin{pmatrix} \theta_1 \\ \vdots \\ \theta_n \\ 2\varepsilon \end{pmatrix}.$$

The goal is to show that

$$\exists_{z \in \mathbb{Z}^{n+1}} : \|b - Az\| \leq \varepsilon. \tag{$*$}$$

This would imply

$$\begin{aligned} \varepsilon &\geq \sqrt{(x_1 - \alpha_1 y - \theta_1)^2 + \cdots + (x_n - a_n y - \theta_n)^2 + (M^{-1} y - 2\varepsilon)^2} \\ &\geq \begin{cases} |x_i - \alpha_i y - \theta_i| & 1 \leq i \leq n, \\ |M^{-1} y - 2\varepsilon| & i = n+1. \end{cases} \end{aligned}$$

The latter is equivalent to $\varepsilon M \leq y \leq 3\varepsilon M$. Hence, we can generate infinitely many $(x_1, \dots, x_n, y) \in \mathbb{Z}^{n+1}$ that satisfy the statement of the theorem. We have

$$(A^t)^{-1} = \begin{pmatrix} & & & 0 \\ & \mathbb{1}_n & & \vdots \\ & & & 0 \\ M\alpha_1 & \dots & M\alpha_n & M \end{pmatrix}.$$

Therefore, our dual lattice is

$$\Lambda_M^* = (A^t)^{-1}\mathbb{Z}^{n+1} = \{(x_1, \ldots, x_n, M(\alpha_1 x_1 + \cdots + \alpha_n x_n + y)) : x_1, \ldots, x_n, y \in \mathbb{Z}\}.$$

Let

$$R = \frac{(n+1)C(n+1)}{2\varepsilon}$$

and

$$\mu = \min_{(x_1,\ldots,x_n,y)\in S} |\alpha_1 x_1 + \cdots + \alpha_n x_n + y|,$$

where

$$S = \{(x_1, \ldots, x_n, y) \in \mathbb{Z}^{n+1} \setminus \{0\} : |x_i| < R \ \forall_{1\le i\le n} \wedge |\alpha_1 x_1 + \cdots + \alpha_n x_n + y| \le 1\}.$$

The set S is clearly finite. Recall that $\alpha_1, \ldots, \alpha_n$ are linearly independent over $\mathbb{Q}$. Hence, $\mu > 0$. We choose

$$M > \max\left\{\frac{R}{\mu}, R\right\}.$$

Then, given any $u = (x_1, \ldots, x_n, y) \in \Lambda_M^* \setminus \{0\}$, we have

$$\|u\|_2 = \sqrt{x_1^2 + \cdots + x_n^2 + M^2(\alpha_1 x_1 + \cdots + \alpha_n x_n + y)^2} \ge R,$$

which we can prove by considering the following three cases:

(i) $\exists_{1\le i\le n} : |x_i| \ge R$. ✓

(ii) $\forall_{1\le i\le n} : |x_i| < R, (x_1, \ldots, x_n, y) \notin S$
Use $M > R$.

(iii) $\forall_{1\le i\le n} : |x_i| < R, (x_1, \ldots, x_n, y) \in S$
Use $M > R/\mu$.

This means that $\lambda_1^* > R$ (first minimum of Λ_M^* with respect to the unit ball). We can then apply Corollary 1.6.2 to deduce that

$$\exists_{v=Az\in\Lambda_M^*} : \|b - Az\|_2 \le \frac{(n+1)C(n+1)}{2R} = \varepsilon.$$

This is the bound $(*)$, which finishes the proof. □

1.7 Lagrange's Four Square Theorem

Theorem 1.7.1 (Lagrange's Four Square Theorem, 1770)

Given $n \in \mathbb{N}_{\ge 1}$, $\exists_{a,b,c,d\in\mathbb{N}}$ such that

$$n = a^2 + b^2 + c^2 + d^2.$$

We'll do a Geometry of Numbers-style proof of this. We begin with a simple lemma:

Lemma 1.7.2 (Two Square Residue Lemma)

Given any $p \in \mathbb{P}$, there exist $a, b \in \mathbb{Z}$ such that

$$a^2 + b^2 \equiv -1 \mod p.$$

Proof. If $p = 2$, then $(a, b) = (1, 0)$ works. For $p > 2$, define

$$\mathfrak{A} = \{a^2 \bmod p : 0 \leq a < p/2\}.$$

The elements of $\mathfrak{A}$ are distinct because

$$a^2 \equiv \widehat{a}^2 \mod p \implies p \mid a^2 - \widehat{a}^2 = (a + \widehat{a})(a - \widehat{a}),$$

from which we can immediately deduce that $p \mid a - \widehat{a}$, possibly by $a = \widehat{a} = 0$, id est $a \equiv \widehat{a} \mod p$. Therefore,

$$\#\mathfrak{A} = \frac{p-1}{2} + 1 = \frac{p+1}{2}.$$

Similarly, we define

$$\mathfrak{B} = \{-1 - b^2 \bmod p : 0 \leq b < p/2\}.$$

By the same argument,

$$\#\mathfrak{B} = \frac{p+1}{2}.$$

Therefore, we can use the pigeonhole principle to deduce that $\mathfrak{A} \cap \mathfrak{B} \neq \varnothing$, which means that

$$\exists_{a \in \mathfrak{A}, b \in \mathfrak{B}} : a^2 \equiv -1 - b^2 \mod p. \qquad \square$$

We also need the following lemma, the proof of which will be on the next exercise sheet.

Lemma 1.7.3 (Lattice Generated by Linear Forms)

Let $k_1, \ldots, k_m \in \mathbb{N}_{\geq 1}$, $m \geq 1$. Assume that $L_1(x), \ldots, L_m(x) \in \mathbb{Z}[x]$ are linear forms. Define

$$\Lambda = \{x \in \mathbb{Z}^n : L_j(x) \equiv 0 \bmod k_j \ \forall_{1 \leq j \leq m}\}.$$

Then Λ is a lattice with $\det \Lambda \leq k_1 \cdots k_m$.

Proof. Exercise. $\square$

These two lemmata allow us to prove Theorem 1.7.1.

Proof of Theorem 1.7.1. For $n = 1$, the statement is trivial. Otherwise, write $n = m^2 p_1 \cdots p_r$, where $p_1 \neq \cdots \neq p_r \in \mathbb{P}$. Writing $\widetilde{a} = ma, \ldots, \widetilde{d} = md$, it suffices to consider $n = p_1 \cdots p_r$. We define

$$\{(u_1, \ldots, u_4) \in \mathbb{Z}^4 : u_1 - a_p u_3 - b_p u_4 \equiv 0 \bmod p, u_2 - b_p u_3 + a_p u_4 \equiv 0 \bmod p \ \forall_{\mathbb{P} \ni p \mid n}\},$$

where a_p, b_p are such that $a_p^2 + b_p^2 \equiv -1 \mod p$ (use Lemma 1.7.2). Then Λ is a lattice with $\det \Lambda \leq n^2$. Let $\varepsilon > 0$ be sufficiently small with respect to n. Define

$$C = B^4_{\sqrt{2(n-\varepsilon)}}(0).$$

Then

$$\mathrm{vol}(C) = \frac{\pi^2}{2}\sqrt{2(n-\varepsilon)}^4 = 2\pi^2(n-\varepsilon)^2 > 16n^2 \geq 2^4 \det \Lambda.$$

Now, Minkowski's First Theorem 1.4.1 implies that there exists $0 \neq u \in \Lambda \cap C$. In particular,

$$0 < u_1^2 + u_2^2 + u_3^2 + u_4^2 \leq 2(n-\varepsilon) < 2n.$$

We also have $u \in \Lambda$, which means that

$$\begin{aligned} u_1^2 + u_2^2 + u_3^2 + u_4^2 &\equiv (a_p u_3 + b_p u_4)^2 + (b_p u_3 - a_p u_4)^2 + u_3^2 + u_4^2 \\ &\equiv (a_p^2 + b_p^2 + 1)u_3^2 + (a_p^2 + b_p^2 + 1)u_4^2 \\ &\equiv 0 \qquad \mod p \end{aligned}$$

for each $\mathbb{P} \ni p \mid n$.
Hence, $p \mid u_1^2 + u_2^2 + u_3^2 + u_4^2 \ \forall_{\mathbb{P} \ni p \mid n}$, which implies $n \mid u_1^2 + u_2^2 + u_3^2 + u_4^2$. Combining this with the upper bound on n yields

$$n = u_1^2 + u_2^2 + u_3^2 + u_4^2. \qquad \square$$

1.8 Short Vectors in a Lattice

Apparently, finding the shortest vector in a lattice is an important (and quite difficult) problem in cryptography. Some cryptographic schemes rely on this fact. The secret corresponds to the shortest vector. Though we cannot do this in general, we can find a „pretty short" vector (LLL-Algorithm). We will do this next week.

Lecture 7 (8th May 2026 - W4L2)

Lecture 8 (13th May 2026 - W5L1)

1.8.1 Reduced Basis

Let $\{b_1, \ldots, b_n\}$ be linearly independent vectors in $\mathbb{R}^n$. The Gram–Schmidt orthogonalisation process produces vectors $\{b_1', \ldots, b_n'\}$ defined recursively by

$$b_1' = b_1,$$

$$b_2' = b_2 - \mu_{2,1} b_1',$$

$$\vdots$$

$$b_i' = b_i - \sum_{j=1}^{i-1} \mu_{i,j} b_j',$$

$$\vdots$$

$$b'_n = b_n - \sum_{j=1}^{n-1} \mu_{n,j} b'_j,$$

where

$$\mu_{i,j} = \frac{\langle b_i, b'_j \rangle}{\langle b'_j, b'_j \rangle} \; \forall_{1 \le i \le n,\, 1 \le j \le i-1}.$$

These vectors satisfy

$$\langle b'_i, b'_j \rangle = 0 \qquad \text{for } i \neq j.$$

Recall: We can write the vector b_i as

$$b_i = b'_i + \sum_{j=1}^{i-1} \mu_{i,j} b'_j,$$

where $\sum_{j=1}^{i-1} \mu_{i,j} b'_j$ is the projection of b_i onto $\mathrm{Span}_{\mathbb{R}}\{b_1, \ldots, b_{i-1}\} = \mathrm{Span}_{\mathbb{R}}\{b'_1, \ldots, b'_{i-1}\}$.

Definition 1.8.1 – Reduced Basis

A basis $\{b_1, \ldots, b_n\}$ of a lattice $\Lambda \subseteq \mathbb{R}^n$ is **reduced** if

(i) $|\mu_{i,j}| \le \frac{1}{2}$ for $1 \le j < i \le n$ and

(ii) $\|b'_i + \mu_{i,i-1} b'_{i-1}\|_2^2 \ge \frac{3}{4} \|b'_{i-1}\|_2^2$ for $2 \le i \le n$.

Remark (The constant 3/4)**:** The choice of the constant $\frac{3}{4}$ is not crucial; any real number in the interval $(\frac{1}{4}, 1)$ works. Also,

$$b_i = b'_i + \mu_{i,i-1} b'_{i-1} + \sum_{j=1}^{i-2} \mu_{i,j} b'_j,$$

so condition (ii) essentially says that the highlighted portion of b_i is longer (up to a constant factor) than b'_{i-1}.

Proposition 1.8.2 (Properties of Reduced Bases)

Let $\{b_1, \ldots, b_n\}$ be a reduced basis of Λ. Let $\{b'_1, \ldots, b'_n\}$ be the Gram-Schmidt orthogonalization. Then

(i) $\|b_l\|_2^2 \le 2^{i-1} \|b'_i\|_2^2$ for $1 \le l \le i \le n$,

(ii) $\det \Lambda \le \|b_1\|_2 \cdots \|b_n\|_2 \le 2^{\frac{n(n-1)}{4}} \det \Lambda$ and

(iii) $\|b_1\|_2 \le 2^{\frac{n-1}{4}} (\det \Lambda)^{1/n}$.

Proof.

(i) This follows from the definition. The statement is trivial for $i = 1$. For $2 \le i \le n$, condition (ii) in Definition 1.8.1 implies:

$$\|b'_i\|_2^2 + \mu_{i,i-1}^2 \|b'_{i-1}\|_2^2 \geq \frac{3}{4}\|b'_{i-1}\|_2^2$$

Since $|\mu_{i,i-1}| \leq \frac{1}{2}$, we have $\mu_{i,i-1}^2 \leq \frac{1}{4}$, which gives:

$$\|b'_i\|_2^2 \geq \left(\frac{3}{4} - \frac{1}{4}\right)\|b'_{i-1}\|_2^2 = \frac{1}{2}\|b'_{i-1}\|_2^2$$

By induction, we have

$$2^{i-l}\|b'_i\|_2^2 \geq \|b'_l\|_2^2 \text{ for } 1 \leq l \leq i. \tag{$\star$}$$

Now, relating $\|b_l\|_2$ to $\|b'_l\|_2$:

$$\|b_l\|_2^2 = \|b'_l\|_2^2 + \sum_{j=1}^{l-1} \mu_{l,j}^2 \|b'_j\|_2^2 \leq \|b'_l\|_2^2 + \sum_{j=1}^{l-1} \frac{1}{4} \cdot 2^{l-j}\|b'_l\|_2^2$$

Summing the geometric series yields $\|b_l\|_2^2 \leq 2^{l-1}\|b'_l\|_2^2$. Combined with ($\star$), we get the result.

(ii) The lower bound is Hadamard's inequality. For the upper bound, we use $\|b_i\|_2 \leq 2^{\frac{i-1}{2}}\|b'_i\|_2$ and the fact that

$$\det \Lambda = \prod_{i \in \{1,\dots,n\}} \|b'_i\|_2.$$

(iii) Applying (i) with $l = 1$ gives $\|b_1\|_2^2 \leq 2^{i-1}\|b'_i\|_2^2$. Taking the product over $i = 1, \dots, n$ and then the $2n$-th root leads to the bound. □

Proposition 1.8.3 (Shortest Vector Approximation)

Let $\{b_1, \dots, b_n\}$ be a reduced basis of Λ. For any $0 \neq x \in \Lambda$:

$$\|b_1\|_2^2 \leq 2^{n-1}\|x\|_2^2$$

Remark: This implies that b_1 is within a factor of $2^{\frac{n-1}{2}}$ of the shortest vector. The LLL algorithm computes such a basis efficiently. In modern cryptography, n often ranges from 256 to over 1024.

Proof. Write

$$\begin{aligned} x = &\ m_1 b_1 + \cdots + m_n b_n \\ = &\ \alpha_1 b'_1 + \cdots + \alpha_n b'_n \end{aligned}$$

with $m_1, \dots, m_n \in \mathbb{Z}$ and $\alpha_1, \dots, \alpha_n \in \mathbb{R}$. Let j_0 be the largest index such that

$$m_{j_0} \neq 0.$$

Then

$$\mathrm{Span}_{\mathbb{R}}\{b_1, \dots, b_{j_0}\} = \mathrm{Span}_{\mathbb{R}}\{b'_1, \dots, b'_{j_0}\}$$

and it follows from the Definition of $b'_1, \dots, b'_n$ that $m_{j_0} = \alpha_{j_0}$. We further have

$$\alpha_{j_0+1} = \dots = \alpha_n = 0.$$

Then

$$\begin{aligned} \|x\|_2^2 &\geq \alpha_{j_0}^2 \|b'_{j_0}\|_2^2 \\ &\geq \|b'_{j_0}\|_2^2 \\ &\geq 2^{-(j_0-1)} \cdot \|b_1\|_2^2 \end{aligned}$$

by Proposition 1.8.2(i). This implies

$$\|b_1\|_2 \leq 2^{\frac{j_0-1}{2}} \|x\|_2 \leq 2^{\frac{n-1}{2}} \|x\|_2. \qquad \square$$

This is a small generalisation of the previous proposition.

Proposition 1.8.4 (Generalised Shortest Vector Approximation)

Let $\{b_1, \dots, b_n\}$ be a reduced basis of a lattice Λ. Let $1 \leq \ell \leq n$ and $x_1, \dots, x_\ell \in \Lambda$ be linearly independent over $\mathbb{R}$.
Then

$$\|b_i\|_2^2 \leq 2^{n-1} \cdot \max\{\|x_1\|_2^2, \dots, \|x_\ell\|_2^2\}$$

for all $1 \leq i \leq \ell$.

Proof. For each x_i, write

$$\begin{aligned} x_i &= m_{i,1} b_1 + \dots + m_{i,n} b_n \\ &= \alpha_{i,1} b'_1 + \dots + \alpha_{i,n} b'_n \end{aligned}$$

with $m_{i,j} \in \mathbb{Z}$ and $\alpha_{i,j} \in \mathbb{R}$. Let $j_0(i)$ be the largest j such that $m_{i,j} \neq 0$.
Then, by the definition of $\{b'_1, \dots, b'_n\}$, it follows that

$$\alpha_{i,j_0(i)+1} = \dots = \alpha_{i,n} = 0$$

and

$$0 \neq m_{i,j_0(i)} = \alpha_{i,j_0(i)},$$

as before.
We have

$$\|x_i\|_2^2 \geq \alpha_{i,j_0(i)}^2 \|b'_{j_0(i)}\|_2^2 \geq \|b'_{j_0(i)}\|_2^2 \tag{1}$$

Without loss of generality (relabel $x_1, \dots, x_\ell$, if necessary) we assume

$$j_0(1) \leq \dots \leq j_0(\ell).$$

Claim

For each $1 \leq i \leq \ell$, we have $i \leq j_0(i)$.

Suppose otherwise. Then $j_0(i) < i$, but

$$\{x_1, \dots, x_i\} \subseteq \bigcup_{h=1}^{i} \mathrm{Span}_{\mathbb{R}}\{b_1, \dots, b_{j_0(h)}\} = \mathrm{Span}_{\mathbb{R}}\{b_1, \dots, b_{j_0(i)}\}$$

The left side is, however, linearly independent over $\mathbb{R}$. ↯
Hence, we have $i \leq j_0(i)$.
Using this, (1) becomes

$$\|x_i\|_2^2 \geq \|b'_{j_0(i)}\|_2^2 \geq 2^{-(j_0(i)-1)}\|b_i\|_2^2$$

by Proposition 1.8.2(i). This implies

$$\begin{aligned} \|b_i\|_2 \leq & \quad 2^{\frac{j_0(i)-1}{2}}\|x_i\|_2 \\ \leq & \quad 2^{\frac{n-1}{2}}\|x_i\|_2 \\ \leq & \quad 2^{\frac{n-1}{2}} \cdot \max\{\|x_1\|_2, \dots, \|x_\ell\|_2\}, \end{aligned}$$

which holds for any $1 \leq i \leq \ell$. □

Lecture 8 (13th May 2026 - W5L1)

Lecture 9 (15th May 2026 - W5L2)

1.9 The LLL Algorithm

The **LLL (Lenstra-Lenstra-Lovász) algorithm** turns a basis into a reduced basis (and thus gives a „short“ vector in Λ, recall Proposition 1.8.3).

Definition 1.9.1 – Property(k)

Let $2 \leq k \leq n+1$. We say a basis $\{b_1, \dots, b_n\}$ satisfies **Property(k)** if:

(i) $|\mu_{i,j}| \leq \frac{1}{2}$ for all $1 \leq j < i < k$.

(ii) $\|b'_i + \mu_{i,i-1}b'_{i-1}\|_2^2 \geq \frac{3}{4}\|b'_{i-1}\|_2^2$ for all $2 \leq i < k$.

Note: Property($n+1$) means the basis is reduced. And Property(2) holds trivially for any basis.

1.9.1 How the Algorithm Works

We start with a basis $B_0 = \{b_1, \dots, b_n\}$ of Λ. In particular, it satisfies Property(2). The algorithm proceeds in steps, producing a sequence of bases:

$$B_0 \xrightarrow{\mathrm{Prop}(2)} B_1 \xrightarrow{\mathrm{Prop}(k_1)} B_2 \xrightarrow{\mathrm{Prop}(k_2)} \cdots \to B_{\text{final}} \xrightarrow{\mathrm{Prop}(n+1)}$$

where $k_{i+1} = k_i \pm 1$. Once we reach Property($n+1$), we get a reduced basis. Done! As we shall see, k can go down only a finite number of times, so the algorithm terminates.

Remark (Efficiency of the Algorithm)**:** The algorithm is efficient. If we start with a basis $\{b_1, \ldots, b_n\}$ with $\|b_i\|_2 \leq B$ $(1 \leq i \leq n)$, then the number of arithmetical operations (addition, subtraction, multiplication, division) needed in the algorithm is $\mathcal{O}(n^4 \log B)$, and the algorithm runs in polynomial time in B.

1.9.2 Steps of the Algorithm

Initial step: Let $2 \leq k \leq n$. Assume we have a partial basis satisfying Property(k). We first achieve

$$|\mu_{k,k-1}| \leq \frac{1}{2}. \tag{$*$}$$

If $(*)$ does not hold, let $r \in \mathbb{Z}$ be such that $|\mu_{k,k-1} - r| \leq \frac{1}{2}$. We replace b_k with $b_k - rb_{k-1}$. This has the effect of changing $\mu_{k,k-1}$ to $\mu_{k,k-1} - r$. Since $|\mu_{k,k-1} - r| \leq \frac{1}{2}$, the new basis satisfies $(*)$.

Additional calculation. If we write $c_k = b_k - rb_{k-1}$, then its Gram-Schmidt vector is:

$$c_k' = c_k - \sum_{j=1}^{k-1} \frac{\langle c_k, b_j' \rangle}{\langle b_j', b_j' \rangle} b_j' = c_k - \sum_{j=1}^{k-1} (\mu_{k,j} - r) b_j'$$

I think that this is incorrect. We have

$$\begin{aligned}
c_k' &= c_k - \sum_{j=1}^{k-1} \frac{\langle c_k, b_j' \rangle}{\langle b_j', b_j' \rangle} b_j' \\
&= c_k - \frac{\langle c_k, b_{k-1}' \rangle}{\langle b_{k-1}', b_{k-1}' \rangle} - \sum_{j=1}^{k-2} \frac{\langle c_k, b_j' \rangle}{\langle b_j', b_j' \rangle} b_j' \\
&= c_k - \mu_{k,k-1} b_{k-1}' + \frac{\langle r b_{k-1}, b_{k-1}' \rangle}{\langle b_{k-1}', b_{k-1}' \rangle} b_{k-1}' - \sum_{j=1}^{k-2} \frac{\langle c_k, b_j' \rangle}{\langle b_j', b_j' \rangle} b_j' \\
&= c_k - \mu_{k,k-1} b_{k-1}' + r b_{k-1}' - \sum_{j=1}^{k-2} \frac{\langle c_k, b_j' \rangle}{\langle b_j', b_j' \rangle} b_j'.
\end{aligned}$$

Note: b_i, b_i' and $\mu_{i,j}$ are not affected for $1 \leq i \leq k-1$. □

Now we have a basis $\{b_1, \ldots, b_n\}$ satisfying Property(k) and $(*)$. We distinguish two main cases:

Case 1 (Swap Step): If $k \geq 2$ and

$$\|b_k' + \mu_{k,k-1} b_{k-1}'\|_2^2 < \frac{3}{4} \|b_{k-1}'\|_2^2 \tag{$\widehat{*}$}$$

Then we switch b_k and b_{k-1} while keeping the rest the same. Our new basis is $c_1, \ldots, c_n$ with $c_{k-1} = b_k$, $c_k = b_{k-1}$, and $c_i = b_i$ otherwise. This changes $\mu_{k-1,j}$ and $\mu_{k,j}$, but it does not affect $b_i, b_i', \mu_{i,j}$ for $1 \leq i \leq k-2$. The new basis still satisfies Property$(k-1)$.
Now c_{k-1}' is the projection of $c_{k-1} = b_k$ onto $(\mathrm{Span}_{\mathbb{R}}\{b_1', \ldots, b_{k-2}'\})^{\perp}$, i.e.

$$c_{k-1}' = b_k' + \mu_{k,k-1} b_{k-1}'.$$

From $(\widehat{*})$, we get $\|c_{k-1}'\|_2^2 < \frac{3}{4}\|b_{k-1}'\|_2^2$. So the length of the „new“ c_{k-1}' is shorter than the „old“ b_{k-1}' by a factor of $3/4$.

Remark (Note)**:** This is what is used to show that k can go down only finitely many times.

Case 2 (Size Reduction Step): If $k \geq 2$ and

$$\|b'_k + \mu_{k,k-1} b'_{k-1}\|_2^2 \geq \frac{3}{4}\|b'_{k-1}\|_2^2 \tag{$\widetilde{*}$}$$

First note that we already have $|\mu_{k,k-1}| \leq 1/2$ (from $(*)$). Let l be the largest integer such that $|\mu_{k,l}| > 1/2$. (Recall $\mu_{i,j}$ is only defined for $1 \leq j < i \leq n$). Let $r \in \mathbb{Z}$ be such that $|\mu_{k,l} - r| \leq 1/2$. We then replace b_k with $b_k - rb_l$. Thus $\mu_{k,l}$ becomes $\mu_{k,l} - r$.
This change affects other $\mu_{k,j}$ for $j < l$, but it does not affect $b_i, b'_i, \mu_{i,j}$ for $i < k$. It also doesn't affect $(\widetilde{*})$, as $l < k-1$. We repeat this process until we obtain that the basis satisfies $|\mu_{k,j}| \leq 1/2$ for all $1 \leq j \leq k-1$. The final basis satisfies Property$(k+1)$. To prove that the algorithm terminates, we need the following lemma, the proof of which will be on the next exercise sheet.

Lemma 1.9.2 (Lower Bound for Δ)

Let $\{b_1, \ldots, b_n\}$ be a basis of Λ. Then

$$\Delta(b_1, \ldots, b_n) = \prod_{i=1}^{n} \det(\Lambda_i)^2 \geq \frac{m(\Lambda)^{\frac{n(n+1)}{2}}}{4^{\frac{n(n+1)}{2}}} \prod_{i=1}^{n} \mathrm{vol}(\mathfrak{B}_i)^2$$

where $m(\Lambda) = \min\{\langle x, x\rangle : x \in \Lambda \setminus \{0\}\}$ and $\mathfrak{B}_i = \{x \in \mathbb{R}^i : |x| \leq 1\}$.

Proof. Exercise sheet 5. □

We can now prove that the algorithm terminates.

Proof that the LLL-Algorithm terminates. It is clear that $D_n = (\det \Lambda)^2$. Furthermore,

$$D_i = \|b'_1\|_2^2 \cdots \|b'_i\|_2^2 = (\det \Lambda_i)^2$$

where $\Lambda_i = [b_1 \ldots b_i v_{i+1} \ldots v_n]\mathbb{Z}^n$ with $\langle b_j, v_t\rangle = 0$ and $\{v_{i+1}, \ldots, v_n\}$ orthonormal (Gram-Schmidt sheet 3).
It can be checked that in both cases 0 and 2, $D_1, \ldots, D_n$ (and hence Δ) do not change. (It only involves modification of the form $b_k \to b_k - cb_i$ for some $i < k$).
In case 1, only D_{k-1} changes:

$$\begin{aligned}
D_{k-1} = \|b'_1\|_2^2 \cdots \|b'_{k-1}\|_2^2 \to \quad & \|c'_1\|_2^2 \cdots \|c'_{k-1}\|_2^2 \\
= \quad & \|b'_1\|_2^2 \cdots \|c'_{k-1}\|_2^2 \\
< \quad & \frac{3}{4}\|b'_1\|_2^2 \cdots \|b'_{k-1}\|_2^2 \\
= \quad & \frac{3}{4} D_{k-1,\mathrm{old}}
\end{aligned}$$

But all the other D_i do not change, which implies that $\Delta_{\mathrm{new}} < \frac{3}{4}\Delta_{\mathrm{old}}$. But Δ cannot get arbitrarily small due to Lemma 1.9.2. □

Conclusion: The RHS depends only on Λ and n. Therefore, in the algorithm, Case 1 can only occur finitely many times (i.e. $k \to k-1$ only bounded number of times), and hence k must reach $n+1$ and the algorithm terminates!

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Lecture 10 (22nd May 2026 - W6)

Example. Let $\alpha_1, \dots, \alpha_n \in \mathbb{R}$, not all in $\mathbb{Q}$. Also let $\varepsilon > 0$ be small. Suppose we want to find $q \in \mathbb{N}_{\geq 1}$ and $x_1, \dots, x_n \in \mathbb{Z}$ such that

$$|q\alpha_i - x_i| \leq \varepsilon \ \forall_{1 \leq i \leq n}. \tag{1}$$

Recall that in the proof of Corollary 1.4.5, we proved that given $Q > 1$, $\exists_{(x_1,\dots,x_n) \in \mathbb{Z}^{n+1}_{\text{prim}}} :$ $0 < q \leq Q \wedge |q\alpha_i - x_i| \leq Q^{-1/n}$. Applying this with $Q^{-1/n} = \varepsilon$, we can solve (1) with

$$0 < q \leq \varepsilon^{-n}. \tag{2}$$

Next, let's see what we get using LLL. Consider

$$\Lambda = \begin{pmatrix} & & & -\alpha_1 \\ & \mathbb{1}_n & & \vdots \\ & & & -\alpha_n \\ 0 & \cdots & 0 & \delta \end{pmatrix} \mathbb{Z}^{n+1},$$

with $\delta > 0$ to be chosen later. Then $\Lambda = \{(x_1 - q\alpha_1, \dots, x_n - q\alpha_n, \delta q) : x_1, \dots, x_n, q \in \mathbb{Z}\}$. Recall that LLL efficiently produces a reduced basis $\{b_1, \dots, b_{n+1}\}$ of Λ, satisfying

$$\|b_1\|_2 \leq 2^{\frac{n}{4}} \det \Lambda^{\frac{1}{n+1}} = 2^{\frac{n}{4}} \delta^{\frac{1}{n+1}}.$$

This was Proposition 1.8.2(iii), one of the properties of a reduced basis. Let us choose δ such that

$$2^{\frac{n}{4}} \delta^{\frac{1}{n+1}} = \varepsilon \iff \delta = \varepsilon^{n+1} 2^{\frac{-n(n+1)}{4}}.$$

Let us denote

$$b_1 = \begin{pmatrix} x_1 - q\alpha_1 \\ \vdots \\ x_n - q\alpha_n \\ \delta q \end{pmatrix}$$

(assume $q \geq 0$ without loss of generality). Then

$$\sqrt{(x_1 - q\alpha_1)^2 + \cdots + (x_n - q\alpha_n)^2 + (\delta q)^2} \leq \varepsilon,$$

which implies

$$|x_i - q\alpha_i| \leq \varepsilon \ \forall_{1 \leq i \leq n} \text{ and } 0 \leq q < \frac{\varepsilon}{\delta} = \varepsilon^{-n} 2^{\frac{n(n+1)}{4}}.$$

Note that if $q = 0$, we get a contradiction. $\square$

Chapter 2

The Bombieri-Pila Determinant Method

We have done a lot with lattice points inside a central symmetric convex body. In this section, we will count lattice points on a curve[1]. Our goal in this chapter is to prove the following.

Theorem 2.0.1 (Bombieri-Pila, 1989)

Let $F \in \mathbb{R}[x, y]$ be an irreducible polynomial of degree $d \geq 2$. If N is sufficiently large with respect only to d, then

$$\#\{(x, y) \in [1, N]^2 \cap \mathbb{Z}^2 : F(x, y) = 0\} \leq (\log N)^{\mathcal{O}(d)} N^{\frac{1}{d}}.$$

Remarks: The implicit constant in $\mathcal{O}(d)$ is absolute. In particular, it does not depend on the coefficients of F. Therefore, the bound holds uniformly over all irreducible $F \in \mathbb{R}[x, y]$ of degree d. Also, it is best possible up to the $(\log N)^{\mathcal{O}(d)}$ term (consider $F(x, y) = y - x^\alpha$).

Very rough proof idea. Draw the rectangle as well as the solution curve. Cover the solution curve with a small number of tiles that look like

$$\{(x, f(x)) : x \in [a, b]\}$$

for some $a < b \in \mathbb{R}$ and f as well as its derivatives decaying rapidly. As we shall see, we will be able to bound the integer points of

$$\#\mathbb{Z}^2 \cap \{(x, f(x)) : x \in [a, b]\}$$

efficiently. $\square$

2.1 Technical Preparation

Some notation for this chapter: The set $\mathcal{M} \subset \mathbb{R}[x, y]$ is a finite set of monomials, id est

$$\mathcal{M} = \{x^{j_1} y^{j_2} : \vec{j} = (j_1, j_2) \in S\}$$

with a finite $S \subset \mathbb{N}^2$. We also denote $S(\mathcal{M}) = S$. Further, we define

$$J_1 = J_1(\mathcal{M}) = \sum_{x^{j_1} y^{j_2} \in \mathcal{M}} j_1 \text{ and } J_2 = J_2(\mathcal{M}) = \sum_{x^{j_1} y^{j_2} \in \mathcal{M}} j_2.$$

[1]Professor Yamagishi: For our intents and purposes, this is just a zero set of a two-variable polynomial. No schemes, sheaves, connected. None of these Algebraic Geometry adjectives. Any Algebraic Geometers here? No one? Yeah, no one's taking that course.

Lemma 2.1.1 (Technical Preparation for Bombieri-Pila)

Let $\mathcal{M}$ be as above and $D = \#\mathcal{M} \geq 2$. Assume that $I \subseteq [0, N]$ is a closed interval and $f \in \mathcal{C}^{D-1}(I)$. Further suppose that $X > 0$ and $\delta \geq \frac{1}{N}$ satisfy that

$$\left| \frac{f^{(i)}(x)}{i!} \right| \leq X\delta^i \ \forall_{0 \leq i < D, x \in I}.$$

If

$$|I| < \frac{1}{4}\delta^{-1} \left((2N)^{J_1} (DX)^{J_2} \right)^{-\frac{1}{\binom{D}{2}}},$$

then

$$\exists_{0 \neq P \in \mathrm{Span}_{\mathbb{R}}(\mathcal{M})} : \{(x, f(x)) : x \in I\} \cap \mathbb{Z}^2 \subseteq V(P).$$

Here,

$$V(P) = \{(x, y) \in \mathbb{R}^2 : P(x, y) = 0\}.$$

Remarks: Given $z_1, \ldots, z_m \in \mathbb{R}^2$ with $m < D$, it's easy to deduce that there exists $P \in \mathrm{Span}_{\mathbb{R}}(\mathcal{M})$ such that $z_1, \ldots, z_m \in V(P)$ from elementary linear algebra results. But this lemma allows us to find $P \in \mathbb{R}[x, y]$ such that $\mathbb{Z}^2 \cap \{(x, f(x)) : x \in I\} \subseteq V(P)$ with good quantitative properties ($\deg P$ and $|I|$).

2.1.1 Preparation and Proof of Lemma 2.1.1

We will need several auxilliary lemmata.

Lemma 2.1.2 (Rank Criterion)

Let $\mathcal{M}$ be as above and $\vec{z}_1, \ldots, \vec{z}_t \in \mathbb{R}^2$. If $\mathrm{rank}((\vec{z}_i^{\,\vec{j}})_{\substack{1 \leq i \leq t \\ \vec{j} \in S(\mathcal{M})}}) < \#\mathcal{M}$, then there exists

$$0 \neq P \in \mathrm{Span}_{\mathbb{R}}(\mathcal{M}) : \vec{z}_1, \ldots, \vec{z}_t \in V(P).$$

Proof. Suppose that the rank of the matrix is r. Without loss of generality, we let

$$\mathrm{rank}((\vec{z}_i^{\,\vec{j}})_{\substack{1 \leq i \leq r \\ \vec{j} \in S(\mathcal{N})}}) = r, \text{ where } \mathcal{N} \subseteq \mathcal{M} \text{ with } \#\mathcal{N} = r.$$

By assumption, we have $r = \#\mathcal{N} < \#\mathcal{M}$, which means that there exists $\vec{k} = (k_1, k_2) \in$

$S(\mathcal{M}) \setminus S(\mathcal{N})$. Consider $P \in \mathbb{R}[x,y]$ defined by

$$\begin{aligned} P(x,y) = & \det A(x,y) \\ := & \det \begin{pmatrix} (x^{j_1}y^{j_2})_{\vec{j}\in S(\mathcal{N})} & x^{k_1}y^{k_2} \\ & \vec{z}_1^{\vec{k}} \\ (\vec{z}_i^{\vec{j}})_{\substack{1\le i\le r \\ \vec{j}\in S(\mathcal{N})}} & \vdots \\ & \vec{z}_r^{\vec{k}} \end{pmatrix} \\ = & \sum_{\vec{j}\in S(\mathcal{N})\cup\{\vec{k}\}} \varepsilon_{\vec{j}} \det(A_{\vec{j}}) x^{j_1}y^{j_2}, \end{aligned}$$

where $\varepsilon_{\vec{j}} \in \{\pm 1\}$ and $A_{\vec{j}}$ is the matrix obtained from $A(x,y)$ by removing the top row and the column $\vec{j}$. We know that at least one of the $\det A_{\vec{j}}$ is nonzero, which ensures that our polynomial is nonzero. Therefore, we have

$$P \in \mathrm{Span}_{\mathbb{R}}(\mathcal{N} \cup \{x^{k_1}y^{k_2}\}) \subseteq \mathrm{Span}_{\mathbb{R}}(\mathcal{M}).$$

Finally, if $1 \le i \le r$, then $A(\vec{z}_i)$ has a repeated row, which means that $P(\vec{z}_i) = 0$. If $r+1 \le i \le t$, then $\det A(\vec{z}_i) = 0$ because if $\mathrm{rank} A(\vec{z}_i) = r+1$, this contradicts that

$$r = \mathrm{rank}((\vec{z}_i^{\vec{j}})_{\substack{1\le i\le t \\ \vec{j}\in S(\mathcal{M})}}).$$

Hence, we have $\vec{z}_1, \ldots, \vec{z}_t \in V(P)$. $\square$

Lecture 10 (22nd May 2026 - W6)

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Lecture 11 (3rd June 2026 - W7L1)

We are in the middle of proving Lemma 2.1.1. Here is the rough idea of how the proof goes. Say

$$\{(x_1, f(x_1)), \ldots, (x_D, f(x_D))\} \subseteq \{(x, f(x)) : x \in I\} \cap \mathbb{Z}^2.$$

If the matrix

$$\left(x_i^{j_1} f(x_i)^{j_2}\right)_{\substack{1\le i\le D \\ \vec{j}=(j_1,j_2)\in S(\mathcal{M})}}$$

has full rank, then its determinant is nonzero. The derivative bounds for f and the shortness of I will show that this determinant is very small, but since all entries are integers, its absolute value is bounded below by 1. This is a contradiction. Hence, the matrix does not have full rank, and Lemma 2.1.2 gives the desired polynomial.

Lemma 2.1.3 (Generalised Vandermonde Bound)

Let $n \ge 1$, let $I \subseteq \mathbb{R}$ be a closed interval, and suppose $f_1, \ldots, f_n \in \mathcal{C}^{n-1}(I)$. Let $A_{i,j} \ge 0$ be such that

$$\max_{x\in I} \left| \frac{f_j^{(i-1)}(x)}{(i-1)!} \right| \le A_{i,j} \ \forall_{1\le i,j\le n}.$$

Then, for any $x_1, \ldots, x_n \in I$, we have

$$|\det(f_j(x_i))_{1\le i,j\le n}| \le \prod_{1\le m<i\le n} |x_i - x_m| \cdot \sum_{\sigma\in S_n} \prod_{i=1}^{n} A_{i,\sigma(i)},$$

where S_n denotes the group of permutations of $\{1, \ldots, n\}$.

Proof. If $x_1, \ldots, x_n$ are not distinct, then the result is trivial. Therefore, assume they are distinct.
First suppose $1 \le i, j \le n$. We can find $\xi_{i,j} \in I$ such that

$$\frac{f_j^{(i-1)}(\xi_{i,j})}{(i-1)!} = \sum_{l=1}^{i} f_j(x_l) \prod_{\substack{1\le m\le i\\ m\ne l}} \frac{1}{x_l - x_m}. \tag{1}$$

To see this, consider

$$g(\xi) = \sum_{l=1}^{i} f_j(x_l) \prod_{\substack{1\le m\le i\\ m\ne l}} \frac{\xi - x_m}{x_l - x_m}.$$

We have $\deg g < i$ and $g(x_\ell) = f_j(x_\ell)$ for all $1 \le \ell \le i$. Hence, by repeated applications of Rolle's theorem, there exists $\xi_{i,j} \in I$ such that

$$\frac{d^{i-1}}{d\xi^{i-1}}(g - f_j)(\xi_{i,j}) = 0.$$

Thus,

$$f_j^{(i-1)}(\xi_{i,j}) = g^{(i-1)}(\xi_{i,j}) = \sum_{l=1}^{i} f_j(x_l)(i-1)! \prod_{\substack{1\le m\le i\\ m\ne l}} \frac{1}{x_l - x_m},$$

as desired.
For $1 \le l \le i$, let

$$G_{l,i}(x_1, \ldots, x_i) = (i-1)! \prod_{\substack{1\le m\le i\\ m\ne l}} \frac{1}{x_l - x_m}.$$

Then from (1) we get

$$f_j^{(i-1)}(\xi_{i,j}) = \sum_{l=1}^{i} f_j(x_l) G_{l,i}(x_1, \ldots, x_i)$$

for some $\xi_{i,j} \in I$. Hence,

$$\begin{aligned}
& \det\left(f_j^{(i-1)}(\xi_{i,j})\right)_{1\le i,j\le n} \\
= \quad & \det\left(\sum_{l=1}^{i} f_j(x_l)G_{l,i}(x_1,\dots,x_i)\right)_{1\le i,j\le n} \\
= \quad & \det\begin{pmatrix} G_{1,1} & 0 & 0 & \cdots \\ G_{1,2} & G_{2,2} & 0 & \cdots \\ G_{1,3} & G_{2,3} & G_{3,3} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \cdot \det(f_j(x_i))_{1\le i,j\le n} \\
= \quad & \prod_{i=1}^{n} G_{i,i}(x_1,\dots,x_i) \cdot \det(f_j(x_i))_{1\le i,j\le n}.
\end{aligned}$$

Therefore,

$$\begin{aligned}
|\det(f_j(x_i))_{1\le i,j\le n}| &= \prod_{i=1}^{n} |G_{i,i}(x_1,\dots,x_i)|^{-1} \cdot \left|\det\left(f_j^{(i-1)}(\xi_{i,j})\right)_{1\le i,j\le n}\right| \\
&\le \prod_{i=1}^{n} \frac{\prod_{1\le m<i}|x_i - x_m|}{(i-1)!} \cdot \sum_{\sigma\in S_n}\prod_{i=1}^{n} |f_{\sigma(i)}^{(i-1)}(\xi_{i,\sigma(i)})| \\
&\le \prod_{1\le m<i\le n} |x_i - x_m| \cdot \sum_{\sigma\in S_n}\prod_{i=1}^{n} A_{i,\sigma(i)}.
\end{aligned}$$

□

We will use this lemma with functions of the form $x^{j_1} f(x)^{j_2}$.

Lemma 2.1.4 (Derivative Bounds for Monomials Along a Graph)

Let $k \ge 1$, let $I \subseteq [0, N]$ be a closed interval, and let $f \in \mathcal{C}^k(I)$. Suppose that $X > 0$ and $\delta \ge \frac{1}{N}$ are such that

$$\max_{x\in I}\left|\frac{f^{(i)}(x)}{i!}\right| \le X\delta^i \ \forall_{0\le i\le k}.$$

Then, for any $\vec{j} = (j_1, j_2) \in \mathbb{Z}^2_{\ge 0}$, the function

$$f_{\vec{j}}(x) = x^{j_1} f(x)^{j_2}$$

satisfies

$$\max_{x\in I}\left|\frac{f_{\vec{j}}^{(i-1)}(x)}{(i-1)!}\right| \le (2N)^{j_1}(iX)^{j_2}\delta^{i-1} \ \forall_{1\le i\le k+1}.$$

Proof. Exercise sheet 6. □

Proof of Lemma 2.1.1. Let

$$\{(x, f(x)) : x \in I\} \cap \mathbb{Z}^2 = \{\vec{z}_1, \dots, \vec{z}_t\},$$

where $\vec{z}_i = (x_i, f(x_i))$. By Lemma 2.1.2, if the matrix

$$A = (\vec{z}_i^{\vec{j}})_{\substack{1\le i\le t\\ \vec{j}\in S(\mathcal{M})}}$$

satisfies rank$A < D$, then we are done. Therefore, let us suppose rank$A = D$. Without loss of generality, we assume

$$0 \neq \Delta = \det(\vec{z}_i^{\vec{j}})_{\substack{1\le i\le D\\ \vec{j}\in S(\mathcal{M})}}.$$

Since each $\vec{z}_i \in \mathbb{Z}^2$, we have $|\Delta| \in \mathbb{Z}_{\ge 1}$.
Let $f_{\vec{j}}(x) = x^{j_1} f(x)^{j_2}$, so that $f_{\vec{j}}(x_i) = \vec{z}_i^{\vec{j}}$. By Lemma 2.1.4 with $k = D-1$ and the hypothesis on the derivatives of f, we get

$$\max_{x\in I}\left|\frac{f_{\vec{j}}^{(i-1)}(x)}{(i-1)!}\right| \le (2N)^{j_1}(DX)^{j_2}\delta^{i-1} =: A_{i,\vec{j}}.$$

By Lemma 2.1.3, we get

$$\begin{aligned}
1 \le |\Delta| &= \left|\det(f_{\vec{j}}(x_i))_{\substack{1\le i\le D\\ \vec{j}\in S(\mathcal{M})}}\right| \\
&\le \prod_{1\le m<i\le D}|x_i - x_m| \cdot \sum_{\sigma}\prod_{i=1}^{D} A_{i,\sigma(i)} \\
&\le |I|^{\binom{D}{2}} \cdot \sum_{\sigma}\prod_{i=1}^{D}(2N)^{\sigma(i)_1}(DX)^{\sigma(i)_2}\delta^{i-1} \\
&\le |I|^{\binom{D}{2}} \cdot D! \cdot (2N)^{J_1(\mathcal{M})}(DX)^{J_2(\mathcal{M})}\delta^{\binom{D}{2}},
\end{aligned}$$

where the sum is over all bijections $\sigma : \{1,\dots,D\} \to S(\mathcal{M})$.
Thus,

$$|I|^{\binom{D}{2}} \ge (D!)^{-1}\left((2N)^{J_1(\mathcal{M})}(DX)^{J_2(\mathcal{M})}\right)^{-1}\delta^{-\binom{D}{2}},$$

and so

$$|I| \ge (D!)^{-\frac{1}{\binom{D}{2}}}\left((2N)^{J_1(\mathcal{M})}(DX)^{J_2(\mathcal{M})}\right)^{-\frac{1}{\binom{D}{2}}}\delta^{-1}.$$

Since $D! \le D^D$ and $D^{-\frac{D}{\binom{D}{2}}} \ge \frac{1}{4}$ for all $D \ge 2$, we obtain

$$|I| \ge \frac{1}{4}\delta^{-1}\left((2N)^{J_1(\mathcal{M})}(DX)^{J_2(\mathcal{M})}\right)^{-\frac{1}{\binom{D}{2}}},$$

which contradicts our assumption. Hence, rank$(A) < D$, as desired. □

Lecture 11 (3rd June 2026 - W7L1)

. . . .

Lecture 12 (5th June 2026 - W7L2)

We will use this in the following manner:

Lemma 2.1.5 (Alternative Formulation of the Main Technical Lemma)

Let $\mathcal{M} \subseteq \mathbb{R}[x,y]$ be a finite set of monomials with $\#\mathcal{M} = D \geq 2$. Also let $I \subseteq [0,N]$ be a closed interval and $f \in \mathcal{C}^{d-1}(I)$. Suppose that $X > 0$ and $\delta \geq \frac{1}{N}$ are such that

$$\max_{x\in I}\left|\frac{f^{(i)}(x)}{i!}\right| \leq X\delta^i \;\forall_{0\leq i<D}.$$

Then $\{(x,f(x)) : x \in I\} \cap \mathbb{Z}^2$ is contained in the union of

$$4\delta\lambda^1(I)((2N)^{J_1(\mathcal{M})}(DX)^{J_2(\mathcal{M})})^{\frac{1}{\binom{D}{2}}} + 1$$

curves $V(P)$ with $0 \not\equiv P \in \mathrm{Span}_{\mathbb{R}}(\mathcal{M})$.

Remark (Trivial Bound)**:** We have the trivial bound $\lambda^1(I) + 1$. So we do better if

$$\delta((2N)^{J_1(M)}(DX)^{J_2(M)})^{\frac{1}{\binom{D}{2}}} < \frac{1}{4}.$$

Proof. Let $\{(x,f(x)) : x \in I\} = \{z_1,\ldots,z_t\}$, where $z_i = (x_i, f(x_i))$ with $x_1 < \cdots < x_t$. Define a sequence $(n_k)_{k\in\mathbb{N}}$ by $n_0 = 1$, and then recursively define n_ℓ to be the largest number such that

$$\{z_i : n_{\ell-1} \leq i < n_\ell\} \subseteq V(P_\ell) \text{ for some } 0 \not\equiv P_\ell \in \mathrm{Span}_{\mathbb{R}}(M).$$

Suppose that the sequence terminates at n_T. Then

$$\{z_1,\ldots,z_t\} \subseteq V(P_1) \cup \cdots \cup V(P_T) \cup \underbrace{\{z_{n_T},\ldots,z_t\}}_{\subseteq V(P_{T+1})},$$

where the statement in the underbrace follows from the definition of n_T. Hence, $\{(x,f(x)) : x \in I\} \cap \mathbb{Z}^2$ is contained in $T+1$ curves in $\mathrm{Span}_{\mathbb{R}}(\mathcal{M})$. For each $0 \leq \ell < T$,

$$\{z_i : n_\ell \leq i \leq n_{\ell+1}\} \not\subseteq V(P_{\ell+1})$$

for any $0 \not\equiv P \in \mathrm{Span}_{\mathbb{R}}(M)$. Use Lemma 2.1.1 with $I = [x_{n_\ell}, x_{n_{\ell+1}}]$. This implies that

$$|x_{n_{\ell+1}} - x_{n_\ell}| \geq \frac{1}{4}\delta^{-1}((2N)^{J_1}(DX)^{J_2})^{-\frac{1}{\binom{D}{2}}}.$$

We therefore have

$$\begin{aligned}\lambda^1(I) &\geq |x_{n_T} - x_1| \\ &= \sum_{\ell=0}^{T-1}|x_{n_{\ell+1}} - x_{n_\ell}| \\ &\geq T\frac{1}{4}\delta^{-1}((2N)^{J_1}(DX)^{J_2})^{-\frac{1}{\binom{D}{2}}}.\end{aligned}$$

Rearrange this to get the bound for T. □

Recall:

Theorem 2.1.6 (Bézout)

Let $F, G \in \mathbb{R}[x,y]$ be non-constant without a common divisor. Then there are at most $\deg(F)\deg(G)$ points $(\widehat{x},\widehat{y}) \in \mathbb{R}^2$, counted with multiplicity, such that $F(\widehat{x},\widehat{y}) = G(\widehat{x},\widehat{y}) = 0$. In particular, if F is irreducible over $\mathbb{R}$ and $F \nmid G$, then

$$\#\{(\widehat{x},\widehat{y}) \in \mathbb{R}^2 : F(\widehat{x},\widehat{y}) = G(\widehat{x},\widehat{y}) = 0\} \leq \deg(F)\deg(G).$$

2.2 Dividing the curve

We begin with the following lemma.

Lemma 2.2.1 (Bound for Points on a Solution Curve)

Suppose that $F \in \mathbb{R}[x,y]$ is irreducible of degree d over $\mathbb{R}$, with $\partial_y F(x,y) \not\equiv 0$. Let $I \subseteq \mathbb{R}$ be a closed interval (not a point), $k \geq 1$ and $f \in \mathcal{C}^\infty(I)$ satisfy $F(x, f(x)) = 0 \ \forall_{x \in I}$. Suppose that f is **NOT** a degree k polynomial. Then, we have

$$\#\{x \in I : f^{(k)}(x) = c\} \leq 2kd^2 \ \forall_{c \in \mathbb{R}^\times}.$$

Proof. We basically take the k-th derivative of $F(x, f(x)) = 0$ and see what happens. We show by induction that for each $k \geq 1$, $\exists_{H_k \in \mathbb{R}[x,y] : \deg(H_k) \leq D_k = (k-1)(2d-3)+d-1}$:

$$H_k(x, f(x)) + (\partial_y F(x, f(x)))^{2k-1} f^{(k)}(x) = 0 \ \forall_{x \in I}. \qquad (*)$$

For $k = 1$, we have

$$\partial_x F(x, f(x)) + \partial_y F(x, f(x)) f^{(1)}(x) = 0 \ \forall_{x \in I}.$$

Therefore, $(*)$ holds with $H_k(x,y) = \partial_x F(x,y)$ and $\deg(H_k) \leq d - 1 = D_1$. Assume that $(*)$ hold for some $k \geq 1$. Take the derivative of $(*)$ to get

$$\begin{aligned} 0 \equiv \ & \partial_x H_k(x, f(x)) + \partial_y H_k(x, f(x)) f'(x) + (\partial_y F(x, f(x)))^{2k-1} f^{(k+1)}(x) + \\ & (2k-1)(\partial_y F(x, f(x)))^{2k-2} f^{(k)}(x)(\partial^2_{xy} F(x, f(x)) + \partial^2_y F(x, f(x)) f'(x)). \end{aligned}$$

Multiply by $(\partial_y F(x, f(x)))^2$ (and simplify the notation) to get

$$0 = (\partial_y F)^2(\partial_x H_k + \partial_y H_k f') + (\partial_y F)^{2k+1} f^{(k+1)} + (2k-1)(\partial_y F)^{2k} f^{(k)}(\partial^2_{xy} F + \partial^2_y f').$$

Now use $(*)$ ($H_k = -(\partial_y F)^{2k-1} f^{(k)}$ and $\partial_x F = -\partial_y F f'$) to deduce that

$$\begin{aligned} 0 = \ & (\partial_y F)^2 \partial_x H_k - \partial_x F \partial_y F \partial_y H_k + (\partial_y F)^{2k+1} f^{(k+1)} - (2k-1) H_k \underbrace{\partial_y F(\partial^2_{xy} F + \partial^2_y F f')}_{= \partial_y F \partial^2_{xy} F - \partial_x F \partial^2_y F} \\ = \ & H_{k+1}(x, f(x)) + (\partial_y F(x, f(x)))^{2k+1} f^{k+1}(x) \ \forall x \in I. \end{aligned}$$

Check that $\deg H_{k+1} \leq D_k + 2d - 3 = D_{k+1}$. This proves $(*)$ for $k+1$. It follows from

$(*)$ that, given $x \in I$ and $c \in \mathbb{R}^\times$,

$$f^{(k)}(x) = c \implies R_c(x, f(x)) = 0,$$

where $R_c(x,y) = H_k(x,y) + (\partial_y F(x,y))^{2k-1}c$. Check that $\deg R_c \leq 2kd$. Therefore,

$$\begin{aligned} & \{x \in I : f^{(k)}(x) = c\} \\ \subseteq \; & \{x \in I : R_c(x, f(x)) = 0\} \\ \subseteq \; & \{x \in I : R_c(x,y) = 0, F(x,y) = 0 \text{ for some } y \in \mathbb{R}\}. \end{aligned}$$

This implies that (under the assumption that $F \nmid R_c$, which we will prove later)

$$\begin{aligned} \#\{x \in I : f^{(k)}(c)\} \leq \; & \#\{(x,y) \in \mathbb{R}^2 : F(x,y) = 0, R_c(x,y) = 0\} \\ \overset{F \nmid R_c,\text{ Thm. 2.1.6}}{\leq} & \deg(F)\deg(R_c) \\ \leq \; & d2kd \\ = \; & 2kd^2, \end{aligned}$$

as desired. It remains to show that $F \nmid R_c$. Let us suppose otherwise. This would mean that $F(x, f(x)) = 0$ implies $R_c(x, f(x)) = 0$, where we have

$$R_c(x, f(x)) = H_k(x, f(x)) + \partial_y F(x, f(x))^{2k-1}c \overset{(*)}{=} -\partial_y F(x, f(x))^{2k-1}(f^{(k)}(x) - c).$$

By the same argument as before,

$$\begin{aligned} & \#\{x \in I : \partial_y F(x, f(x)) = 0\} \\ \leq \; & \{(x,y) \in \mathbb{R}^2 : F(x,y) = \partial_y F(x,y) = 0\} \\ \overset{F \nmid \partial_y F f}{\leq} & \deg(F)\deg(\partial_y F) \\ \leq \; & d(d-1). \end{aligned}$$

Hence, $f^{(k)}(x) = c \neq 0 \; \forall_{x \in I}$, except for at most d^2 points. Thus, f is of degree k. $\lightning$
This completes the proof. $\square$

Lecture 12 (5th June 2026 - W7L2)

Lecture 13 (10th June 2026 - W8L1)

Lemma 2.2.2 (Partitioning the Interval for a Solution Curve)

Suppose $F \in \mathbb{R}[x,y]$ is irreducible over $\mathbb{R}$ and of degree $d \geq 2$. Let I be a closed interval and $f \in \mathcal{C}^\infty(I)$ such that $F(x, f(x)) = 0 \; \forall_{x \in I}$. Let $k \geq 1$ and $A_1, \ldots, A_k > 0$. Then, we may partition I into $\mathcal{O}(k^2 d^2)$ intervals I_v such that for each $1 \leq i \leq k$ and I_v, we have either

(i)

$$\left| f^{(i)}(x) \right| \leq A_i \; \forall_{x \in I_v} \text{ or}$$

(ii)

$$\left| f^{(i)}(x) \right| \geq A_i \; \forall_{x \in I_v}.$$

Proof. We consider two cases separately:

(1) **The function f is NOT a polynomial of degree $\leq k$.**
Let

$$\{x_1 < \cdots < x_R\} = \bigcup_{1 \leq i \leq k} \{x \in I : f^{(i)}(x) = \pm A_i\}.$$

By Lemma 2.2.1, we have

$$R \ll k^2 d^2.$$

Denote $I \coloneqq [x_0, x_{R+1}]$. Then $I_v = [x_v, x_{v+1}]$ for each $0 \leq v \leq R$ and these I_v satisfy the property.

(2) **The function f is a polynomial of degree $h \leq k$.**
We have

$$f^{(i)}(x) \equiv 0 \ \forall_{h < i \leq k}$$

and

$$f^{(h)}(x) \equiv c \neq 0.$$

Therefore, we can apply the previous argument with $h-1$ in place of k and the result follows. $\square$

We use this lemma to divide up the curve based on the derivative of f. If all derivatives are small, then we use Lemma 2.1.5. If not, we prove that $\lambda(I_v)$ is small (this is the easy case).

Lemma 2.2.3 (The Case of Small Derivatives)

Let $\ell \geq d \geq 2$ and $I \subseteq [0, N]$ be a closed intervall. Fix $f \in \mathcal{C}^\infty(I)$ with $F(x, f(x)) = 0 \ \forall_{x \in I}$, F being of degree d and irreducible over $\mathbb{R}$. Suppose that $N > 0$ and $\delta \geq \frac{1}{\lambda(I)}$ satisfy

$$\max_{x \in I} \left| \frac{f^{(i)}(x)}{i!} \right| \leq N\delta^i \ \forall_{0 \leq i \leq k = d(\ell - d + 1)}.$$

Then

$$\#\{(x, f(x)) : x \in I\} \cap \mathbb{Z}^2 \ll (d\ell)^c N^{\frac{1}{d} + \mathcal{O}\left(\frac{1}{\ell}\right)} \delta \lambda(I)$$

for some absolute $c, \ell, d > 0$.

Proof. Let i_0 be the largest i such that $x^{d-i}y^i$ is a monomial appearing in F. Define $\mathcal{M} = \{x^{j_1}y^{j_2} : d \leq j_1 + j_2 \leq \ell, x^{d-i_0}y^{i_0} \nmid x^{j_1}y^{j_2}\}$. We shall now compute $\#\mathcal{M}$. We have

$$\#\mathcal{M} = \sum_{d \leq h \leq \ell} \#\{(j_1, j_2) \in S(\mathcal{M}) : h = j_1 + j_2\}.$$

Fix $d \leq h \leq \ell$. Clearly, $j_2 = h - j_1$. Now $x^{d-i_0}y^{i_0} \nmid x^{j_1}y^{j_2}$ implies that $d - i_0 > j_1$ or $i_0 > j_2$. Hence, $0 \leq j_1 < d - i_0$ or $h - i_0 < j_1 \leq h$. Since $h \geq d \geq i_0$, the number of choices for j_1 is $d - i_0 + i_0 = d$. Therefore,

$$\#\mathcal{M} = \sum_{d \le h \le \ell} d = d(\ell - d + 1) = k = d\ell + \mathcal{O}(d^2).$$

Also,

$$\begin{aligned} J_1(\mathcal{M}) + J_2(\mathcal{M}) &= \sum_{d \le h \le \ell} dh \\ &= d\left(\frac{\ell(\ell+1)}{2} - \frac{d(d-1)}{2}\right) \\ &\le \frac{\ell^2 d}{2} + \mathcal{O}(d\ell) \\ &= \frac{k^2}{2d} + \mathcal{O}\left(\frac{k^2}{\ell}\right). \end{aligned}$$

This implies that

$$\frac{J_1(M) + J_2(M)}{\binom{k}{2}} \le \frac{1}{d} + \mathcal{O}\left(\frac{1}{\ell}\right). \tag{1}$$

Next, we show that $F \nmid P$ for any $0 \not\equiv P \in \mathrm{Span}_{\mathbb{R}}(\mathcal{M})$. Suppose that $F \mid P$. This implies that $P = FG$ with $\deg G = d'$. Let i_0' be the largest i such that $x^{d-i}y^i$ is a monomial appearing in G. Then $x^{d+d'-i_0-i_0'}y^{i_0+i_0'}$ is a monomial appearing in P. Therefore, we have

$$x^{d+d'-i_0-i_0'}y^{i_0+i_0'} \in \mathcal{M},$$

which is a contradiction because $x^{d-i_0}y^{i_0}$ divides this monomial. $\lightning$
Under the hypothesis of this lemma, it follows from Lemma 2.1.5 that

$$\{(x, f(x)) : x \in I\} \cap \mathbb{Z}^2 \subseteq \bigcup_{P \in \mathfrak{P}} V(P), \tag{$*$}$$

where the union is over $\mathfrak{P} \subseteq \mathrm{Span}_{\mathbb{R}}(\mathcal{M})$ of size

$$\begin{aligned} &\le 4\delta\lambda(I)\left((2N)^{J_1(M)}(kN)^{J_2(M)}\right)^{\frac{1}{\binom{k}{2}}} + 1 \\ &\overset{(1)}{\le} 4\delta\lambda(I)N^{\frac{1}{d}+\mathcal{O}\left(\frac{1}{\ell}\right)}k^{\frac{1}{d}+\mathcal{O}\left(\frac{1}{\ell}\right)} + 1. \end{aligned}$$

We therefore have

$$\begin{aligned} \#\{(x, f(x)) : x \in I\} \cap \mathbb{Z}^2 &\le \sum_{P \in \mathfrak{P}} \#\{(x, y) \in \mathbb{Z}^2 : F(x, y) = P(x, y) = 0\} \\ &\overset{\text{Bézout, } F \nmid P}{\le} \sum_{P \in \mathfrak{P}} \deg(F)\deg(P) \\ &\ll \left(4\delta\lambda(I)N^{\frac{1}{d}+\mathcal{O}\left(\frac{1}{\ell}\right)} + k^{\frac{1}{d}+\mathcal{O}\left(\frac{1}{\ell}+1\right)}\right)d\ell \\ &\ll (d\ell)^c\delta\lambda(I)N^{\frac{1}{d}+\mathcal{O}\left(\frac{1}{\ell}\right)}, \end{aligned}$$

where we used that the left hand side of $(*)$ is contained in $V(F) \cap \mathbb{Z}^2$. Also recall that $k = d\ell + \mathcal{O}(d^2)$. $\square$

Lemma 2.2.4 (The Short Interval Bound)

Let $k \geq 1$, $0 < \delta < 1$, and $X > 0$. Further assume that I is a closed bounded interval and $f \in \mathcal{C}^k(I)$. If

$$\max_{x \in I} \left| \frac{f^{(i)}(x)}{i!} \right| \leq X\delta^i \ \forall_{0 \leq i < k}$$

and

$$\min_{x \in I} \left| \frac{f^{(k)}(x)}{k!} \right| \geq X\delta^k,$$

we have $\lambda(I) \leq \frac{2}{\delta}$.

Proof. Let $I = [a, b]$. Then there exists $\xi \in I$ such that

$$f(b) - f(a) = \sum_{i=1}^{k-1} \frac{f^{(i)}(a)}{i!}(b-a)^i + \frac{f^{(k)}(\xi)}{k!}(b-a)^k.$$

Therefore, we have

$$X\delta^k \lambda(I)^k \leq \left| \frac{f^{(k)}(\xi)}{k!}(b-a)^k \right| \leq 2X + \sum_{i=1}^{k-1} X\delta^i \lambda(I)^i.$$

This implies that

$$\delta^k \lambda(I)^k \leq 2 + \sum_{i=1}^{k-1} \delta^i \lambda(I)^i.$$

Write $\lambda = \delta\lambda(I)$. Then

$$2 + \sum_{i=1}^{k-1} \lambda^i = 2 + \frac{\lambda^k - \lambda}{\lambda - 1}.$$

If $\lambda > 2$, then $\lambda^k \leq 2 + \lambda^k - \lambda$, which implies $\lambda \leq 2$ ↯. Therefore, we have $\lambda \leq 2$, i.e. $\lambda(I) \leq \frac{2}{\delta}$. $\square$

Lecture 13 (10th June 2026 - W8L1)

Lecture 14 (12th June 2026 - W8L2)

2.3 Number of lattice points on a curve

Theorem 2.3.1 (Number of lattice points on a curve)

Let $d \geq 2$ and $N \geq 1$ be sufficiently large in terms of d. Let $I = [0, N]$ and $f \in \mathcal{C}^\infty(I)$ satisfy

$$|f'(x)| \leq 1 \ \forall_{x \in I}.$$

Suppose that there exists an irreducible polynomial $F \in \mathbb{R}[x,y]$ of degree d such that

$$F(x, f(x)) = 0 \ \forall_{x \in I}.$$

Then

$$\#\left(\{(x, f(x)) : x \in I\} \cap \mathbb{Z}^2\right) \leq (\log N)^{\mathcal{O}(d)} N^{\frac{1}{d}}.$$

Proof. For $T > 1$, define $g(T)$ to be the maximum of 2 and the supremum of

$$\#\left(\{(x, h(x)) : x \in J\} \cap \mathbb{Z}^2\right)$$

over all closed intervals $J \subseteq [0,T]$ with $\lambda(J) \geq 1$, all $h \in \mathcal{C}^\infty(J)$ with $|h'(x)| \leq 1$ for all $x \in J$, and all irreducible $G \in \mathbb{R}[x,y]$ of degree d such that $G(x, h(x)) = 0$ on J. For $0 < T \leq 1$, set $g(T) = 2$.
We will later prove the following recursive estimate.

Claim 2.3.2

Given any integer $\ell \geq d$, there exists $K = K(d, \ell)$ with

$$2 \leq K \leq \ell^{\mathcal{O}(1)}$$

such that, for all $T > 0$,

$$g(T) \leq K^{\mathcal{O}(d)} T^{\frac{1}{d} + \mathcal{O}\left(\frac{1}{\ell}\right)} + K g(K^{-2d} T),$$

where the implicit constants are absolute.

We show that this claim implies the theorem. After increasing the implicit constants, the claim gives absolute constants $c_0, c_1, c_2 > 0$ such that

$$g(T) \leq K^{c_1 d} T^{\frac{1}{d} + \frac{c_2}{\ell}} + K g(K^{-2d} T) \quad \text{and} \quad K \leq \ell^{c_0}.$$

Since N is sufficiently large in terms of d, we may choose $\ell = \lceil \log N \rceil \geq d$. Iterating the estimate, we obtain

$$g(N) \leq K^{c_1 d} N^{\frac{1}{d} + \frac{c_2}{\ell}} \sum_{j=0}^{n-1} K^{-j} + K^n g(K^{-2nd} N),$$

where n is chosen such that $K^{-2nd} N \leq 1 < K^{-2d(n-1)} N$. Thus $g(K^{-2nd} N) \leq 2$ and $K^n \leq K N^{\frac{1}{2d}}$. Hence

$$g(N) \ll K^{c_1 d} N^{\frac{1}{d} + \frac{c_2}{\ell}} + K N^{\frac{1}{2d}} \ll \ell^{\mathcal{O}(d)} N^{\frac{1}{d} + \frac{c_2}{\ell}} \ll (\log N)^{\mathcal{O}(d)} N^{\frac{1}{d}}.$$

It remains to prove the claim.

Proof of the Claim. Let $J \subseteq [0,T]$ be a closed interval with $\lambda(J) = X$, where $1 \leq X \leq T$, and let $h \in \mathcal{C}^\infty(J)$ be such that $|h'(x)| \leq 1$ for all $x \in J$ and $G(x, h(x)) = 0$ for some irreducible $G \in \mathbb{R}[x,y]$ of degree d.
Since the range of h on J has length at most X, after replacing h by $h + y_0$ for a suitable

$y_0 \in \mathbb{Z}$, and $G(x, y)$ by $G(x, y - y_0)$, we may assume

$$|h(x)| \leq X \ \forall_{x \in J}.$$

This replacement does not change the number of lattice points on the graph.
Fix $\ell \geq d$, put $k = d(\ell - d + 1)$, and choose a parameter $\delta \in \left(\frac{2}{X}, 1\right)$ for the moment. Apply Lemma 2.2.2 with $A_i = i! X \delta^i$ for $1 \leq i \leq k$. This partitions J into $\mathcal{O}(k^2 d^2)$ intervals J_v such that, for each $0 \leq i \leq k$ and each J_v, either

$$\left| \frac{h^{(i)}(x)}{i!} \right| \leq X \delta^i \ \forall_{x \in J_v}$$

or

$$\left| \frac{h^{(i)}(x)}{i!} \right| \geq X \delta^i \ \forall_{x \in J_v}.$$

Here the assertion for $i = 0$ is always in the first case.
Pick one of the intervals J_v. If $\lambda(J_v) \geq \frac{1}{\delta}$ and the first alternative holds for all $0 \leq i \leq k$, then Lemma 2.2.3, after translating J_v horizontally into $[0, X]$, gives

$$\#\left(\{(x, h(x)) : x \in J_v\} \cap \mathbb{Z}^2\right) \ll (d\ell)^c X^{\frac{1}{d} + \mathcal{O}\left(\frac{1}{\ell}\right)} \delta \lambda(J_v).$$

Otherwise, either $\lambda(J_v) < \frac{1}{\delta}$, or there exists $1 \leq k_0 \leq k$ such that

$$\left| \frac{h^{(i)}(x)}{i!} \right| \leq X \delta^i \ \forall_{x \in J_v}, \ 0 \leq i < k_0,$$

and

$$\left| \frac{h^{(k_0)}(x)}{k_0!} \right| \geq X \delta^{k_0} \ \forall_{x \in J_v}.$$

In the latter case, Lemma 2.2.4 gives $\lambda(J_v) \leq \frac{2}{\delta}$. Hence every interval in this second case contributes at most $g\left(\frac{2}{\delta}\right)$ lattice points.
Combining the two cases, we get

$$\#\left(\{(x, h(x)) : x \in J\} \cap \mathbb{Z}^2\right) \ll \quad (d\ell)^c X^{\frac{1}{d} + \mathcal{O}\left(\frac{1}{\ell}\right)} \delta X + C_1 d^2 (d\ell)^2 g\left(\frac{2}{\delta}\right)$$

for some absolute constant $C_1 > 0$.
Choose

$$K = C_1 d^2 (d\ell)^2.$$

Increasing C_1 if necessary, we have $2 \leq K \leq \ell^{\mathcal{O}(1)}$. If $2K^{2d} < X$, set

$$\delta = \frac{2K^{2d}}{X}.$$

Then

$$\#\left(\{(x, h(x)) : x \in J\} \cap \mathbb{Z}^2\right) \leq K^{\mathcal{O}(d)} X^{\frac{1}{d} + \mathcal{O}\left(\frac{1}{\ell}\right)} + K g(K^{-2d} X).$$

Since $X \leq T$ and g is increasing, this is

$$\leq K^{\mathcal{O}(d)} T^{\frac{1}{d}+\mathcal{O}\left(\frac{1}{\ell}\right)} + K g(K^{-2d}T).$$

On the other hand, if $2K^{2d} \geq X$, then

$$\#\left(\{(x, h(x)) : x \in J\} \cap \mathbb{Z}^2\right) \leq X + 1 \leq 2K^{2d} + 1,$$

and therefore

$$\#\left(\{(x, h(x)) : x \in J\} \cap \mathbb{Z}^2\right) \leq K^{\mathcal{O}(d)} T^{\frac{1}{d}+\mathcal{O}\left(\frac{1}{\ell}\right)} + K g(K^{-2d}T),$$

since $T \geq 1$. Taking the supremum over all admissible J, h, G proves the claim. $\square$

This completes the proof of our theorem. $\square$

Lecture 14 (12th June 2026 - W8L2)

Lecture 15 (17th June 2026 - W9L1)

Last class we proved Theorem 2.3.1.

Exercise 2.1

By modifying the proof, show that we can replace $I = [0, N]$ with any closed subinterval of $[0, N]$.

2.4 Final Preparation for the Proof of Theorem 2.0.1

Let $F \in \mathbb{R}[x, y]$ be of degree $d \geq 2$ and irreducible over $\mathbb{R}$. Further assume that $\partial_x F, \partial_y F \not\equiv 0$ because the statement would otherwise be trivial as

$$\begin{aligned} & \#\{(x, y) \in \mathbb{Z}^2 \cap [0, N]^2 : F(x, y) = 0\} \\ = \; & (N+1)\#\{x \in \mathbb{Z} \cap [0, N] : F(x) = 0\} \\ = \; & 0. \end{aligned}$$

Let $V = V(F) = \{(x, y) \in \mathbb{R}^2 : F(x, y) = 0\}$ and $\mathrm{Sing}(V) = \{(x, y) \in \mathbb{R}^2 : F(x, y) = \partial_x F(x, y) = \partial_y F(x, y) = 0\}$. The implicit function theorem can be combined with the fact that V is algebraic to deduce that $V \setminus \mathrm{Sing}(V) = \bigsqcup_{i=1}^n C_i$, where the C_i are the connected components (and there are only finitely many of them) and $C_i \cong_{\mathcal{C}^1} \mathbb{R}$ or $\mathbb{S}^1$. Therefore, we have

$$\begin{aligned} & V \cap [0, N]^2 \\ = \; & (\mathrm{Sing}(V) \cap [0, N]^2) \cup \bigcup_{i=1}^n C_i \cap [0, N]^2, \end{aligned}$$

where each i such that $C_i \cap [0, N]^2 \neq \varnothing$ satifies one of the following:

(i) $\overline{C_i}$ intersects $\partial[0, N]^2$,

(ii) $\overline{C_i}$ has an end point at $\mathrm{Sing}(V)$ and

(iii) $\overline{C_i}$ is a loop inside $(0, N)^2$.

Fact 2.4.1

Given any $p \in V$,

$$\#\{i \in \{1, \ldots, n\} : p \in \overline{C_i}\} \leq 2d.$$

Example where this works. To see this, by translating if necessary assume that $p = (0, 0)$ and $F = F_d + F_{d-1} + \cdots + F_4 + (x-y)y^2$ (where the F_i are the components of the respective degrees). Note that this is an exemplary choice of F. Near $(0, 0)$, $F = 0$ looks like the plot of $x = y$ and $y = 0$ with the singular point 0. $\square$

Then the number of i satisfying (i) is (explanation below – credit to Gemini)

$$\begin{aligned} &\leq 2d\#(V(F) \cap \partial[0, N]^2) \\ &\leq 2d4d \\ &\leq 8d^2. \end{aligned}$$

Note that we can fix the ordinate or abscissa along every edge of the rectangle. Then, this becomes a one-variable polynomial with at most d zeroes.
The number of i satisfying (ii) is

$$\begin{aligned} &\leq 2d\#(V(F) \cap V(\partial_x F) \cap V(\partial_y F)) \\ &\overset{\text{Bézout, } F \text{ irr.}}{\leq} 2dd^2 \\ &= 2d^3. \end{aligned}$$

Finally, the number of i satisfying (iii) is (if C_i is a loop, then there exists $(x_0, y_0) \in C_i$ such that $\partial_x F(x_0, y_0) = 0$ or $\partial_y F(x_0, y_0) = 0$)

$$\begin{aligned} &\leq 2d(\#(V(F) \cap V(\partial_x F))) + \#(V(F) \cap V(\partial_y F)) \\ &\leq 4d^3. \end{aligned}$$

Hence, the number of connected components of $V \setminus \mathrm{Sing}(V) \cap [0, N]^2$ is $\mathcal{O}(d^3)$.

2.5 The Proof of Theorem 2.0.1

Proof of Theorem 2.0.1. Take any connected component C_i such that $\overline{C_i} \cap [0, N]^2 \neq \varnothing$. Assume that this is C_1 without loss of generality. We first remove all the points in $(V(\partial_x F) \cup V(\partial_y F)) \cap V(F)$ from C_1. The removed set has cardinality $\mathcal{O}(d^2)$. Suppose that $\partial_x F \not\equiv \pm\partial_y F$ so that $\partial_x F \pm \partial_y F$ is a polynomial of degree $\leq d - 1$. By Bézout, we have

$$\#(C_1 \cap V(\partial_x F + \partial_y F)) + \#(C_1 \cap V(\partial_x F - \partial_y F)) \leq 2d^2.$$

Furthermore, we remove all the points where $\partial_x F = \pm\partial_y F$. We denote the remaining set as

$$\bigcup_{j=1}^{M_1} C_{1,j}.$$

Then, for each j, we either have

(i) $|\partial_x F(x,y)| \leq |\partial_y F(x,y)| \ \forall_{(x,y)\in C_{1,j}}$ or

(ii) $|\partial_x F(x,y)| \geq |\partial_y F(x,y)| \ \forall_{(x,y)\in C_{1,j}}$.

If $\partial_x F \equiv \pm\partial_y F$, then these are automatically satisfied, without removing any more after $\mathcal{O}(d^2)$ points. Pick a j. Suppose that (i) holds without loss of generality (sligthly modify the argument otherwise). Let U_j be a sufficiently small open set such that $C_1 \cap U_j = C_{1,j}$ and we can apply the implicit function theorem. Then there exists an open interval $I_j \subseteq [0,N]$ and a smooth function $f_j \in \mathcal{C}^\infty(I_j)$ satisfying $C_{1,j} = \{(x, f_j(x)) : x \in I_j\}$.and

$$f_j'(x) = -\frac{\partial_x F(x, f_j(x))}{\partial_y F(x, f_j(x))} \ \forall_{x\in I_j}.$$

In particular,

$$\begin{aligned} |f_j'(x)| &= \frac{|\partial_x F(x, f_j(x))|}{|\partial_y F(x, f_j(x))|} \\ &\leq 1 \ \forall_{x\in I_j}. \end{aligned}$$

From Theorem 2.3.1, we obtain that

$$\begin{aligned} &\#C_{1,j} \cap [0,N]^2 \cap \mathbb{Z}^2 \\ \leq \ &\#\{(x, f_j(x)) : x \in \overline{I_j}\} \cap \mathbb{Z}^2 \\ \leq \ &\log^{\mathcal{O}(d)}(N) N^{\frac{1}{d}}. \end{aligned}$$

Therefore,

$$\begin{aligned} &\#\{(x,y) \in \mathbb{Z}^2 \cap [0,N]^2 : F(x,y) = 0\} \\ \ll \ &d^3 \left(\mathcal{O}(d^2) + d^2 \log^{\mathcal{O}(d)}(N) N^{\frac{1}{N}}\right) \\ \ll \ &d^5 \log^{\mathcal{O}(d)}(N) N^{\frac{1}{d}}, \end{aligned}$$

where the first estimate uses that the number of C_i such that $\overline{C_i} \cap [0,N]^2 \neq \varnothing$ is in $\mathcal{O}(d^3)$, that the number of removed points is $\mathcal{O}(d^2)$ and the number of j is d^2. □

In a similar manner, we can prove the following:

Theorem 2.5.1 (Bombieri-Pila II)

Let $k \geq 2$ and

$$D = \frac{(k+1)(k+2)}{2}.$$

Further assume that $f : [0,N] \to [0,N]$ is stricly convex. If $f \in \mathcal{C}^D([0,N],[0,N])$ and $f^{(D)}(x) \neq 0 \ \forall_{x\in[0,N]}$, then

$$\#\{(x, f(x)) : x \in [0,N]\} \cap \mathbb{Z}^2 \ll_k N^{\frac{1}{2}+\frac{8}{3(k+3)}+o(1)}.$$

The implicit constant depends only on k.

Proof. Next exercise sheet. □

Chapter 3

Roth's Theorem on Diophantine Approximation

The goal of this chapter is to prove:

Theorem (Roth, 1955)

Let $\alpha \in \mathbb{R} \cap \mathbb{A}$ and $\varepsilon > 0$. Then the set

$$\left\{(a, q) \in \mathbb{Z}^2_{\text{prim}} : q > 0, \left|\alpha - \frac{a}{q}\right| < \frac{1}{q^{2+\varepsilon}}\right\}$$

is finite.

Remark: Let $\alpha \in \mathbb{A} \cap \mathbb{R}$. Then α has a minimal polynomial, which we can modify to get $\mu_\alpha(t) = a_n t^n + \cdots + a_0 \in \mathbb{Z}[t]$. Then $a_n\alpha$ is a root of

$$t^n + \sum_{j=0}^{n-1} a_n^{n-j-1} a_j t^j,$$

i.e. an algebraic integer. Take a sufficiently large q. Then

$$\begin{aligned} & \left|\alpha - \frac{a}{q}\right| < \frac{1}{q^{2+2\varepsilon}} \\ \implies \quad & \left|a_n\alpha - \frac{a a_n}{q}\right| < \frac{|a_n|}{q^{2+2\varepsilon}} < \frac{1}{q^{2+\varepsilon}}. \end{aligned}$$

Therefore, it suffices to prove Theorem 1.1.5 for algebraic integers.

Lecture 15 (17th June 2026 - W9L1)

Lecture 16 (19th June 2026 - W9L2)

3.1 Notation and the Auxiliary Polynomial

Notation: Let $\alpha \in \mathbb{R}$ be an algebraic integer with the minimal polynomial

$$t^n + a_{n-1}t^{n-1} + \cdots + a_0 \in \mathbb{Z}[t].$$

Then we define

$$H(\alpha) = \max\{1, |a_{n-1}|, \ldots, |a_0|\}.$$

We will make use of polynomials of the following form:

$$0 \neq P(x_1, \dots, x_m) = \sum_{\vec{j} \in J} c(\vec{j}) x_1^{j_1} \cdots x_m^{j_m},$$

where $\vec{j} = (j_1, \dots, j_m)$,

$$J = J(r_1, \dots, r_m) = \{0, \dots, r_1\} \times \cdots \times \{0, \dots, r_m\} \quad \text{and} \quad c(\vec{j}) \in \mathbb{R}.$$

We let

$$\|P\| = \max_{\vec{j} \in J} |c(\vec{j})|,$$

and

$$P^{[\vec{k}]}(x_1, \dots, x_m) = \frac{1}{k_1!} \cdots \frac{1}{k_m!} \cdot \frac{\partial^{k_1}}{\partial x_1^{k_1}} \cdots \frac{\partial^{k_m}}{\partial x_m^{k_m}} P(x_1, \dots, x_m).$$

In particular, by Taylor's theorem,

$$P(x_1 + y_1, \dots, x_m + y_m) = \sum_{\vec{k} \in J} P^{[\vec{k}]}(x_1, \dots, x_m) \cdot y_1^{k_1} \cdots y_m^{k_m}.$$

Lemma 3.1.1 (Properties of $P^{[\vec{k}]}$)

The polynomial $P^{[\vec{k}]}$ satisfies the following properties:

(i) If $P(x_1, \dots, x_m) \in \mathbb{Z}[x_1, \dots, x_m]$, then $P^{[\vec{k}]}(x_1, \dots, x_m) \in \mathbb{Z}[x_1, \dots, x_m]$ for any $\vec{k} \in (\mathbb{Z}_{\geq 0})^m$.

(ii) Given any $\vec{k} \in (\mathbb{Z}_{\geq 0})^m$, we have

$$\deg_{x_i} P^{[\vec{k}]} = \begin{cases} \deg_{x_i} P - k_i & \text{if } k_i \leq \deg_{x_i} P, \\ -\infty & \text{otherwise.} \end{cases}$$

(iii) For any $\vec{k} \in (\mathbb{Z}_{\geq 0})^m$, we have

$$\|P^{[\vec{k}]}\| \leq 2^{r_1 + \cdots + r_m} \|P\|.$$

Proof. Exercise with derivatives and binomial coefficients. □

Let $\vec{\alpha} \in \mathbb{R}^m$. We say $0 \neq P$ has **index** I at $\vec{\alpha}$ with respect to $\vec{s} = (s_1, \dots, s_m) \in \mathbb{N}_{\geq 1}^m$ if

$$I = \min \left\{ \sum_{i=1}^m \frac{k_i}{s_i} : P^{[\vec{k}]}(\vec{\alpha}) \neq 0, \vec{k} \in J \right\}.$$

If $P \equiv 0$, we set $I = \infty$. We denote the index of P at $\vec{\alpha}$ with respect to $\vec{s}$ by

$$\mathrm{ind}_{\vec{s}}(P, \vec{\alpha}) = \mathrm{ind}_{\vec{s}}(P).$$

Lemma 3.1.2 (Properties of the Index)

Let $\vec{\alpha} \in \mathbb{R}^m$ and $\vec{s} \in \mathbb{N}^m_{\geq 1}$. Suppose $P, P_1, P_2 \not\equiv 0$. Then

(i) $\mathrm{ind}_{\vec{s}}(P^{[\vec{k}]}) \geq \mathrm{ind}_{\vec{s}}(P) - \sum_{i=1}^m \frac{k_i}{s_i}$ for any $\vec{k} \in J$,

(ii) $\mathrm{ind}_{\vec{s}}(P_1 + P_2) \geq \min(\mathrm{ind}_{\vec{s}}(P_1), \mathrm{ind}_{\vec{s}}(P_2))$ and

(iii) $\mathrm{ind}_{\vec{s}}(P_1 \cdot P_2) = \mathrm{ind}_{\vec{s}}(P_1) + \mathrm{ind}_{\vec{s}}(P_2)$.

Proof. Exercise sheet. □

Lemma 3.1.3 (Representation of Powers of Algebraic Integers)

Let α be an algebraic integer of degree n and let $\ell \in \mathbb{Z}_{\geq 0}$. Then there exist $a_0^{(\ell)}, \ldots, a_{n-1}^{(\ell)} \in \mathbb{Z}$ such that

$$\alpha^\ell = a_{n-1}^{(\ell)} \alpha^{n-1} + \cdots + a_0^{(\ell)},$$

with $|a_j^{(\ell)}| \leq (H(\alpha) + 1)^\ell$.

Proof. Using the minimal polynomial of α, we have

$$\alpha^n = -a_{n-1}\alpha^{n-1} - \cdots - a_0,$$

with $a_0, \ldots, a_{n-1} \in \mathbb{Z}$ (since α is an algebraic integer of degree n). Therefore the statement holds for $0 \leq \ell \leq n$ (and trivially for $\ell \leq n-1$, choose $a_\ell^{(\ell)} = 1$ and set the rest to zero). Use induction to complete the proof (exercise). (Done by ChatGPT – Sorry, no time :)) Assume that for some $\ell \geq n$ we have

$$\alpha^\ell = \sum_{j=0}^{n-1} a_j^{(\ell)} \alpha^j, \qquad |a_j^{(\ell)}| \leq (H(\alpha) + 1)^\ell$$

for all $0 \leq j \leq n-1$. Multiplying by α, we obtain

$$\alpha^{\ell+1} = \sum_{j=0}^{n-2} a_j^{(\ell)} \alpha^{j+1} + a_{n-1}^{(\ell)} \alpha^n.$$

Using the relation

$$\alpha^n = -a_{n-1}\alpha^{n-1} - \cdots - a_0,$$

it follows that

$$\alpha^{\ell+1} = \sum_{j=1}^{n-1} a_{j-1}^{(\ell)} \alpha^j - a_{n-1}^{(\ell)} \sum_{j=0}^{n-1} a_j \alpha^j.$$

Hence

$$\alpha^{\ell+1} = \sum_{j=0}^{n-1} b_j \alpha^j,$$

where

$$b_0 = -a_0 a_{n-1}^{(\ell)},$$

and, for $1 \le j \le n-1$,

$$b_j = a_{j-1}^{(\ell)} - a_j a_{n-1}^{(\ell)}.$$

Since $a_j, a_j^{(\ell)} \in \mathbb{Z}$, we have $b_j \in \mathbb{Z}$. Moreover,

$$|b_0| \le H(\alpha)(H(\alpha)+1)^\ell \le (H(\alpha)+1)^{\ell+1},$$

and for $1 \le j \le n-1$,

$$\begin{aligned} |b_j| &\le |a_{j-1}^{(\ell)}| + |a_j|\,|a_{n-1}^{(\ell)}| \\ &\le (H(\alpha)+1)^\ell + H(\alpha)(H(\alpha)+1)^\ell \\ &= (H(\alpha)+1)^{\ell+1}. \end{aligned}$$

Therefore

$$|b_j| \le (H(\alpha)+1)^{\ell+1} \qquad (0 \le j \le n-1),$$

which proves the induction step. □

Lemma 3.1.4 (Cardinality Bound for a Set of Small Indices)

Let $r_1, \dots, r_m \in \mathbb{N}_{\ge 1}$. Given $\lambda > 0$,

$$\#\left\{\vec{k} \in J(r_1, \dots, r_m) : \sum_{i=1}^m \frac{k_i}{r_i} \le \tfrac{1}{2}(m-\lambda)\right\} \le \frac{\sqrt{2m}}{\lambda}(r_1+1)\cdots(r_m+1).$$

Proof. Next exercise sheet (induction on m). □

Theorem 3.1.5 (Construction of an Auxiliary Polynomial)

Let $0 < \varepsilon < 1$. Let α be an algebraic integer of degree n with the minimal polynomial $f(t) = t^n + a_{n-1}t^{n-1} + \cdots + a_0 \in \mathbb{Z}[t]$. Let $m > 8n^2\varepsilon^{-2}$ be an integer, and let $r_1, \dots, r_m \in \mathbb{N}_{\ge 1}$. Then there exists $P(x_1, \dots, x_m) \in \mathbb{Z}[x_1, \dots, x_m]$ with $J = J(r_1, \dots, r_m)$ such that

(i) $P \not\equiv 0$,

(ii) $\operatorname{ind}_{\vec{r}}(P, (\alpha, \dots, \alpha)) \ge \frac{1}{2}m(1-\varepsilon)$ and

(iii) $\|P\| \le (4(H(\alpha)+1))^{r_1+\cdots+r_m}$.

In other words, we have constructed a polynomial $P(x_1, \dots, x_m)$ with a zero of large order at $(\alpha, \dots, \alpha)$!

Proof. Let us denote

$$P(x_1, \dots, x_m) = \sum_{\vec{j} \in J(r_1, \dots, r_m)} c(j_1, \dots, j_m) x_1^{j_1} \cdots x_m^{j_m},$$

with $c(j_1, \dots, j_m)$ integers to be determined. Set

$$N = (r_1 + 1) \cdots (r_m + 1) = \#J(r_1, \dots, r_m).$$

We want $P^{[\vec{k}]}(\alpha, \dots, \alpha) = 0$ for all $\vec{k} \in (\mathbb{Z}_{\geq 0})^m$ satisfying

$$\sum_{i=1}^m \frac{k_i}{r_i} < \tfrac{1}{2}(m - \varepsilon m),$$

since this implies $\mathrm{ind}_{\vec{r}}(P, (\alpha, \dots, \alpha)) \geq \frac{1}{2}(m - \varepsilon m)$. From Lemma 3.1.3, we know

$$\alpha^\ell = a_{n-1}^{(\ell)} \alpha^{n-1} + \cdots + a_0^{(\ell)}, \tag{$*$}$$

with integers $|a_j^{(\ell)}| \leq (H(\alpha) + 1)^\ell$. Then

$$P^{[\vec{k}]}(\alpha, \dots, \alpha) = \sum_{\substack{k_i \leq j_i \leq r_i \\ (1 \leq i \leq m)}} \binom{j_1}{k_1} \cdots \binom{j_m}{k_m} c(j_1, \dots, j_m) \alpha^{(j_1 - k_1) + \cdots + (j_m - k_m)}.$$

By substituting $(*)$ and collecting the powers of α, we obtain

$$= \sum_{h=0}^{n-1} \alpha^h \underbrace{\sum_{\vec{j}} \binom{j_1}{k_1} \cdots \binom{j_m}{k_m} c(j_1, \dots, j_m) a_h^{(j_1 - k_1 + \cdots + j_m - k_m)}}_{= Q_{\vec{k}, h} \in \mathbb{Z}}.$$

Since $\{1, \alpha, \dots, \alpha^{n-1}\}$ is linearly independent over $\mathbb{Q}$, we have

$$P^{[\vec{k}]}(\alpha, \dots, \alpha) = 0 \iff 0 = Q_{\vec{k}, 0} = \cdots = Q_{\vec{k}, n-1}.$$

Also, $|a_h^{(\ell)}| \leq (H(\alpha) + 1)^\ell$, $\binom{j_i}{k_i} \leq 2^{j_i} \leq 2^{r_i}$ and $j_1 - k_1 + \cdots + j_m - k_m \leq r_1 + \cdots + r_m$. Now we view $0 = Q_{\vec{k}, 0} = \cdots = Q_{\vec{k}, n-1}$ as a system of linear equations in the $c(\vec{j})$, with integer coefficients whose absolute values are

$$\leq (H(\alpha) + 1)^{r_1 + \cdots + r_m} \cdot 2^{r_1 + \cdots + r_m}.$$

It follows from Lemma 3.1.4 that

$$\begin{aligned} \#\left\{ \vec{k} \in J(r_1, \dots, r_m) : \sum_{i=1}^m \frac{k_i}{r_i} \leq \tfrac{1}{2}(m - m\varepsilon) \right\} &\leq \frac{\sqrt{2m}}{m\varepsilon} (r_1 + 1) \cdots (r_m + 1) \\ &= \frac{\sqrt{2}}{\sqrt{m} \cdot \varepsilon} N < \frac{N}{2n}, \end{aligned}$$

since $m > 8n^2 \varepsilon^{-2}$. Then

$$0 = Q_{\vec{k}, 0} = \cdots = Q_{\vec{k}, n-1} \qquad \left(\forall \vec{k} \in J(r_1, \dots, r_m) \text{ s.t. } \sum_{i=1}^m \frac{k_i}{r_i} \leq \tfrac{1}{2}(m - m\varepsilon) \right)$$

is a system of M linear equations over $\mathbb{Z}$ in N variables, where (for each $\vec{k}$ there are n

equations)

$$M < \frac{N}{2n} \cdot n = \frac{N}{2} < N = \#J(r_1, \dots, r_m),$$

and $|\text{coeff.}| \leq H = (2 \cdot (H(\alpha) + 1))^{r_1 + \cdots + r_m}$. We now invoke the following result, which we will prove next class.

Lemma (Siegel)

Given a system of M linear equations over $\mathbb{Z}$ in N variables with $N > M$, whose coefficients satisfy $|\text{coeffs}| \leq H$, there exists a solution $\vec{x} \in \mathbb{Z}^N \setminus \{0\}$ such that

$$\max_{1 \leq i \leq N} |x_i| \leq (N \cdot H)^{\frac{M}{N-M}}.$$

Therefore, we can choose $c(\vec{j}) \in \mathbb{Z}$ for all $\vec{j} \in J(r_1, \dots, r_m)$, not all of them zero, such that

$$\underbrace{0 = Q_{\vec{k},0} = \cdots = Q_{\vec{k},n-1}}_{\iff P^{[\vec{k}]}(\alpha, \dots, \alpha) = 0} \quad \left(\forall \vec{k} \in J(r_1, \dots, r_m) \text{ s.t. } \sum_{i=1}^{m} \frac{k_i}{r_i} \leq \tfrac{1}{2}(m - m\varepsilon) \right)$$

and

$$\begin{aligned} 0 < \max |c(\vec{j})| &\leq (NH)^{\frac{M}{N-M}} \\ &\leq N \cdot H && (2M < N) \\ &\leq (4 \cdot (H(\alpha) + 1))^{r_1 + \cdots + r_m}. \end{aligned}$$

$\square$

Lecture 16 (19th June 2026 - W9L2)

. . . .

Lecture 17 (24th June 2026 - W10L1)

Theorem 3.1.6 (Siegel's lemma)

Let $N > M$ and $A = (a_{ij}) \in M_{M \times N}(\mathbb{Z})$ with $\max_{i,j} |a_{ij}| \leq H$. Then there exists $x \in \mathbb{Z}^N$ such that $Ax = 0$ with $0 < \max_{1 \leq i \leq N} |x_i| \leq (NH)^{\frac{M}{N-M}}$.

Proof. For each $1 \leq i \leq M$, define $L_i(x) = \sum_{j=1}^{N} a_{ij} x_j$. Denote

$$-C_i = \sum_{j=1}^{N} \min\{a_{ij}, 0\}$$

and

$$D_i = \sum_{j=1}^{N} \max\{a_{ij}, 0\}.$$

Clearly, $0 \leq C_i + D_i \leq NH$. Given any $y \in \mathbb{Z}^N \cap [0, B]^N$, we have $-C_i B \leq L_i(y) \leq$

$D_i B \ \forall_{1 \leq i \leq M}$. Let $B \in \mathbb{N}_{\geq 1}$. Then

$$\begin{aligned} & \#\{(L_1(y), \ldots, L_M(y)) : y \in \mathbb{Z}^N \cap [0, B]^N\} \\ \leq \ & \prod_{i=1}^{M} D_i B + C_i B + 1 \\ \leq \ & \prod_{i=1}^{M} (NHB + 1) \\ = \ & (NHB+1)^M. \end{aligned}$$

If $\#\{y \in \mathbb{Z}^N \cap [0, B]^N\} = (B+1)^N > (NHB+1)^M$, then there exist distinct $y_1, y_2 \in \mathbb{Z}^N \cap [0, B]^N$ such that $L_i(y_1) = L_i(y_2) \ \forall_{1 \leq i \leq M}$. Hence, $L_i(y_1 - y_2) = 0 \ \forall_{1 \leq i \leq M}$ and $Ax = 0$ with $0 \neq x = y_1 - y_2$ and $x \in \mathbb{Z}^N \cap [-B, B]^N$. Finally, check that

$$B = \left\lfloor (NH)^{\frac{M}{N-M}} \right\rfloor$$

does the job. □

Let's recall some notation.

$$P(x_1, \ldots, x_m) = \sum_{j \in J} C(j) x_1^{j_1} x_m^{j_m},$$

$$J = J(r_1, \ldots, r_m) = \{0, \ldots, r_1\} \times \cdots \times \{0, \ldots, r_m\},$$
$$\|P\| = \max_{j \in J} |c(j)|$$

and

$$P^{[k]}(x_1, \ldots, x_m) = \frac{1}{k_1!} \cdots \frac{1}{k_m!} \frac{\partial^{k_1 + \cdots + k_m} p}{\partial_1^{k_1} \cdots \partial_m^{k_m}}.$$

Also define

$$\mathrm{ind}_s(P, \alpha) = \min \left\{ \sum_{i=1}^{m} \frac{k_i}{s_i} : P^{[k]}(\alpha) \neq 0, k \in J \right\}$$

for $s \in \mathbb{N}_{\geq 1}^m$. Last time we proved Theorem 3.1.5. That is, we constructed a polynomial over $\mathbb{Z}$ with a zero of „large" order at $(\alpha, \ldots, \alpha)$. We now deduce this for $\left(\frac{a_1}{q_1}, \ldots, \frac{a_m}{q_m}\right)$, where each $\frac{a_n}{q_n} \in \mathbb{Q}$ is „close" to α, from the information at $(\alpha, \ldots, \alpha)$.

Theorem 3.1.7

Let $0 < \delta < \frac{1}{20}$ and $0 < \varepsilon < \frac{\delta}{100}$ and α an algebraic integer as in Theorem 3.1.5. Suppose that $\frac{a_h}{q_h} \in \mathbb{Q}$ with $1 \leq h \leq m$ fulfil

$$\left| \alpha - \frac{a_h}{q_h} \right| < \frac{1}{q_h^{2+\delta}}$$

and $q_h^{\varepsilon} > 64(H(\alpha)+1) \max\{1, |\alpha|\} \ \forall_{1 \leq h \leq m}$. Also suppose thta $m > 8n^2 \varepsilon^{-2}$ and $r_1, \ldots, r_m \in \mathbb{N}_{\geq 1}$ are such that $r_1 \log q_1 \leq r_h \log q_h \leq (1+\varepsilon) r_1 \log q_1 \ \forall_{1 \leq h \leq m}$. Then the integer polynomial $P(x_1, \ldots, x_m)$ constructed in Theorem 3.1.5 satisfies $\mathrm{ind}_r \left(P, \left(\frac{a_1}{q_1}, \ldots, \frac{a_m}{q_m} \right) \right) \geq \frac{\delta_m}{8}$.

Proof. We begin by applying Theorem 3.1.5 to get $P(x_1, \dots, x_m) \in \mathbb{Z}[x_1, \dots, x_m]$ with $\mathrm{ind}_r(P, (\alpha, \dots, \alpha)) \geq \frac{1}{2}m(1-\varepsilon)$. We want to show: Let $k_1, \dots, k_m \in \mathbb{N}$ fulfil

$$\sum_{h=1}^{m} \frac{k_i}{r_i} < \frac{\delta_m}{8}. \tag{$*$}$$

Then $P^{[k]}\left(\frac{a_1}{q_1}, \dots, \frac{a_m}{q_m}\right) = 0$ $\left(\text{because this implies } \mathrm{ind}_r\left(p, \left(\frac{a_1}{q_1}, \dots, \frac{a_m}{q_m}\right)\right) \geq \frac{\delta_m}{8}\right)$. Let k satisfy $(*)$ and denote $T(x_1, \dots, x_n) = P^{[k]}(x_1, \dots, x_m)$. It follows from Lemma 3.1.1 and Theorem 3.1.5 that $T(x_1, \dots, x_m) \in \mathbb{Z}[x_1, \dots, x_m]$, $\|T\| = (8H(\alpha)+8)^{r_1 + r_m}$ and $\deg_{x_h} T \leq r_h \ \forall_{1 \leq h \leq m}$. Given any $i = (i_1, \dots, i_m) \in J(r_1, \dots, r_m)$, we have

$$\begin{aligned} \left|T^{[i]}(\alpha, \dots, \alpha)\right| \leq & \ 2^{r_1 + \dots + r_m} \|T^{[i]}| \max\{1, |\alpha|\}^{r_1 + \dots + r_m} \\ \leq & \ (32H(\alpha)+32)^{r_1 + \dots + r_m} \max\{1, |\alpha|\}^{r_1 + \dots + r_m}. \end{aligned}$$

because $\|T^{[i]}\| \leq 2^{r_1 + \dots + r_m}\|T\|$. Next, by Lemma 3.1.2, we have

$$\begin{aligned} \mathrm{ind}_r(T, (\alpha, \dots, \alpha)) \geq & \ \mathrm{ind}_r(P, (\alpha, \dots, \alpha)) - \sum_{n=1}^{m} \frac{k_h}{r_h} \\ \overset{\text{Thm. 3.1.5}}{>} & \ \frac{1}{2}m(1-\varepsilon) - \frac{\delta m}{8} \\ = & \ \frac{1}{2}m\left(1 - \varepsilon - \frac{\delta}{4}\right) \\ > & \ \frac{1}{2}m\left(1 - \frac{\delta}{3}\right). \end{aligned}$$

Therefore, $T^{[i]}(\alpha, \dots, \alpha) = 0$ if

$$\sum_{h=1}^{m} \frac{i_h}{r_h} \leq \frac{1}{2}m\left(1 - \frac{\delta}{3}\right).$$

Now, let us put $\beta_h = \alpha - \frac{a_h}{q_h} \ \forall_{1 \leq h \leq m}$. Then $|\beta_h| < \frac{1}{q_h^{2+\delta}}$. By the Taylor series expansion, we get

$$\begin{aligned} T\left(\frac{a_1}{q_1}, \dots, \frac{a_m}{q_m}\right) = & \sum_{i \in J(r_1, \dots, r_m)} T^{[i]}(\alpha, \dots, \alpha)\beta_1^{i_1} \cdots \beta_m^{i_m} \\ = & \sum_{\substack{i \in J \\ \sum_{h=1}^{m} \frac{i_h}{r_h} > \frac{1}{2}m(1-\frac{\delta}{3})}} T^{[i]}(\alpha, \dots, \alpha)\beta_1^{i_1} \cdots \beta_h^{i_h}. (**) \end{aligned}$$

For any i in the sum,

$$\begin{aligned}
-\log|\beta_1^{i_1}\cdots\beta_m^{i_m}| &> (2+\delta)\sum_{h=1}^{m}\frac{i_h}{r_h}(r_h\log q_h) \\
&\geq (2+\delta)r_1\log(q_1)\underbrace{\sum_{h=1}^{m}\frac{i_h}{r_h}}_{>\frac{1}{2}m\left(1-\frac{\delta}{3}\right)} \\
&> \left(1+\frac{\delta}{2}\right)\left(1-\frac{\delta}{3}\right)mr_1\log(q_1) \\
&\geq \left(1+\frac{\delta}{2}\right)\left(1-\frac{\delta}{3}\right)\sum_{h=1}^{m}r_h\log(q_h)\frac{1}{1+\varepsilon} \\
&> (1+\varepsilon)^2\sum_h r_h\log(q_h)\frac{1}{1+\varepsilon}
\end{aligned}$$

for $0<\delta<\frac{1}{20}$ and $0<\varepsilon<\frac{\delta}{100}$, which implies $|\beta_1\cdots\beta_m|<(q_1^{r_1}\cdots q_m^{r_m})^{-1-\varepsilon}$. Hence, $(**)$ becomes

$$\begin{aligned}
&\mathbb{N}\ni\left|q_1^{r_1}\cdots q_m^{r_m}T\left(\frac{a_1}{q_1},\ldots,\frac{a_m}{q_m}\right)\right| \\
<\ & \sum_{i\in J}|T^{[i]}(\alpha,\ldots,\alpha)|(q_1^{r_1}\cdots q_m^{r_m})^{-\varepsilon} \\
\leq\ & 2^{r_1+\cdots+r_m}\left[(32H(\alpha)+32)\max\{|\alpha|,1\}\right]^{r_1+\cdots+r_m}(q_1^{r_1}\cdots q_m^{r_m})^{-\varepsilon} \\
\leq\ & 1.
\end{aligned}$$

This proves the assumption. $\square$

Lecture 17 (24th June 2026 - W10L1)

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Lecture 18 (28th June 2026 - W10L2)

Wronskians. Let Δ denote an operator of the form

$$\frac{\partial^{i_1}}{\partial x_1^{i_1}}\cdots\frac{\partial^{i_m}}{\partial x_m^{i_m}}$$

and denote $\mathrm{ord}(\Delta)=i_1+\cdots+i_m$. Suppose we have $\Delta_1,\ldots,\Delta_h$ with

$$\mathrm{ord}(\Delta_i)\leq i-1\quad(\forall 1\leq i\leq h).$$

Let $g_1,\ldots,g_h$ be sufficiently many times differentiable functions in $x_1,\ldots,x_m$. Then

$$\det\left[\Delta_i g_j\right]_{1\leq i,j\leq h}$$

is called a **(generalised) Wronskian**.

Proposition 3.1.8 (Wronskian criterion)

Let $m\geq 1$. Let $h\geq 1$ and let $g_1,\ldots,g_h$ be rational functions over $\mathbb{Q}$ in variables $x_1,\ldots,x_m$. Suppose

$$c_1g_1+\cdots+c_hg_h\equiv 0,\quad c_1,\ldots,c_h\in\mathbb{Q}$$

implies $c_1 = \cdots = c_h = 0$. Then there exists a Wronskian

$$\det\,[\Delta_i g_j]_{1\le i,j\le h} \not\equiv 0.$$

Proof. We prove the statement by induction on h. Suppose first that $m = 1$. In this case, consider

$$\Delta_i = \frac{d^{i-1}}{dx_1^{i-1}} \quad (\ \forall 1 \le i \le h).$$

Then

$$\det\,[\Delta_i g_j] \not\equiv 0$$

by an exercise.
Now suppose the statement holds for $< h$, and prove it for h. The case $h = 1$ is trivial, since $\Delta_1 g_1 = g_1 \not\equiv 0$. Thus we assume $g_1, \dots, g_h$ are linearly independent over $\mathbb{Q}$; in particular $g_1 \ne 0$.
We may assume without loss of generality that $g_1 \equiv 1$. Indeed, let

$$\widetilde{g}_j = \frac{1}{g_1} g_j \quad (\ \forall 1 \le j \le h).$$

Then

$$\Delta_i \widetilde{g}_j = \sum_a Q_{i,a}(g_1)\Delta_i^{(a)} g_j,$$

where each $Q_{i,a}(g_1)$ is a rational function whose numerator is a polynomial in the partial derivatives of g_1 (including g_1 itself) and whose denominator is a power of g_1. This is independent of g_j. Thus, by multilinearity of the determinant, $\det[\Delta_i \widetilde{g}_j]$ is a sum of Wronskians of $g_1, \dots, g_h$ multiplied by rational functions in the partial derivatives of g_1. Hence, if $\det[\Delta_i \widetilde{g}_j] \not\equiv 0$, at least one of the summands is nonzero. So we assume without loss of generality that $g_1 \equiv 1$.
If $g_h \equiv c \in \mathbb{Q}$, then $cg_1 - g_h \equiv 0$, a contradiction. Therefore, without loss of generality,

$$\frac{\partial g_h}{\partial x_1} \not\equiv 0.$$

Suppose there exist $c_2, \dots, c_h \in \mathbb{Q}$, not all zero, such that

$$\frac{\partial}{\partial x_1}(c_2 g_2 + \cdots + c_h g_h) \equiv 0.$$

If not, we set $k_1 = 1$. Otherwise, without loss of generality $c_2 = 1$; replacing g_2 by

$$g_2 + c_3 g_3 + \cdots + c_h g_h$$

does not change the Wronskian by the properties of the determinant. Thus, without loss of generality,

$$\frac{\partial g_2}{\partial x_1} \equiv 0.$$

Continuing in this manner, without loss of generality we obtain $1 \leq k_1 < h$ such that

$$\frac{\partial g_1}{\partial x_1} \equiv \cdots \equiv \frac{\partial g_{k_1}}{\partial x_1} \equiv 0,$$

and every nontrivial $\mathbb{Q}$-linear combination of $g_{k_1+1}, \dots, g_h$ depends on x_1.
By the inductive hypothesis, applied to $g_1, \dots, g_{k_1}$, there exist $\widetilde{\Delta}_1, \dots, \widetilde{\Delta}_{k_1}$ of orders $\leq 0, 1, \dots, k_1 - 1$, respectively, such that

$$W_1 = \det \left[\widetilde{\Delta}_i g_j\right]_{1 \leq i,j \leq k_1} \neq 0.$$

Moreover, there are no nontrivial $\mathbb{Q}$-linear relations among

$$\frac{\partial g_{k_1+1}}{\partial x_1}, \dots, \frac{\partial g_h}{\partial x_1}.$$

Thus, again by the inductive hypothesis, there exist $\widetilde{\Delta}_{k_1+1}, \dots, \widetilde{\Delta}_h$ of orders $\leq 0, 1, \dots, h - k_1 - 1$, respectively, such that

$$W_2 = \det \left[\widetilde{\Delta}_i \frac{\partial g_j}{\partial x_1}\right]_{k_1+1 \leq i,j \leq h} \neq 0.$$

Let

$$\Delta_i = \begin{cases} \widetilde{\Delta}_i & 1 \leq i \leq k_1, \\ \widetilde{\Delta}_i \frac{\partial}{\partial x_1} & k_1 + 1 \leq i \leq h. \end{cases}$$

In particular, $\operatorname{ord}(\Delta_i) \leq i - 1$ for $1 \leq i \leq h$. Then

$$W = \det \left[\Delta_i g_j\right]_{1 \leq i,j \leq h} = \det \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} = W_1 W_2 \neq 0. \qquad \square$$

Slight digression. Note Roth's theorem is trivial for non-real algebraic numbers. Let $\alpha = \lambda + i\beta$ with $\lambda, \beta \in \mathbb{R}$ and $\beta \neq 0$. Then clearly

$$|\beta| \leq \left|\frac{a}{q} - \alpha\right| < \frac{1}{q^{2+\varepsilon}}$$

only holds for at most finitely many $\frac{a}{q} \in \mathbb{Q}$ with $q > 0$ and $\gcd(a, q) = 1$.

Lecture 18 (28th June 2026 - W10L2)

Lecture 19 (1st July 2026 - W11L1)

Theorem 3.1.9

Let $m \in \mathbb{N}_{\geq 1}$, $0 < \varepsilon < \frac{1}{12}$, and set $\omega = \omega(m, \varepsilon) = 24 \cdot 2^{-m} \left(\frac{\varepsilon}{12}\right)^{\exp(\ln(2)(m-1))} < 1$. Let $r_1, \dots, r_m \in \mathbb{N}_{\geq 1}$ be such that $\omega r_h \geq r_{h+1}$ $\forall_{1 \leq h \leq m-1}$. Let $q_1, \dots, q_m \in \mathbb{N}_{\geq 1}$ be such that $q_h^{r_h} \geq q_1^{r_1}$ and $q_h^{\omega} \geq 2^{3m}$ $\forall_{1 \leq h \leq m}$. Suppose $P \in \mathbb{Z}[x_1, \dots, x_m] \setminus \{0\}$ is such that $\deg_{x_h} P \leq r_h$ $\forall_{1 \leq h \leq m}$ and $\|P\| \leq q_1^{\omega r_1}$. Then

$$\operatorname{ind}_r \left(P, \left(\frac{a_1}{q_1}, \dots, \frac{a_m}{q_m}\right)\right) \leq \varepsilon,$$

provided $\gcd(a_h, q_h) = 1\ \forall_{1\leq h\leq m}$.

Rough idea of the proof. Say $P(x_1, x_2)$ has large order of vanishing at $\left(\frac{a_1}{q_1}, \frac{a_2}{q_2}\right)$. Write $P(x_1, x_2) = \varphi_0(x_2) + \varphi_1(x_2)x_1 + \cdots + \varphi_r(x_2)x_1^r$. Define

$$0 \not\equiv W(x_1, x_2) = \underbrace{\det(\Delta_i\varphi_j(x_2))}_{U(x_2)\in\mathbb{Z}[x_2]}\underbrace{\det(\Delta_i x_1^{j-1})}_{V(x_1)\in\mathbb{Z}[x_1]}.$$

A large order of vanishing of P at $\left(\frac{a_1}{q_1}, \frac{a_2}{q_2}\right)$ would imply the same for W, which in turn implies the same for V at $\frac{a_1}{q_1}$ or U at $\frac{a_2}{q_2}$. Hence, $(q_1x_1 - a_1)^N \mid V(x_1)$. But if $\|V\| < q_1^N$, then $V \equiv 0$. Thus, we have $W \equiv 0$, which contradicts our assumptions. □

We shall now give a real proof.

Proof. We prove this by induction on m. For $m = 1$, suppose that

$$0 = P\left(\frac{a_1}{q_1}\right) = \cdots = P^{(t-1)}\left(\frac{a_1}{q_1}\right) \text{ and } P^{(t)}\left(\frac{a_1}{q_1}\right) \neq 0.$$

Here $\gcd(a_1, q_1) = 1$ and $q_1 > 0$. Write $P(x) = \left(x - \frac{a_1}{q_1}\right)^t T(x)$ for some $T \in \mathbb{Q}[x]$. Hence,

$$P(x) = (q_1x - a_1)^t \underbrace{(q_1^{-t}T(x))}_{\in\mathbb{Z}[x]}$$

by Gauß's lemma. Thus, $q_1^t \leq \|P\| \leq q_1^{\omega r_1}$, which implies $t \leq \omega r_1$. Since $m = 1$, $\omega(1, \varepsilon) = \varepsilon$. Since $m = 1$, $\omega(1, \varepsilon) = \varepsilon$, we have

$$\operatorname{ind}_{r_1}\left(P, \frac{a_1}{q_1}\right) = \frac{t}{r_1} \leq \omega = \varepsilon,$$

which completes the base case. Now suppose that the whole thing holds for all all $m' < m \in \mathbb{N}_{\geq 1}$. Let us write

$$P(x_1, \ldots, x_m) = \sum_{1\leq j\leq k} \varphi_j(x_1, \ldots, x_{m-1})\psi_j(x_m),$$

where φ_j, ψ_j are $\mathbb{Q}$-polynomials. We fix such a representation with k minimal. We know that $k \leq r_m + 1$ by choosing $\psi_j(x_m) = x_m^{j-1}$. Let us prove that φ_j and ψ_j have two properties.

Claim 3.1.10

The set $\{\varphi_1, \ldots, \varphi_k\}$ is linearly independent over $\mathbb{Q}$.

Proof of the claim. Suppose that there exist $c_1, \ldots, c_k \in \mathbb{Q}$, not all zero, such that

$$c_1\varphi_1 + \cdots + c_k\varphi_k \equiv 0.$$

Assume that $c_k = -1$ without loss of generality. This implies $\varphi_k = c_1\varphi_1 + \cdots + c_{k-1}\varphi_{k-1}$. Now substitute this into P and get

$$P(x_1, \ldots, x_m) = \sum_{j=1}^{k-1} \varphi_j(x_1, \ldots, x_{m-1})(\psi_j(x_m) + c_j\psi_k(x_m)),$$

which contradicts the minimality of k ↯. □

Claim 3.1.11

The set $\{\psi_1, \ldots, \psi_k\}$ is linearly independent over $\mathbb{Q}$.

Proof of this claim. The same argument. □

Since $\psi_1, \ldots, \psi_k$ are polynomials over $\mathbb{Q}$ and linearly independent over $\mathbb{Q}$, it follows from Proposition 3.1.8 that

$$U_1(x_m) = \left(\frac{\mathrm{d}^i}{\mathrm{d}(x_m)^i}\psi_j(x_m)\right)_{1\leq i,j\leq k} \not\equiv 0.$$

Similarly, there exist $\widetilde{\Delta}_1, \ldots, \widetilde{\Delta}_k$ such that

$$\widetilde{\Delta}_i = \frac{\partial^{i_1+\cdots+i_{m-1}}}{\partial x_1^{i_1}\cdots\partial_{m-1}^{i_{m-1}}}$$

with $\mathrm{ord}\widetilde{\Delta} = i_1 + \cdots + i_{m-1} \leq i-1 \leq k-1 \leq r_m$. Then $V_1(x_1, \ldots, x_{m-1}) = \det(\widetilde{\Delta}_i\varphi_j)_{1\leq i,j\leq k} \not\equiv 0$. We also denote

$$\iota(i) = (i_1, \ldots, i_{m-1}).$$

Next, we define

$$W_1(x_1, \ldots, x_m) = \det\left(\widetilde{\Delta}_i\frac{\partial^{j-1}}{\partial x_m^{j-1}}P(x_1, \ldots, x_m)\right)_{1\leq i,j\leq k}.$$

Since

$$\begin{aligned} &\widetilde{\Delta}_i\frac{\partial^{j-1}}{\partial x_m^{j-1}}P(x_1, \ldots, x_m) \\ = \;&\widetilde{\Delta}_i\frac{\partial^{j-1}}{\partial x_m^{j-1}}\sum_{h=1}^{k}\varphi_h(x_1, \ldots, x_{m-1})\varphi_h(x_m) \\ = \;&\sum_{h=1}^{k}\widetilde{\Delta}_i\varphi_h\frac{\mathrm{d}^{j-1}}{\mathrm{d}(x_m)^{j-1}}\psi_h(x_m), \end{aligned}$$

we have

$$\begin{aligned} W_1(x_1,\dots,x_m) = & \det\left((\widetilde{\Delta}_i\varphi_h)\left(\frac{\mathrm{d}^{j-1}}{\mathrm{d}(x_m)^{j-1}}\psi_h\right)^t\right) \\ = & V_1(x_1,\dots,x_{m-1})U_1(x_m) \\ \not\equiv & 0. \end{aligned}$$

Then let

$$\begin{aligned} W(x_1,\dots,x_m) = & \frac{W_1(x_1,\dots,x_m)}{\prod_{i=1}^k \iota(i)!\prod_{j=1}^k (j-1)!} \\ = & \det(P^{[\iota(i),j-1]})_{1\le i,j\le k}. \end{aligned}$$

This si still a polynomial over $\mathbb{Z}$ because $P^{[j]}\in\mathbb{Z}[x_1,\dots,x_m]$ for any j. We can deduce from Gauß's lemma that

$$W(x_1,\dots,x_m) = V(x_1,\dots,x_{m-1})U(x_m)$$

with U,V polynomials over $\mathbb{Z}$ (in fact $c_1U_1 = U$ and $c_2V_1 = V$ with $c_1,c_2\in\mathbb{Q}$). Since $\deg_{x_h} P^{[j]}\le r_h\ \forall_{1\le h\le m}, j\in\mathbb{N}^m$, $\deg_{x_h} w\le r_hk\ \forall_{1\le h\le m}$. Similarly, $\deg_{x_h} V\le r_hk\ \forall_{1\le h\le m-1}$ and $\deg_{x_m} U\le r_mk$. Also,

$$\|P^{[j]}\|\le 2^{r_1+\dots+r_m}\|P\|\le 2^{r_1+\dots+r_m}q_1^{r\omega_1}.$$

It follows that

$$\|W\|\le k!\left(2^{r_1+\dots+r_m}\max_{i,j}\|P^{[(\iota(i),j-1)]}\|\right)^k$$

by the bound for the number of summand in $P^{[j]}$. This is in turn less than or equal to

$$\begin{aligned} & k!\left(4^{r_1+\dots+r_m}q_1^{r\omega}\right)^k \\ \le\ & 2^{kr_m}\left(4^{r_1+\dots+r_m}q_1^{r\omega}\right)^k \\ \le\ & \left(8^{r_1+\dots+r_m}q_1^{r_1\omega}\right)^k \\ \le\ & 8^{mr_1k}q_1^{r_1\omega k} \\ \le\ & (q_1^{\omega})^{r_1k}q_1^{r_1\omega k} \\ =\ & q_1^{2\omega r_1k}. \end{aligned}$$

Recall that

$$\begin{aligned} W = & U(x_m)V(x_1,\dots,x_{m-1}) \\ = & \sum_\ell u_\ell x_m^\ell\sum_k v_kx_1^{k_1}\cdots x_{m-1}^{k_{m-1}} \\ = & \sum_{(\ell,k)} u_\ell v_kx_1^{k_1}\cdots x_{m-1}^{k_{m-1}}x_m^\ell. \end{aligned}$$

This implies that $\|U\|,\|V\|\le\|W\|\le q_1^{2\omega r_1k}$. Index of V: We consider

$\mathrm{ind}_{kr}\left(V, \left(\frac{a_1}{q_1}, \dots, \frac{a_{m-1}}{q_{m-1}}\right)\right)$. By the inductive hypothesis, we have

$$\mathrm{ind}_{kr}\left(V, \left(\frac{a_1}{q_1}, \dots, \frac{a_{m-1}}{q_{m-1}}\right)\right) \leq \varepsilon_0,$$

provided that $\|V\| \leq q_1^{2\omega r_1 k} \leq q_1^{\omega(m-1,\varepsilon_0) r_1 k}$. Since $\omega = \omega(m, \varepsilon) = \frac{1}{2}\omega\left(m-1, \frac{\varepsilon^2}{12}\right)$, we choose $\varepsilon_0 = \frac{\varepsilon^2}{12}$. Note that all of the other hypotheses are satisfied. Thus,

$$\begin{aligned} &\mathrm{ind}_{kr}\left(V, \left(\frac{a_1}{q_1}, \dots, \frac{a_{m-1}}{q_{m-1}}\right)\right) \leq \frac{\varepsilon^2}{12} \\ \implies &\mathrm{ind}_{r}\left(V, \left(\frac{a_1}{q_1}, \dots, \frac{a_{m-1}}{q_{m-1}}\right)\right) \leq \frac{k\varepsilon^2}{12}, \end{aligned}$$

Lecture 19 (1st July 2026 - W11L1)

Lecture 20 (3rd July 2026 - W11L2)

Recall that we are proving Theorem 3.1.9 by induction. We have

$$W(x_1, \dots, x_m) = U(x_m)V(x_1, \dots, x_{m-1})$$

with $U \in \mathbb{Z}[x_m]$, $V \in \mathbb{Z}[x_1, \dots, x_{m-1}]$, and

$$\|U\|, \|V\| \leq q_1^{2\omega r_1 k}.$$

We showed

$$\mathrm{ind}_{r}\left(V, \left(\frac{a_1}{q_1}, \dots, \frac{a_{m-1}}{q_{m-1}}\right)\right) \leq \frac{k\varepsilon^2}{12}.$$

This time we consider

$$\mathrm{ind}_{kr_m}\left(U, \frac{a_m}{q_m}\right)$$

and apply the inductive hypothesis with $m = 1$. We choose ε_0 such that

$$\|U\| \leq q_1^{2\omega r_1 k} \leq q_m^{2\omega r_m k} \leq q_m^{\omega(1,\varepsilon_0) r_m k}.$$

Again set $\varepsilon_0 = \frac{\varepsilon^2}{12}$, since, using $m \geq 2$,

$$2\omega = 2\omega(m, \varepsilon) \leq \omega\left(1, \frac{\varepsilon^2}{12}\right).$$

Hence

$$\mathrm{ind}_{kr_m}\left(U, \frac{a_m}{q_m}\right) \leq \frac{\varepsilon^2}{12},$$

and therefore

$$\mathrm{ind}_{r_m}\left(U, \frac{a_m}{q_m}\right) \leq \frac{k\varepsilon^2}{12}.$$

By Lemma 3.1.2, it follows that

$$\mathrm{ind}_{\vec{r}}\left(W, \left(\frac{a_1}{q_1}, \dots, \frac{a_m}{q_m}\right)\right) \leq \frac{k\varepsilon^2}{12} + \frac{k\varepsilon^2}{12} = \frac{k\varepsilon^2}{6}.$$

We use this to bound

$$I = \mathrm{ind}_{\vec{r}}\left(P, \left(\frac{a_1}{q_1}, \dots, \frac{a_m}{q_m}\right)\right).$$

By Lemma 3.1.2, we have

$$\begin{aligned}
\mathrm{ind}_{\vec{r}}\left(P^{[\iota(i), j-1]}, \left(\frac{a_1}{q_1}, \dots, \frac{a_m}{q_m}\right)\right) &\geq I - \left(\frac{i_1}{r_1} + \dots + \frac{i_{m-1}}{r_{m-1}} + \frac{j-1}{r_m}\right) \\
&\geq I - \frac{i_1 + \dots + i_{m-1}}{r_{m-1}} - \frac{j-1}{r_m} \\
&\geq I - \frac{r_m}{r_{m-1}} - \frac{j-1}{r_m} \\
&\geq I - \omega - \frac{j-1}{r_m} \\
&\geq I - \frac{\varepsilon^2}{24} - \frac{j-1}{r_m},
\end{aligned}$$

recalling that $\iota(i) = (i_1, \dots, i_{m-1})$, $i_1 + \dots + i_{m-1} \leq i - 1 \leq k - 1 \leq r_m$, $r_m \leq \omega r_{m-1}$, and $\omega \leq \omega(2, \varepsilon) = \frac{\varepsilon^2}{24}$.
Recall

$$W(x_1, \dots, x_m) = \det\left(P^{[\iota(i), j-1]}\right)_{1 \leq i, j \leq k}.$$

Applying Lemma 3.1.2 to the determinant expansion gives

$$\begin{aligned}
\frac{k\varepsilon^2}{6} &\geq \mathrm{ind}_{\vec{r}}\left(W, \left(\frac{a_1}{q_1}, \dots, \frac{a_m}{q_m}\right)\right) \\
&\geq \min_{\sigma \in S_k} \mathrm{ind}_{\vec{r}}\left(\prod_{i=1}^{k} P^{[\iota(i), \sigma(i)-1]}, \left(\frac{a_1}{q_1}, \dots, \frac{a_m}{q_m}\right)\right) \\
&\geq \sum_{j=1}^{k} \max\left(I - \frac{\varepsilon^2}{24} - \frac{j-1}{r_m}, 0\right) \\
&\geq -\frac{k\varepsilon^2}{24} + \sum_{j=1}^{k} \max\left(I - \frac{j-1}{r_m}, 0\right).
\end{aligned}$$

Hence

$$\frac{1}{k} \sum_{j=1}^{k} \max\left(I - \frac{j-1}{r_m}, 0\right) \leq \frac{5\varepsilon^2}{24} < \frac{\varepsilon^2}{4}.$$

Finally, we consider two cases. Suppose first that $I \geq \frac{k-1}{r_m}$. Recall that $k-1 \leq r_m$. Then

$$\begin{aligned}
\frac{\varepsilon^2}{4} &> \frac{1}{k}\sum_{j=1}^{k} \max\left(I - \frac{j-1}{r_m}, 0\right) \\
&= \frac{1}{k}\sum_{j=1}^{k} \left(I - \frac{j-1}{r_m}\right) \\
&= I - \frac{k-1}{2r_m} \\
&= \frac{I}{2} + \frac{1}{2}\left(I - \frac{k-1}{r_m}\right) \\
&\geq \frac{I}{2}.
\end{aligned}$$

Hence $I < \frac{\varepsilon^2}{2} < \varepsilon$, as desired.
On the other hand, suppose $I < \frac{k-1}{r_m}$. Then

$$\begin{aligned}
\frac{\varepsilon^2}{4} &> \frac{1}{k}\sum_{j=1}^{k} \max\left(I - \frac{j-1}{r_m}, 0\right) \\
&= \frac{1}{k}\sum_{1 \leq j \leq \lfloor I r_m \rfloor + 1} \left(I - \frac{j-1}{r_m}\right) \\
&= \frac{1}{k}(\lfloor I r_m \rfloor + 1)\left(I - \frac{\lfloor I r_m \rfloor}{2r_m}\right) \\
&\geq \frac{1}{k}(\lfloor I r_m \rfloor + 1)\frac{I}{2} \\
&\geq \frac{I^2 r_m}{2k} \\
&\geq \frac{I^2}{4},
\end{aligned}$$

since $k \leq r_m + 1 \leq 2r_m$. Thus $I \leq \varepsilon$ in this case as well. □

Proof of Roth's theorem. We have a real algebraic integer α of degree n. Let $0 < \delta < \frac{1}{20}$ and $0 < \varepsilon < \frac{\delta}{100}$. Suppose

$$\mathcal{R} = \left\{ \frac{a}{q} \in \mathbb{Q} : \left|\alpha - \frac{a}{q}\right| < \frac{1}{q^{2+\delta}},\ q > 0,\ \gcd(a,q) = 1 \right\}$$

is infinite. Set $m > 8n^2\varepsilon^{-2}$ and $\omega = \omega(m, \varepsilon)$ as in Theorem 3.1.9.
First we fix $\frac{a_1}{q_1} \in \mathcal{R}$ such that

(i) $q_1^{\varepsilon} > 64(H(\alpha) + 1)\max\{1, |\alpha|\}$,

(ii) $q_1^{\omega} \geq 2^{3m}$ and

(iii) $q_1^{\omega} \geq 4^m (H(\alpha) + 1)^m$.

Next we choose $\frac{a_2}{q_2}, \ldots, \frac{a_m}{q_m} \in \mathcal{R}$ such that

$$\frac{1}{2}\omega \log q_{h+1} \geq \log q_h \quad (\ \forall 1 \leq h \leq m-1).$$

Note, in particular, that $q_{h+1} > q_h$.
We then fix $r_1 \in \mathbb{N}_{\geq 1}$ satisfying

$$\varepsilon r_1 \log q_1 \geq \log q_m$$

and set

$$r_h = \left\lfloor \frac{r_1 \log q_1}{\log q_h} \right\rfloor + 1 \quad (\ \forall 2 \leq h \leq m).$$

Then we have

$$q_h^\varepsilon \geq q_1^\varepsilon > 64(H(\alpha) + 1) \max\{1, |\alpha|\},$$

and

$$r_1 \log q_1 \leq r_h \log q_h \leq r_1 \log q_1 + \log q_h \leq (1 + \varepsilon) r_1 \log q_1.$$

Hence the conditions of Theorem 3.1.7 are satisfied, and we get a nonzero

$$P \in \mathbb{Z}[x_1, \dots, x_m]$$

with the desired properties.
We also have the conditions of Theorem 3.1.9:

$$q_h^\omega \geq q_1^\omega \geq 2^{3m} \quad (\ \forall 1 \leq h \leq m),$$
$$q_h^{r_h} \geq q_h^{\frac{r_1 \log q_1}{\log q_h}} = q_1^{r_1},$$

and

$$\frac{r_{h+1}}{r_h} \leq \frac{\frac{2 r_1 \log q_1}{\log q_{h+1}}}{\frac{r_1 \log q_1}{\log q_h}} = \frac{2 \log q_h}{\log q_{h+1}} \leq \omega.$$

Finally, by Theorem 3.1.7,

$$\begin{aligned} \|P\| &\leq (4H(\alpha) + 4)^{r_1 + \dots + r_m} \\ &\leq (4H(\alpha) + 4)^{m r_1} \\ &\leq q_1^{\omega r_1}. \end{aligned}$$

Thus the conditions of Theorem 3.1.9 are satisfied. We obtain

$$\frac{\delta m}{8} \leq \operatorname{ind}_{\vec{r}} \left(P, \left(\frac{a_1}{q_1}, \dots, \frac{a_m}{q_m} \right) \right) \leq \varepsilon,$$

where the left inequality comes from Theorem 3.1.7 and the right inequality from Theorem 3.1.9. This is a contradiction, because

$$\frac{\delta m}{8} \leq \varepsilon < \frac{\delta}{100} \implies m < \frac{8}{100}.$$

Therefore, $\mathcal{R}$ is finite. $\square$

Comment. The proof does not lead to an effective bound for $\#\mathcal{R}$. The proof of Roth's theorem goes something like this: if we have good approximations

$$\frac{a_1}{q_1}, \dots, \frac{a_m}{q_m}$$

with $q_1 > Q_0(\delta, \alpha, m)$ and

$$q_m > q_{m-1} > \cdots > q_1$$

with sufficiently large gaps between each q_{i+1} and q_i, then we get a contradiction. The proof does not give some effectively computable $Q_{\text{eff}} > 0$ such that any good approximation $\frac{a}{q}$ satisfies $q \leq Q_{\text{eff}}$.

Lecture 20 (1st July 2026 - W11L1)

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Lecture 21 (8th July 2026 - W12L1)

Remark (Rough sketch of a proof of Thue's Theorem 1.1.6)**:** This proves that

$$\mathcal{R} = \left\{ \frac{a}{q} \in \mathbb{Q} : \left| \alpha - \frac{a}{q} \right| < q^{-\gamma}, q > 0, \gcd(a, q) = 1 \right\}$$

is finite, provided that $\gamma > \frac{n+2}{2}$ where $\deg \alpha = n$. Let $\varepsilon > 0$ be small. We construct $P \in \mathbb{Z}[X_1, X_2]$ of the form $P(x_1, x_2) = P_1(x_1)x_2 + P_0(x_1)$, with

$$\frac{\partial^j P}{\partial (x_1)^j}(\alpha, \alpha) = 0 \ \forall_{1 \leq j \leq L-1},$$

$$\|P\| \leq C(\alpha, \varepsilon)^L \text{ and}$$

$$\deg_{x_1} P \leq (1 + \varepsilon)\frac{n}{2}L.$$

(The proof that such a P exists is similar to the proof of Theorem 3.1.5.) We suppose that $\mathcal{R}$ is infinite and choose $\frac{a_1}{q_1} \in \mathcal{R}$ with q_1 sufficiently large in comparison to α as well as ε. Also choose $\frac{a_2}{q_2} \in \mathcal{R}$ with q_2 sufficiently large in comparison to q_1. Let $z = \left(\frac{a_1}{q_1}, \frac{a_2}{q_2}\right)$. The bound $\|P\| \leq C(\alpha, \varepsilon)^L$ gives a bound on the order of vanishing of P at z. With L chosen appropriately, we get that, for some $1 \leq j \leq \varepsilon L$, $Q = P^{[(i_1, 0)]}$ satisfies

$$\frac{\partial^j Q}{\partial (x_1)^j}(\alpha, \alpha) = 0 \ \forall_{1 \leq j \leq (1-\varepsilon)(L-1)}$$

and

$$\deg_{x_1}(Q) \leq (1 + \varepsilon)\frac{n}{2}L, \|Q\| \ll C(\alpha, \varepsilon)^L, Q(z) \neq 0.$$

It follows that

$$q_1^{-\frac{n}{2}L-1} \leq Q(z) \leq q_1^{-\gamma L},$$

where the lower estimate uses the non-vanishing of the integer polynomial at $z \in \mathbb{Q}$ and the upper bound use the Taylor series at (α, α) and z near (α, α). Thus,

$$\frac{n+1}{2} \geq \gamma \ \lightning.$$

For details, we refer to *Polynomial Methods*[2] by Larry Guth.
Recall that in our proof of Roth's theorem, we made use of simultaneous approximation, where m can be arbitrarily large. Roth was able to impose strong vanishing conditions on the polynomial P by making use of these m simultaneous approximations.

Chapter 4

Schmidt's Subspace Theorem

Notational remark. We write

$$\|\cdot\|_\infty : \mathbb{R}^n \to \mathbb{R} : x = (x_1, \dots, x_n) \mapsto \max_{i \in \{1,\dots,n\}} |x_i|.$$

Theorem 4.0.1 (Schmidt)

Suppose $L_1(x), \dots, L_n(x)$ are linearly independent linear forms in $x = (x_1, \dots, x_n)$ with real algebraic coefficients. Then, given any $\delta > 0$, there are finitely many rational subspaces (defined by linear equations with coefficients in $\mathbb{Q}$) $T_1, \dots, T_N \subset \mathbb{R}^n$ such that

$$\{x \in \mathbb{Z}^n \setminus \{0\} : \|L_1(x) \cdots L_n(x)\|_\infty < \|x\|_\infty^{-\delta}\} \subseteq \bigcup_{i=1}^N T_i.$$

We shall now state a stronger theorem. We then prove that this implies Theorem 4.0.1. After that, we shall turn to the proof of this theorem instead.

Theorem 4.0.2 (Strong Subspace Theorem)

Suppose that $L_1, \dots, L_n$ are linearly independent in $x = (x_1, \dots, x_n)$ with real algebraic coefficients $c_1, \dots, c_n \in \mathbb{R}$ such that $c_1 + \cdots + c_n = 0$. Given $Q > 0$, define

$$\Pi(Q) = \{x \in \mathbb{R}^n : \|L_j(x)\|_\infty \leq Q^{c_j} \ \forall_{1 \leq j \leq n}\}.$$

Let $\lambda_1(Q), \dots, \lambda_n(Q)$ be the successive minima of $\mathbb{Z}^n$ with respect to $\Pi(Q)$. Suppose there exists $\delta > 0$, $1 \leq d \leq n-1$ and an unbounded set $\mathfrak{N} \subseteq \mathbb{R}_{>0}$ such that every $Q \in \mathfrak{N}$ satisfies $\lambda_d(Q) < \lambda_{d+1}(Q) Q^{-\delta}$. Then, there exists a rational subspace $S_0 \subseteq \mathbb{R}^n$ of dimension d and an unbounded subset $\mathfrak{N}_0 \subseteq \mathfrak{N}$ such that given any $Q \in \mathfrak{N}_0$, the first d successive minima are achieved by points in S_0.

This corollary supports us in deducing Theorem 4.0.1

Corollary 4.0.3

Let $L_1, \dots, L_n$ and $c_1, \dots, c_n$ be as in Theorem 4.0.2. Let $\delta > 0$. Then there are finitely many rational subspaces $T_1, \dots, T_N \subset \mathbb{R}^n$ such that if $0 \neq x \in \mathbb{Z}^n$ satisfies

$$\|L_i(x)\|_\infty \leq \|x\|_\infty^{c_i - \delta} \ \forall_{1 \leq i \leq n}, \tag{$*$}$$

then $x \in \bigcup_{i=1}^N T_i$.

Proof by Contradiction. Suppose that this is not the case (i.e. x is not contained in finitely many proper rational subspaces of $\mathbb{R}^n$). Let $x_1, \dots, x_n \in \mathbb{Z}^n \setminus \{0\}$ be linearly independent over $\mathbb{Q}$ satisfying ($*$). Then, we recursively define $x_{n+1}, x_{n+2}, \dots$ such that any n of $\{x_j\}_{j=1}^{\infty}$ are linearly independent over $\mathbb{Q}$. We then let $Q = \|x_j\|_\infty$ for some $j \geq 1$ with $\|x_j\|_\infty$ sufficiently large. Let $g_1, \dots, g_n \in \mathbb{Z}^n$ be such that $g_i \in \lambda_i(Q)\Pi(Q)$ $\forall_{1 \leq i \leq n}$. We know that these are linearly independent over $\mathbb{Q}$. Let $S_i(Q) = \mathrm{Span}_{\mathbb{R}}\{g_1, \dots, g_i\}$. Since x_i satisfies ($*$), we have

$$\|L_i(x_j)\|_\infty \leq \|x_j\|_\infty^{c_i - \delta} = Q^{-\delta}|x_j|^{c_i} = \lambda |x_j|^{c_i}$$

with $\lambda = Q^{-\delta}$ $\forall_{1 \leq j \leq n}$. Then,

$$x_j \in \mathbb{Z}^n \cap \lambda\Pi(Q).$$

Therefore, $\lambda_1(Q) \leq \lambda = Q^{-\delta}$. Since $\mathrm{vol}(\Pi(Q))$ is a constant that depends only on $L_1, \dots, L_n$ and n, by Minkowski's Second Theorem 1.5.2, we have

$$1 \ll \lambda_1(Q) \cdots \lambda_n(Q) \ll 1.$$

Thus, $\lambda_n(Q) > 1$ (because otherwise, $1 \ll \lambda_1(Q) \cdots \lambda_n(Q)$ and $\lambda_1(Q) \cdots \lambda_n(Q) \leq Q^{-\delta}$, i.e. $1 \ll Q^{-\delta}$ ↯). In particular, it follows that $x_{j-1} \in S_{n-1}(Q)$. We let $k = k(j)$ be the smallest number such that $x_j \in S_k(Q)$. Then $\lambda_k(Q) \leq Q^{-\delta}$, and combined with $\lambda_n(Q) > 1$, we can deduce that

$$\exists_{d \in \{k, \dots, n-1\}} : \lambda_d(Q) < \lambda_{d+1}(Q) Q^{-\frac{\delta}{n}}.$$

By the pigeonhole principle, there exists a subsequence $x_{j_1}, \dots$ and $d \in \{1, \dots, n-1\}$ such that $\lambda_d(Q) < \lambda_{d+1}(Q)Q^{-\frac{\delta}{n}}$ with $Q = \|x_{j_i}\|_\infty$. We can now apply Theorem 4.0.2 with $\mathfrak{N} = \{|x_{j_i}| : i \geq 1\}$. But $x_{j_i} \in S_k(Q) \subseteq S_d(Q) = S_0$, which contradicts the linear independence of the $x_{j_1}, \dots$. ↯ □

We are now ready to deduce Theorem 4.0.1.

Proof of Theorem 4.0.1. First, the statement is clear for $x \in \mathbb{Z}^n$ with $L_1(x), \dots, L_n(x) = 0$.

Exercise 4.1 – L

t $1 \leq j \leq n$. Then there exists a rational subspace $\widetilde{T}_j \subset \mathbb{R}^n$ such that

$$\{x \in \mathbb{Z}^n : L_j(x) = 0\} \subset \widetilde{T}_j.$$

(Just use an integral basis of $\mathbb{Q}(\alpha_{j_1}, \dots, \alpha_{j_n})$, where $L_j(x) = \alpha_{j_1}x_1 + \cdots + \alpha_{j_n}x_n$.) Thus, we only consider $x \in \mathbb{Z}^n$ with $L_1(x), \dots, L_n(x) \neq 0$. Then there exists $C > 0$ such that

$$\|x\|_\infty \leq \|L_j(x)\|_\infty \leq \|x\|_\infty^2 \ \forall_{1 \leq j \leq n}$$

for all $\|x\|_\infty$ sufficiently large. We divide the interval $[-c, 2)$ into

$$[-c, 2) = \bigcup_{t=0}^{M} [a_t, b_t)$$

with $0 < b_t - a_t \leq \frac{\delta}{2n}$. Take any n of these intervals and consider

$$\|x\|_\infty^{a_{t_j}} \leq \|L_j(x)\|_\infty < \|x\|_\infty^{b_{t_j}} \ \forall_{1\leq j\leq n}.$$

Since

$$\|L_1(x)\cdots L_n(x)\|_\infty < \|x\|_\infty^{-\delta},$$

we only need to consider $a_{t_1} + \cdots + a_{t_n} < -\delta$. Then set $c_j = b_{t_j} - \frac{1}{n}(b_{t_1} + \cdots + b_{t_n})$, and this satisfies $c_1 + \cdots + c_n = 0$. Thus, $b_{t_j} \leq c_j - \frac{\delta}{2n}$, which implies $\|L_j(x)\|_\infty < \|x\|^{b_{t_j}} \leq \|x\|^{c_j - \frac{\delta}{2n}}$. By Corollary 4.0.3 (and since $(M+1)^n \ll 1$), we get the finite number of rational subspaces. $\square$

Lecture 21 (8th July 2026 - W12L1)

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Lecture 22 (10th July 2026 - W12L2)

There will be no questions about the following in the exams.

Lecture 22 (10th July 2026 - W12L2)

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Lecture 23 (15th July 2026 - W13L1)

Appendix

List of Definitions

List of Theorems

List of Lemmata

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Bibliography

[1] John William Scott Cassels. *An introduction to the geometry of numbers*. Classics in Mathematics. Springer Science & Business Media, 1997. ISBN: 978-3-540-61788-4.

[2] Larry Guth. *Polynomial methods in combinatorics*. eng. University lecture series vol. 64. Providence R.I: American Mathematical Society, 2016. ISBN: 9781470428907. URL: https://toc.library.ethz.ch/objects/pdf03/e01_978-1-4704-2890-7_01.pdf.